LONGITUDINAL TIME-TO-EVENT GRAPH MINING PIPELINE FOR MUSCULOSKELETAL INJURY FORECASTING

by

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A thesis submitted in partial fulfillment of the
requirements for the Doctor of Philosophy
degree in Informatics (Health Informatics)
in the Graduate College of
The University of Iowa

August 2021

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To my endlessly supportive parents.

ACKNOWLEDGEMENTS

This thesis was far from a solo effort. I would first like to thank Iowa Technology Institute and Malum Terminus Technology Incorporated for funding parts of this project. I am also grateful for University of Iowa Department of Intercollegiate Athletics who graciously allowed me to use their data as the lifeblood of this thesis.

I would like to express gratitude to my advisor, Dr. Karim Abdel-Malek, for taking me under his wing and supporting me up in time of need. I also thank Dr. Kevin Kregel for his thoughtful questions which steered the focus of this thesis, Dr. Daniel Sewell for inaugurating me to the statistical analysis of graphs, Dr. Guadalupe Canahuate for her support, constructive feedback and attentive revisions, and Dr. Bijaya Adhikari for guiding me into the fascinating field of graph mining.

I wish to express sincere thanks to Dr. Matt Andre for his mentorship, friendship, and for teaching me what it means to be a scientist. A debt of gratitude is also owed to Bill Maxwell for his open arms and widsom, and Terry Noonan for always having my back. I place on record my appreciation for Landon Evans who showed me my first graph of which sparked my burning obsession for graph theory. He consistently urged me to run with my passion and questioned me when he thought necessary – I thank him for that. I also want to acknowledge Scott Keplinger for his patience and unceasing encouragement.

Finally, I must express my very profound gratitude to my parents, significant other, and friends, for providing me with unfailing support along this venture. Thank you.

ABSTRACT

Injury is pervasive to sport and is an important yet challenging forecasting problem with high practical value. Athlete monitoring technologies are deployed for health and well-being surveillance, but incidence rates have not decreased in proportion to their widespread use. Existing injury prediction algorithms have deployed off-the-shelf machine learning solutions, leading to mis-formulated approaches that often fail to appreciate the complexities of an athlete's ever-evolving physiology. This has resulted in a gap between machine learning capabilities and their actual effectiveness in mitigating athletic injuries. The focus of this thesis is to contribute a pragmatic forecasting framework to mitigate sports-related, non-contact musculoskeletal injuries. Chapter 1 positions my research perspective under the theory of complexity. I formulate injury forecasting as a longitudinal time-to-event problem and specify three specific aims: 1) select an intra-individual dynamic graph construction approach, 2) develop a longitudinal time-to-event forecasting model for dynamic graphs, and 3) design an architecture to identify temporal edges contributing to an athlete's injury forecast.

Chapter 2 addresses Specific Aim #1 by scouring for intra-individual dynamic graph construction methods which fit the philosophical pillars of complexity theory. A range of synthetic datasets are then simulated to validate temporal edge reconstruction performance of the most competitive solutions which assume temporal smoothness. The experimental results are meant to guide the decision of which dynamic graph estimation procedures will be deployed in the grand pipeline in later chapters.

Chapter 3 seeks to contribute a longitudinal time-to-event forecasting model from dynamic graphs as inputs for Specific Aim #2. The proposed solution braids deep learning and graph mining domains into a custom time-to-event formulation. The proposed solution exhibited superior discriminative forecasting performance over two benchmark algorithms, especially with long-range graph trajectories.

Chapter 4 addresses Specific Aim #3 focuses on selecting temporally important edges to help explain the forecast provisioned to a particular dynamic graph. A temporal perturbation is formulated inside an optimization scheme. Extensive simulations test whether the local temporal perturbations can detect chronologically relevant edges compared to local static methods and global temporal methods.

Chapter 5 coalesces the three Specific Aims into an end-to-end application for sport epidemiology. An authentic athlete monitoring dataset from University of Iowa Department of Intercollegiate Athletics is used as an operational example. Athlete-specific dynamic graphs are constructed from longitudinal force plate data using the methodology from Chapter 2. The longitudinal time-to-event model from Chapter 3 is trained and tested to forecast non-contact musculoskeletal injury from the dynamic graph dataset. Athlete-specific injury forecasts are then explained using the architecture from Chapter 4, where novel idiographic inferences into the temporal interactions leading to injury are unveiled.

PUBLIC ABSTRACT

Injury is pervasive to sport and is an important yet challenging forecasting problem with high practical value. Athlete monitoring technologies are deployed for health and well-being surveillance, but incidence rates have not decreased in proportion to their widespread use. Existing injury prediction algorithms have deployed off-the-shelf machine learning solutions, leading to mis-formulated approaches that often fail to appreciate the complexities of an athlete's ever-evolving physiology. This has resulted in a gap between machine learning capabilities and their actual effectiveness in mitigating athletic injuries. The focus of this thesis is to contribute a practical and accurate forecasting framework to mitigate sports-related, non-contact musculoskeletal injuries. Chapter 1 positions my research perspective under the theory of complexity. I formulate injury forecasting as a longitudinal time-to-event problem and specify three specific aims addressed in Chapters 2-4: 1) select an intra-individual dynamic graph construction approach, 2) develop a longitudinal time-to-event forecasting model for dynamic graphs, and 3) design an architecture to identify temporal edges contributing to an athlete's injury forecast. This thesis ends by threading Chapters 2-4 into an end-to-end graph mining pipeline for sport epidemiology. An athlete monitoring dataset from University of Iowa Department of Intercollegiate Athletics serves as an applied example. Athlete-specific dynamic graphs are constructed from longitudinal force plate data. The longitudinal time-to-event model was trained and tested to forecast musculoskeletal injury from the dynamic graph dataset. Athlete-specific injury forecasts are then explained where novel inferences into the temporal interactions leading to injury are unveiled.

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CHAPTER 1

INTRODUCTION

1.1 The Problem

It's the biggest day of his life, the 400 meter semi-finals at the Barcelona '92 Games of the XXV Olympiad. Derek Redmond, the favored quarter-miler of the Games, is rounding the first turn in lane five. As he reaches top speed in the middle of the straightaway he is forced to clutch his right glute and gimp to a halt, watching the heat dart ahead without him. Derek's hamstring partially tore off his ischial tuberosity while running in his own lane, free of contact. His proximal hamstring tissue was unable to cope with the intrinsic demands to impart such velocity. Yet, Derek has ran his entire life. He sprinted at the international level seven years prior to the '92 Games. Why was his hamstring intolerant of the requisite torque on this important day?

With one out in the eighth inning, Detriot Tigers reliever, Joel Zumaya, let another one of his blazing fastballs fly at Target Field in Minneapolis, MN. This 99 miles per hour full count pitch left Zumaya collapsed behind the mound writhing in pain. The violent angular velocity buckled his elbow with a non-displaced olecranon fracture. What's peculiar is that Zumaya was asymptomatic in the pitches leading up to the blowout. In fact, his humeroulnar joint had supported over 1000 pitches above 100 miles per hour before this throw; his elbow was prepared to perform the task. But in that moment, his elbow's readiness was compromised. The mechanical load suddenly exceeded the limits of his connective tissue like the end of a sparkling fuse.

Musculoskeletal injuries are not confined to the competitive spotlight. One of the

most decorated athletes in basketball history, Sue Bird, tore her anterior cruciate ligament performing a routine practice drill in her freshman year at University of Connecticut. Sport-related injuries are also not confined to the elite – injury is omnipresent from Little League to Olympians. But being an athlete is commonly viewed as positive for both mind and body, and that sport is practiced by healthy bodies experiencing healthy outcomes. While the sphere of sport is crowded by people on and off the field who wish this were true, anyone who has played, coached, administered or watched from the sidelines can attest that this view is partly a misconception. Nearly every athlete has a story to tell about injury, whether it be themselves or about another teammate. Between 2009-2014, sport medicine had become a burgeoning billion-dollar industry [14, 53], where orthopedists, athletic trainers, and physical therapists treat hundreds-of-thousands of sport-related injuries each year.

The common denominator from the stories above is that these injuries happened in isolation from others. Such injuries are termed non-contact, meaning they were not a result of body collisions. This type of sports injury primarily occurs during the physical preparation or competition of a variety of individual or team sports such as baseball, volleyball, soccer, or tennis, *inter alia*. Cyclic sports, such as rowing, swimming, or long-distance running, are also culturally accustomed to overuse syndrome [56], which results from chronic internal stressors generated during training. Generally speaking, non-contact musculoskeletal injuries cause structural damage from mechanical stress to which the tissue cannot adapt to in time, and are presumably possible to eliminate by means of appropriate training management guided by consistent athlete monitoring [64, 83].

The commercialization of non-contact sports injury mitigation has spawned a "Cam-

brian explosion" of athlete monitoring technologies. Teams are now fully integrated with assorted devices, inundated with high-dimensional data, and rely on companies' trite algorithms to manage training prescriptions. But incidence rates have not decreased in proportion to the vertical market success or plethora of research. New methods are needed. The focal point of this thesis is to mitigate sports-related, non-contact musculoskeletal injuries by contributing a practical and accurate forecasting framework.

Conversely, contact sport athletes, who are are sanctioned to weaponize their bodies in the form of mock battles, are not only exposed to non-contact injuries but also injuries resulting from body contact. Contact sports such as American football, rugby, or ice hockey, have become enclaves of socially accepted violence which engender another gamut of musculoskeletal injuries that naturally come with the territory. While contact sports are not excluded from this thesis, unavoidable injuries from body contact are unrealistic to forecast via athlete monitoring and are outside the scope of this thesis.

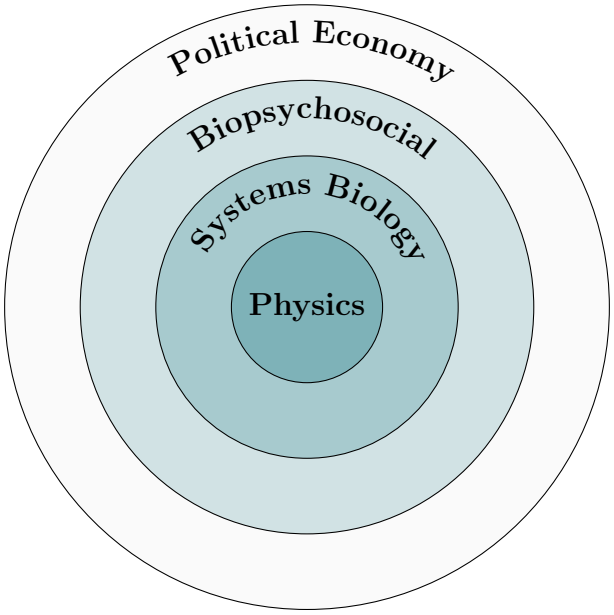
Athletes suffer from a wide variety of aliments that are not commonplace within the general population, although they may arise in the enthusiastic individual who haphazardly engages in unaccustomed activity. Non-contact musculoskeletal injuries also occur in industries involving people who undertake new jobs requiring unaccustomed or repetitive activity, such as assembly line workers. Occupations where physical performance is consequential, for example the military, also suffer from increased rates of musculoskeletal injuries since developing elite physical fitness is imperative. Musculoskeletal injuries from these settings are not the primary narrative of this thesis, but the applications are certainly germane.

1.1.1 Worldview

Academia is guilty of demarcating researchers into neatly trimmed bins. I argue that the complexity of musculoskeletal injury requires interdisciplinarity, where siloed perspectives may hinder our scientific progress. In the same breath, I must carve a boundary for myself to insert my sliver of contributions into the research landscape to avoid sounding amorphous. Figure 1.1 illustrates how I view the fluid layers of inquiry into the sports injury predicament, and defines the worldview for which this thesis embraces.

Among the most important questions any researcher must ask when examining sport-related injuries are "what causes it?" and "why does this happen?". Unsurprisingly, there is a range of appropriate answers that are striated across disciplinary lines.

Figure 1.1. Layers of Inquiry



At the highest level, an injury can be understood as a structural breakdown of the human organism, a breakdown that affects its function. But to explain the causal mechanisms of such breakdown is utterly intertwined among multiple disciplines. A biomechanist's explanation will sound very different from a neurologist's explanation. And a "biomechanical" explanation for an injury is merely an interpretive convenience for approaching the multifaceted problem.

A season-ending musculoskeletal injury just occurred. Why did it happen? Our first layer of inquiry boils down to a problem of physics: connective tissue severance occurs when the forces exerted exceed the tissue's capacity. This is a generalized statement, but incontestable nonetheless. What was the joint angle microseconds before the tissue ruptured? Was this a hazardous kinetic signature that could have been avoided if the athlete was trained properly? Subdisciplines such as biomechanics, ergonomics, and engineering have produced copious volumes of epidemiological research trying to answer these types of questions. I place this perspective at the nucleus because it is the closest to the injury. Not only close in time, but also closest with respect to a conceptual framework – the most irrefutable explanation.

If we rewind a little further from the injury, we step back into a slightly larger conceptual framework, a new round of questions emerge. Why was the strength of their tissue compromised today? Think of the stories like Derek Redmond or Joel Zumaya; their tissue had successfully performed the task before. What has the athlete done, or not done, that lead their system into a maladaptive state? These questions incorporate the systematic view of an athlete's biology and concern biomedical domains such as physiology or systems biol-

ogy. This layer does not dismiss the physics, but is concerned with the influences preceding the physics event. Such a time window could include days, weeks, or months.

If we keep rewinding, our zoomed out view now encompasses factors outside the individual athlete. This next layer embodies the nexus of anthropology, sociology, and psychology, peering into the biopsychosocial motivations of why this athlete exerted themselves beyond their limits. These disciplines analyze the curious paradox of the positive imagery associated with sacrificing your body "for the good of the team". Our social structures tend to valorize sport injuries; playing a game while injured is somehow attributed to positive qualities in one's character. Fans cloak the pain by heroizing fallen athletes who rise from the court or turf. Derek Redmond received a standing ovation while hobbling to the finish line. He likely promoted further hamstring damage by finishing the race, but there was an internal motive that superseded his personal health.

The widest conceptual framework generates questions that could perhaps precede the athlete themself. What political events accelerated the commercialization of sport to a point where athletes yearn for a hazardous occupation? Athletes can easily become embroiled in social hierarchies that acquiesce to the norms of injury and pain. What organizational dynamics entrench injury risk into the power relations of sport? It appears that athletes are trapped in a swarm of political, financial, social, and personal pressures to relinquish their health in order to attain merit and victory through physical performance. This level of inquiry taps into disciplines such as history, philosophy, or socioeconomics.

By weaving through the entire research continuum now hints at something more subtle – that to explain, and thus reduce, injuries from a single discipline implicitly invokes

all the preceding disciplines. With that said, my hopes for societal reform are minimal. But one feasible strategy is to empower sport practitioners to prognostically identify biomedical risks sooner. To equip coaches and other sport staff with accurate, interpretable, and data-supported training management tools could help us retard this endemic – ennoble fitness not sacrifice, safeguard athletes not risk, and seek vitality not pain.

Referring back to Figure 1.1, this thesis resides within the systems biology layer of inquiry. While the studies and contributions inside this thesis are of the computational domain, it is important to remark the worldview of the intended application. My overarching aspiration for this culminating work is to mitigate non-contact musculoskeletal sport injuries by contributing an end-to-end forecasting architecture. Many efforts have been made for sport with similar goals but with scant progress. Approaching such a polarizing debate leads me to delineate and assert my positionality and epistemological research assumptions towards sports-related injuries. What follows is a brief contexualization of the numerous pathways a researcher can travel along and which route I chose to take.

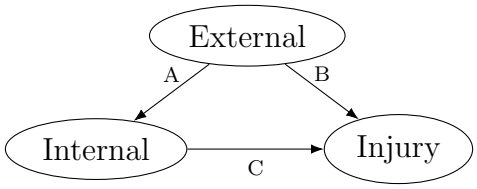
1.1.2 Research Positionality

Figure 1.2 is a crude graphical characterization of the injury research landscape. The pathways are arranged as a causal diagram to depict directionality of plausible hypothesis testing. Injury is the target node which has two incoming edges, B and C. This purposefully portrays that Injury is influenced by, and thus studied, through multiple angles. This opens up the opportunity to substantiate the structure of Figure 1.2.

Physical training has been considered the process of systematically improving physical abilities to refine sport-specific skills and capabilities [182]. A prominent theoretical

framework to conceptualize quantifiable constructs of the training management process is to split training load into two components: internal and external load [103, 83]. External load has been described as the input variable that is manipulated to elicit the desired physiological (internal) response [34]. Coaches plan and prescribe external load to elicit desired physiological responses. Consequently, internal load quantifies and reflects the instantiated physiological coping mechanisms in response to the elicited external load [163].

Figure 1.2. Research Pathways.



Certain key words in these definitions, such as 'manipulate' or 'instantiate', help position the logical flow of Figure 1.2. "There is no causation without manipulation" [98]. According to the interventionalist account of causation [190], causal effects are fleshed out on the basis of interventions: X causes Y if an intervention on X leads to a change in Y. *Prescribing* external loads *elicits* internal responses. External load variables thus capture experiences that athletes undertake, not attributes that are observed, and therefore, should be treated as interventions, not random variables. This entire concept is concisely represented by edge A, External $\rightarrow$ Internal, encompassing much of the exercise physiology research domain.

While intense training is necessary to augment an athlete's physical capacity, exposing an athlete to demanding external loads when their system is not ready to receive the stimuli can lead to musculoskeletal injury [46]. Initial physiological coping mechanisms attempt to protect the internal system through strategic adaptations [183], but can be overridden by the athlete from poor training management [65]. This has activated a research artery which investigates how external load patterns influence injury occurrences, External → Injury. For concision, edge B also includes domains within the 'physics' layer of Figure 1.1, (e.g., biomechanists mitigating injury by researching mechanical torques via external loads).

Monitoring external load is now widely practiced from commercialized athlete surveillance technologies, such as wearables quantifying real-time velocity and accrued distances. Due to their availability and implementation ease, external load variables serve as a proxy for coaches to gauge the intensity of their training prescriptions, even in chaotic team-based sports. And practitioners are often tempted to implicitly estimate internal load based on these external load derivatives. But this approach is conceptually restricted and resorts to extrapolating the External → Internal → Injury pathway.

Including edge C authorizes my assertion of the research positionality herein. Internal → Injury has been my primary research vehicle, and is expressed through my culminating work of this thesis. My contributions are intended to fill methodological gaps inside edge C. I acknowledge this singular angle is not the whole story and that incorporating the chain of edge A and C would provide a more holistic framework. Perhaps this narrow bandwidth is out of necessity; there are limits to how much one researcher can obsess over. The section that follows offers a chronological scrutiny of previous literature and more thoroughly justifies

the theoretical underpinnings that shape my specific research objectives.

1.2 Background

1.2.1 Modern Theories

Perhaps it is due the implementation ease of external load measures has triggered sport medicine professionals to be fixated on the idea that mismanaging external training loads is the sole culprit of damaging an athlete's health through non-contact injuries. Over four decades ago, [25] applied Hans Selye's General Adaptation Syndrome (GAS) theory [155] to the performance of an athlete in response to training. What is now known as the 'Banister Model', proposed that an athlete's response to training can be estimated from the difference between a negative function ('fatigue') and a positive function ('fitness'). The ideal training stimulus 'sweet spot' is the one that maximizes net performance potential by imposing an appropriate training load while limiting the negative consequences of training. But this is merely a conceptual framework adorned onto the GAS theory – not a pragmatic solution.

Based on Banister's work, researchers introduced the concept of the acute:chronic workload ratio to model the relationship between changes in load and injury risk [64, 65]. This ratio describes the acute training load (e.g., the external training load of the last week) to the chronic load (e.g., the four-week rolling average of load). If chronic load has been systematically increased to high levels (i.e., the athlete has developed fitness) and the acute load is low (i.e., the athlete is experiencing minimal fatigue), then the athlete is considered well prepared. Conversely, if acute load exceeds the chronic (i.e., acute loads have rapidly increased, or training over the last four-weeks has been inadequate to develop fitness), then

the athlete is considered unprepared and likely at an increased risk of injury.

This method attempts to quantify Banister's positive and negative functions of training prescription. Empirical evidence from three different sports (Australian football, cricket and rugby league) has demonstrated that injury likelihood is low ($< 10\%$) when the acute:chronic ratio is within the range of 0.8-1.3 [99, 64, 20]. Whereas, when the acute:chronic ratio exceeds 1.5 (i.e., the load in the most recent week is 1.5 times greater than the average of the last four weeks), the likelihood of injury more than doubles. The data from these retrospective cohorts imply that athletes respond better to relatively small increases (and decreases) rather than larger fluctuations in loading patterns. Provided that an athlete reaches these loads in a gradual and controlled fashion, high loads and physically hard training appear to offer a protective effect against injuries, due to the mediating effect on adaptation and the development of physical capacities.

Early critics stressed that this acute:chronic ratio method might comprise research inferences and practical utility due to the mathematical coupling producing spurious correlations irrespective of biological association [129]. Others reproached the method's subjectivity, where the performance exhibits non-trivial differences when using different arbitrary ratio (e.g., 1:5 week, 1:3 week, 3:10 day, etc) [26]. [55] conducted a prospective cohort and found that although associations were present, injury prediction (forecasting) accuracy was paltry via generalized estimating equations. And although [138] estimated that large increases ($> 60\%$) in weekly handball load increased shoulder injury rates in elite youth handball players, the authors cautioned the field that whatever method is chosen (acute:chronic or not), must not be interpreted in isolation. Rather, it must be interpreted in conjunction

with numerous dimensions as injury is a multifaceted concoction of external and internal load.

A pair of epidemiologist then teamed with the developer of the workload ratio method to model injury data as a survival analysis problem. Their three-year prospective cohort gave an example of time-to-event modeling of sports injuries [67]. A Cox proportional hazards regression model was used to identify pre-season factors associated with a high hazard ratio of injury. Athletes whom were heavier, faster, and with poorly developed high-intensity intermittent running ability, had a higher incidence of injuries. A follow-up prospective cohort observed supporting evidence that greater distance covered at low speeds in pre-season were associated with lower injury risks during season, whereas greater amounts of high-velocity running increased the risk of injury [66]. While time-to-event methodology is well-suited for the problem at hand, their tactic of only using baseline (point in time) measurements crumbles when trying to physiologically substantiate that pre-season fitness levels are held constant throughout the season; an athlete's physiological state continuously evolves over time. We will return to the applicability of time-to-event modeling in later sections.

A team of Canadian physical therapists further vexed sport practitioners by shedding light on the complex reality of injury manifestation and challenged the profession to adopt complexity theory when generating injury prediction models [19]. Their emphasis was that if sports medicine professionals wish to curtail injury occurrences in such complex nonlinear beings then the predictive models must appreciate both the erratic internal milieu of the athlete as well as their multifarious external mediators. They crafted an alternative

conceptual framework that allows for patterns to emerge from a recursive web of determinants. Although this team advanced the thinking of injury prediction by recommending more frequent screening, their model was purely theoretical, did not offer quantitative solutions, and therefore, cannot be applied. [33], another physical therapist, supported this new framework by writing an editorial that spitballed ostensibly applicable modeling strategies to reflect dynamic environments, such as stochastic time-series or dynamic Bayesian networks for simulation.

A more recent group of physical therapists debated that the complexity of predicting musculoskeletal injuries is analogous to predicting the formation and trajectory of hurricanes [168]. They proposed that, as with tracking a volatile weather pattern like a hurricane, frequent sampling of biomarkers through athlete testing is a feasible prerequisite for detecting when the human system is susceptible – or resistant to – sustaining an injury. Their proposition that athletes, like hurricanes, are nonlinear dynamical systems closely aligns with my theoretical lens of complexity, to an extent. My strong reservation is that the continuous-time nature of predicting hurricane strength and path is inherently a time-series problem [121], whereas an athlete sustaining an injury is a discrete event; the authors are over abusing this analogy.

[60] then offered an experimental paradigm for constructing injury forecasting models that comply with complexity theory objectives. Instead of concentrating on identifying risk factors related to injury occurrence, they argued that a complex systems approach searches for high-order (latent) variables that describe the macroscopic dynamic behavior of an athlete. The authors presented four steps to formulate a complex systems approach for sports

injury forecasting, as: define the time resolution; define the parameters that characterize the state of the human system; capture the time evolution of selected parameters that sufficiently conforms to the time resolution; and to select methods to analyze the captured dynamics and forecast future states. Although their seemingly simplistic recipe helps sport medicine researchers enter the complexity framework, there was zero mention of the pragmatic landscape of current statistical or computational solutions. I argue that such techniques are not yet developed. Rather, I intend to draw these circuitous abstract discussion down to a tangible, theory-inspired modeling solution. But first, I must define the theory of complex systems which guides my construction of analytical strategies.

1.2.2 Complexity Theory

The theory of complex systems stems from a medley of biology, sociology, and physics [85]. By delineating how each scholastic tentacle shaped such theory I can arrive at a formal definition for which I aspire to abide by.

First, the study of complex interdependent systems is part of the core mission of biology. It is near impossible to describe biological phenomena without describing how agents (e.g., systems, organs, cells) interact with, respond to, adapt to, and influence each other in dynamic ways. For example, proteins, metabolites, and nucleic acids within any living organism are governed by a temporal sequence of genetic regulation [49]. Genetic regulation can be viewed as discrete interactions which can be well-described by graphs. The Krebs cycle, for example, is often represented as a directed graph where nodes are substances and edges are (catalytic) reactions [39]. But what biology has particularly contributed to complexity theory is the concept of adaptation. Many complex systems have the ability to

adapt to, while also being robust against, changing environments [8]. Meaning, the internal environment can shift from stable to chaotic phases in response to the external conditions. When a system is stable, it can easily return to an attractor state upon perturbations. If a system transits to a chaotic phase under perturbations, it will quickly sample large regions of phase states to locate a new stable 'solution' compatible with the perturbed environment [117]; a system adapts by sampling other optima.

Second, social systems can be thought of as a time-varying (sometimes multiplex) graph, where individuals are nodes and their interactions are described by edges. Such edges naturally change over time, whether it be friendships, familial ties, exchange of goods, etc. [164]. Graphs characterizing social systems exhibit a wide spectrum of growth patterns and high levels of plasticity; graphs restructure through edge formation, edge dissolution, or edge reformation [17]. Nodes can also possess states, such as gender, political opinion, wealth, etc. Nodes states typically have an influence on the edge dynamics of the graph; a close connection exists between network topology and node states. Quantitative sociologists and statisticians have developed tools for analyzing co-evolution of graph dynamics and node states [115, 157], where recent work has even flipped the analysis to edge dynamics and node states [156]. A crisp summary of how the social sciences have influenced complexity theory, is that nodes and edges co-evolve across a graph.

Lastly, we consider key features of complex systems that were adopted from physics. The domain of physics has a long history of being an experimental and predictive science of matter and its interactions [135]. Physics tries to understand and predict the macroscopic properties of a system on the basis of its interacting microscopic components. However,

macrostates usually cannot be inferred from properties of the microscopic components alone. Water, for example, is a macrostate from the aggregate interactions of hydrogen and oxygen. This phenomenon of collective properties arising that were unexpected from the components alone is referred to as emergence [144]. This is primarily a consequence of generalized interactions, which can be described, again, as time-dependent graphs.

To summarize, the theory of complex systems is the theory of generalized time-varying interactions between nodes characterized by states, where systems can be described, illustrated, and analyzed, by graphs. The fundamental principle of a complex system is that a graph structure (edge arrangement) may rewire as a consequence of changes in node states. Thus, complex systems are systems whose states change as a result of interactions and whose interactions change synchronously as a result of states. If we agree that an athlete's organism is a complex adaptive system, where various internal physiological systems adapt to external loads, then empirical injury forecasting methodology should follow graph-theoretic ideology. Therefore, the inferential apparatus for the present thesis is graph mining, where graphs serve as the central observational unit. But how does graph mining translate to injury forecasting?

Pulling from the disciplines described above, the set of internal load variables collected for an athlete can be represented by nodes, where dependencies can be portrayed by edges. The search for an athlete's graph structure should permit temporal evolution when deemed necessary (i.e., adaptation). A time-evolving graph signature could reveal (mal)adaptive states over time (i.e., indicating stability/chaos in respond to external loading patterns). The goal from here would be to detect hazardous signals (i.e., emergence) to forecast – and

hopeful evade – an upcoming injurious event.

Said another way, not only are nodes stochastic, but edges between nodes are also stochastic and time-dependent. These layers of stochasticity may seem opaque, but if the developed methods are inspired by the pillars of complexity theory, I believe there is hope for making empirical forecasts about the dynamics of injury manifestation. But progress in applying the science of complex systems to injury forecasting will remain idling without a clear methodology for empirically estimating the dynamics of how an athlete's internal systems interact with each other over time, and translating these emergent properties into quantified risks. This is a sizeable request which will require rigorous formulations.

1.3 Problem Formulation

Mitigating injury via forecasting is far from solved. Repeated failed attempts in the literature has caused threadbare debates of whether or not injuries are even predictable. The consensus has been that the problem is either too complex for traditional machine learning [172, 48], poor performance due to noisy or inadequate covariates [50] or due to data mining issues such as class imbalance [150]. [19] advertised pre-existing complexity theories and urged the field to pivot our lens from identifying risk factors to recognizing risk patterns. [60] described how injury forecasting must implement tools which estimate self-organizing system states and articulated an incremental progression for selecting analytical techniques.

I argue that the profession has struggled for two reasons. First, musculoskeletal injury forecasting from internal load markers has remained unformulized, and therefore, unsolved. Artificial intelligence inherently relies on mathematics. Mathematics depends on axioms, definitions, and assumptions. If these assumptions are not appropriately established before

confronting data, the nature of the situation becomes distorted, and the real-world problem cannot be solved. The process of setting up assumptions is perhaps the most important step in modeling. And as of yet, the field is guilty of grabbing off-the-shelf machine learning algorithms, injecting them with athlete monitoring data, and reporting meager discriminative performance – XGBoost cannot solve all of our problems. Instead, we must convert the orthopedic narrative into strict problem formulations that can be solved. It is only once we coherently formulate the problem that we can reap the benefits artificial intelligence has to offer. As such, every chapter herein has a dedicated section to rigorously defined and formulate the respective problem statements.

Perhaps the hindrance of formulation is due to the many intricate ways in which sports medicine describes musculoskeletal pathology, such as acute versus chronic onset, category of ailment, or gradation of condition. Such nomenclature that is acquired through professional training functions well to convey information between practitioners, organizes modalities for clinicians, and discerns training prescription for coaches. The challenge is to reduce this language into mathematical formalism. And predicting these events is dependent on the way we collect covariate information of our athletes, which governs how we define the problem. Unfortunately, athlete monitoring abounds with ambiguity and lacks regulatory practices. Defining injuries and acquiring data are thus context specific and may alter between teams or researchers.

For the sake of this thesis, musculoskeletal injury is defined as a discrete event. The word discrete easily converts to discrete or categorical dependent variable (class label). If the injuries are homogenized, the injurious event can be viewed as binary – it either happened or

not. Further, it is assumed that athlete monitoring data is collected over time per athlete as a proxy indicator for physiological readiness. Athlete monitoring can capture numerous systems under consideration which usually creates a multivariate, sometimes high-dimensional, space per athlete. This equates to a multivariate time-dependent (longitudinal) setting. But this is not a multivariate time series problem because the majority of athletes do not suffer an injury. This creates a large proportion of missing values in the dependent variable, which would wreak havoc on most time series techniques. Missing data of a discrete dependent variable can be instead formulated as a time-to-event problem in the form of censoring. The ensuing subsection will dissect and substantiate these claims in greater detail, as well as provide color for where this thesis resides in the time-to-event literature.

Aside from methodological formulations, my second argument for unsuccessful injury mitigation across the field is because the scientific approach in sport epidemiology has, in my opinion, overemphasized cohort-level findings and underappreciated the individual as the unit of analysis. This follows my first argument at a distance, but generalizes to the philosophical paradigm of how research is performed for guiding sport practitioners.

Sport epidemiological research is primarily based on analyzing (moderately) large samples such that the results can be generalized to the grand population from which the sample was taken. This type of research is termed nomothetic [188], pitted against idiographic research, which is concerned with the study of a particular individual. The idiographic-nomothetic debate has thick roots in philosophy and has been harnessed by the social sciences, but the ideology holds for physiology research and application, where each person can be modeled as their own high-dimensional physiological space.

1.3.1 Time-To-Event Modeling

One significant challenge in training longitudinal prospective prediction models is the necessity to wait for sufficient number of events to occur: it takes time to model time. Training conventional supervised learning algorithms in their purest form is called batch learning; a practitioner supplies a large batch of labeled training instances to allow a function to map known inputs to known outputs. Popular variants of supervised learning for prediction problems, such as transfer learning, imbalance learning, semi-supervised learning, and multi-task learning, are unfit for this prediction problem on the premise of warranting vast amounts of training data. This leaves practitioners waiting to accumulate enough injury instances for an algorithm to learn a concept. The second challenge is that not every athlete suffers an injury. This results in a large proportion of the dependent variable to be missing, and training prospective forecasting models with missing data is of high practical value.

Statistical time-to-event modeling, such as survival analysis, offer a powerful solution to missing data (e.g., death, admission to hospital, emergence of disease, musculoskeletal injury), called censoring [119]. Most of its rich history has approached the problem by assuming a specific form for the underlying stochastic process, leveraging available data to estimate the relationship between the covariates and parameters of the model, and then deducing these relationships and the distribution of the risk of the event. The Cox proportional hazards model [35] is the most widely-used censored regression model, but it makes heavy assumptions about the underlying stochastic process, about the relationship between the covariates and the parameters, and requires point in time data (single measurements per subject). Rather, to capture time-varying dynamics, the modeling framework must include

a temporal component.

Similar to survival data, time series methods deal with slightly different kind of time-centered analysis [87, 123]. Time series analysis tackles the problem of studying experimental data that have been observed at different points of time [84]. Although time series may be applicable to estimate evolving systems, the standard time-series methods are primarily used for forecasting the future values for existing time-series [199, 21]. In the present problem, survival estimation is used to summarize the survival times of a group of athletes while the response variable in time-series methods are outcomes depending on time which is an independent variable. Hence, although these two problems appear similar, the problem at hand is significantly different and cannot be solved using time-series methods. In the presence of censoring and when the goal is to predict an occurrence of an event (which is usually binary in nature), time-series methods in isolation are not applicable. The only common theme that connects time-series methods to survival analysis is their ability to forecast in the future based on the events that occurred until a given time point.

There has been an increasing interest in adapting machine learning techniques to survival data to escape the linearity assumptions of the Cox model [130, 122]. Decision trees [154, 176] and artificial neural networks [28, 44] for censored data represent some of the earliest works in this field. Well-known support vector machine algorithms have been adopted to accommodate censored data. More recently, advanced machine learning methods such as active learning and regularized learning have also been incorporated into survival models [181, 180]. However, comparison of the performance of these approaches show that these methods do not yield any significant improvements over the standard Cox model.

There are also few other studies which aim at handling censored data as pre-processing step by giving some weights to the censored observations [203, 167].

Bayesian approaches including Naïve Bayes and Bayesian Networks have also been widely used for classification [62] and successfully applied in many domains. However, there has been only few works in the literature using Bayesian methods for survival data [154, 189, 11]. Bayesian networks can visually represent all the relationships between the variables which makes it interpretable for the end users. This is in contrast to the empirical Naïve Bayes method that makes the independence assumption between all the independent variables [62]. Despite the applicability of Bayesian network in the survival analysis domain, only a limited number of research efforts exist for tackling the censored data challenges. The authors of [126] developed a Bayesian neural network approach to model censored data. [166] gave weights to censored instances in order to learn Bayesian networks from survival data. More recently, [11] adapts a Bayesian network for survival data using an approach called inverse probability of censored weighting for each of the record in the dataset to handle the censoring issue.

As of yet, the recurring limitation of the aforementioned time-to-event models is that they can only use a small portion of available data in a longitudinal setting. In particular, even if biomarkers are repeatedly measured per athlete, conventional survival models, ranging from regression to deep learning, are based on the last available measurement before the event indicator label. In order to detect an indicative signal of future injury given an athlete's complex physiology over time, is it vital to incorporate the longitudinal trajectory of biomarkers, rather than discarding valuable information over time.

The most common time-to-event method in the longitudinal setting is referred to joint models, which jointly estimate the repeated measurements with the time-to-event process [92]. A joint model consists of two sub-models, typically a Cox model and a linear mixed-effect model, to estimate the joint distribution of the longitudinal time-to-event data. Modeling the joint distribution has been shown to reduce bias in estimation and enable event predictions at any point throughout the time window of consideration [101]. But estimating the joint distribution involves optimizing the joint likelihood, which depends on fixed specifications; any misspecification could limit overall performance. The joint likelihood is also vulnerable to computational roadblocks when applied to high-dimensional data [93]. [13] considered nonparametric alternatives for the longitudinal sub-model through various splines, but at the cost of higher computational complexity. [128] explored the integration of latent classes to account for potentially heterogenous sub-populations, but they still retained the proportional hazard assumption which contravenes the theory of complex systems.

A different technique for handling longitudinal time-to-event data is called landmarking [90]. The idea of landmarking was to fit a survival model from subjects who are still at risk at the pre-specified landmarking time. [200] claim landmarking is partially conditional since each survival model is conditioned on the available data collected by the landmarking time, instead of including each person's full longitudinal history. Despite the fact that longitudinal measurements are not entirely modeled, which results in loss of information when making predictions, [178] showed that landmarking predictive performance was competitive with full joint modeling. Nonetheless, landmarking is unrealistic for injury forecasting; event predictions are only allowed at pre-specified landmarking times, not at which new measure-

ments are obtained. Furthermore, this method requires *a priori* assumptions regarding the underlying stochastic process of the survival model [177], consequently impeding the model to estimate the relationships between the captured biomarkers and injury.

To summarize my theoretical framework, static deep learning time-to-event methods are inappropriate for sports injury forecasting. The two most competitive time-to-event methods which incorporate longitudinal measurements either 1) impose strict assumptions and are computationally burdensome or 2) require pre-specified arbitrary landmark times and discards data from before landmark time. Furthermore, the available methods discussed thus far are not network-based and thus do not adhere to the theory of complex systems. The described lacuna motivates the need to construct algorithms that are specifically tailored to this prediction problem.

1.3.2 Idiography

The value of large-scale nomothetic research for establishing fundamental laws or causal relationships is irrefutable. But the results from nomothetic research perspectives are limited for our understanding of actual athlete experiences because the translation from the laboratory to practical settings is not always straightforward. Comparing means across groups – a typical approach in nomothetic research – provides a rather shallow picture of the underlying process where the data is being generated: the person. A common means to sport epidemiology research has been studying differences between athletes under the nomothetic perspective, typically cross-sectionally but sometimes longitudinally. These designs are deployed for the purpose of analyzing intra-individual processes and assume the distribution of variables in the population reflects the distribution within an athlete. The

fundamental question is whether this assumption is tenable; is it possible to generalize from the population level to the within-person level?

Acknowledging that cross-sectional approaches are limited to infer about intra-individual processes has surely led to alternative approaches. Longitudinal mixed-effect modeling, for example, enables inference about intra-individual processes through random effects. Hierarchical (multilevel) models are also tenable options with strong theoretical properties. These is a step in the right direction, but the random effects are not entirely within-person; the model fits the random effects weighted by the fixed effects [42]. Further, conventional longitudinal data collection scenarios are between eight and twelve repeated measurements per person [58]. And athlete monitoring collection frequencies can be on a daily basis with periods up to several consecutive months. This data collection structure is more suitable for a time series formulation.

Time series analysis is concerned with the relationships of one or more variables to themselves and each other over time within a single entity. It is a class of techniques that are frequently visited in econometrics but also in biological disciplines. The latter applications are based on viewing the individual as a (complex) stochastic system that varies over time. In this view, time series analysis is a technique that allows one to uncover the structure of intra-individual variability of a particular person. While this makes time series analysis a powerful tool for the study of physiological processes, relatively few studies in terms of prospective athlete monitoring of internal loads are formulated as such.

Apart from data collection issues, a final analytical challenge is to determine whether we can make generalizations from a single data point (realization). The idiographic single-

case study has been criticized for having little to no generalizability value. This requires us to distinguish between statistical generalizations versus analytic generalizations.

Most epidemiological researchers are imbued with statistical generalizations. We are taught that each case represents a sampling point from some larger population and cannot understand how a small set of cases can generalize to any larger population. Indeed a single realization cannot generalize in this manner, nor is it intended to. There perhaps is a strong assumption that statistical generalization are the only way of generalizing findings from research.

In contrast, analytical generalizations depend on the theoretical framework to establish a logic that might be applicable to other situations. Again, an appealing parallel exists in experimental science where generalizing about the findings from a single or small set of experiments does not usually follow any statistical path to a pre-defined population or universe. Rather, the objective for generalizing the findings is the same two-step process. The first step involves a conceptual claim whereby investigators show how their study's findings have informed the relationships among a particular set of theoretical constructs. The second step involves applying the same theoretical propositions to implicate other situations where similar constructs may be relevant.

Making analytical generalization requires carefully constructed claims. The ultimate generalization is not likely to achieve the status of geometrical "proof", per se, but the claims must be presented soundly. Statistical generalizations also do not achieve such "proof" status but by definition are probabilistic statements. [36] clarified that the sought-after generalization is not that of a conclusion but rather more like a working hypothesis. In like

manner, a working hypothesis as an analytical generalization is also a probabilistic statement just not expressed in numerical terms.

Finally, we must also recognize that biological variation potentially participates in two different realms: special and individual. Physiological responses to external stimuli are then, potentially at least, a simultaneous expression of humanity as well as of an individual's repertoire of variability. Failure to represent this theme in statistical operations for research or applied practice may lead to a confounding of these two sources of biological variation, thus hindering the progress of sport epidemiology. If we accept the proposition that peculiarities which characterize variation between individuals are not equal to within individuals, then such confounding shall be studiously averted. Appreciating this dual aspect of physiology inspired the complementary idiographic-nomothetic model of this thesis.

1.4 Specific Aims

My research aims to address the aforementioned limitations is to develop a longitudinal time-to-event method which dynamically forecasts musculoskeletal injuries from athlete internal load measurements. Such methodological development is primarily inspired by complexity theory. My secondary goal is to provide complementary levels of idiogrpahic-nomothetic inference. This entails estimating time-evolving (dynamic) graphs for each athlete based on their internal load measurements over time. The time-to-event model to forecast risk of injury will need to take dynamic graphs as inputs. An athlete's entire graph trajectory should be incorporated when assigning future risks to appreciate the emergent properties of their estimated temporal graph, without the restriction of arbitrary landmarking times. To preserve the pragmatism, an interpretability module is warranted to reduce the

time-to-event model into human-understandable outputs. The resultant methodology will equip sport practitioners with a tool to combat the tragic event of musculoskeletal injuries in a pragmatic and prospective manner.

Below are the three primary aims, in sequential order, which explicitly address the above problems:

1) Select a model to estimate intra-individual dynamic graphs from longitudinal internal load data to supply idiographic inference into an athlete's evolving physiology.

2) Develop a longitudinal time-to-event model which takes dynamic graphs as inputs. This requires a nomothetic perspective to train on historical athletes.

3) Design an architecture to enable a coach to query which temporal edges are contributing to an athlete's provisioned risk. This will supply idiographic-level explanations regarding the forecast provisioned from the nomothetic apparatus.

1.5 Notation

There are a lot of moving pieces when assembling an end-to-end computation pipeline. And data must transform into disparate structures when traversing along the graph mining pipeline herein. This section organizes and tabulates the notational scheme of this thesis for clarity and concision.

Generally speaking, matrix notation will be followed whenever possible. Scalars, vectors, an matrices are italicized, boldface, and boldface capital, respectively. The entries of a matrix $\mathbf{X}$ and a (column) vector $\boldsymbol{x}$ are denoted as X_{ij} and x_{ij}. Sets are represented by calligraphic capital letters and $|\cdot|$ represents the cardinality of a set. Let $[x_1 : x_2]$

denote a set of numbers between elements x_1 and x_2, respectively, where $x_1 < x_2$. Transpose is notated as $(\cdot)^\intercal$; $\mathbf{0}$ and $\mathbf{1}$ refer to vectors of zeros or ones, respectively; while $\mathbb{1}$ denotes the indicator function. The operators $\oplus$, $\odot$, $\circ$, $\mathrm{tr}(\cdot)$, $\det(\cdot)$, $\mathbb{E}[\cdot]$, stand for the direct sum between two vectors, element-wise product, Hadamard product, matrix trace, determinant, and expectation, respectively. An ℓ_p-norm is denoted as $\|\cdot\|_p$. Please refer to Table 1.1 for a comprehensive glossary of Chapter specific notations.

Table 1.1. Glossary of notation.

Notation	Description
k	unique observational unit
ij	node indices $\in \{1, \ldots, V\}$
t	time index $\in \{1, \ldots, T\}$
n	sample size of k units
P	internal load random variables
$\boldsymbol{X}$	raw data matrix $\in \mathbb{R}^{T \times P}$
V	number of nodes; size of graph (P)
L	unique node labels
E	set of graph edges $\subseteq V \times V$
W	set of edge weights $\in \mathbb{R}^E$
$\mathcal{G}$	weighted adjacency tensor $\mathcal{A}^{V \times V \times T}$
$\boldsymbol{\Theta}$	precision matrix $\boldsymbol{\Sigma}^{-1}$
τ	$tn \times 2$ matrix of time intervals $\in \mathbb{R}^+$
δ	event status indicator $\in \{\varnothing, 1\}$
$\mathcal{D}$	training dataset $\{\mathcal{A}, \tau, \delta\}$
$F(\cdot)$	cumulative event probability
h	neural network hidden layer
$\mathbf{a}$	$1 \times V$ vector of node neighborhood
$\mathbf{z}$	$1 \times d$ vector, latent embedding of $\mathbf{a}$
$\mathbf{e}$	latent graph embedding for time t
$\mathcal{E}$	dynamic graph embedding $\in \mathbb{R}^{T \times V}$
θ	parameterized neural network
$\mathbf{M}$	mask matrix $\in \mathbb{R}^{T \times V}$
$\boldsymbol{\Psi}$	temporal perturbation mechanism
B	perturbation window width

1.6 Prior Contributions

This thesis is a continuation of my prior work in sport injury mitigation methodology. [147] was my first contribution during my graduate studies. Three female teams at the University of Iowa were regularly monitored with common athlete monitoring technologies as a part of routine health and well-being surveillance throughout the 2016-2017 competitive season. Internal training load included daily heart rate variability, omegametry, session rate of perceived exertion and subjective wellness questionnaires. External training load were captured via tria-xial accelerometers worn during practice and competition. An empirical dynamic Bayesian network was constructed for nomothetic inference into the temporal multivariate dependencies leading to musculoskeletal injury occurrence, and to investigate the discriminative performance of injury prediction in a temporal dimension. The classifier was able to, in part, learn a concept as the predictive performance surpassed the naïve baseline threshold. Although the network incorporated the element of time, the main limitations were that the topology was static and was not formulated as a time-to-event problem.

[145] further extended the prior project, which primarily contributed computational speed advancements for dynamic Bayesian network construction through various approximation steps. The computational claims were tested with a much larger dataset than [147] with over 100 variables and almost 300 athlete time series. The architecture was flexible enough to evolve over time and better conformed to the theory of complex systems. However, this project also embraced a nomothetic view, constructed one topology for the cohort, and was formulated as a time series classification instead of time-to-event modeling with censorship.

1.7 Thesis Overview

Chapter 1 has delineated the historical origins of sport epidemiology and justified the need for computational solutions that enable idiographic inference under a systems biology worldview. Such considerations positioned this thesis within the theory of complex systems. The problem was formulated into a time-to-event setting, where covariate information shall take the form of intra-individual dynamic graphs, longitudinally forecast injury from the dynamic graphs, and convey the temporal edges contributing to an athlete's predicted injury risk. This objective was decomposed into three Specific Aims that are addressed throughout the remainder of this thesis.

Chapter 2 corresponds to Specific Aim #1: intra-individual dynamic graph construction. There is rich literature concerning dynamic graph estimation from raw temporal observations. This chapter formally surveys the landscape of graph estimation methods ranging from static to dynamic, node-based to covariance selection, and from maximum likelihood to Bayesian. A range of challenging synthetic datasets are then simulated to validate the temporal edge reconstruction performance of the most competitive solutions which assume temporal smoothness. The results from the synthetic experiments will guide the decision of which dynamic graph estimation procedures will be deployed in the grand pipeline in later chapters.

Chapter 3 addresses Specific Aim #2: longitudinal time-to-event forecasting from dynamic graphs as inputs. Unlike Chapter 2, there is a considerable dearth of literature in time-to-event modeling of dynamic graphs. The proposed contributions serve as an experimental platform that braids deep learning and graph mining into a custom time-to-event

formulation. The forecasting performance is validated with synthetic data against two longitudinal time-to-event benchmark algorithms after graph-theoretic pre-processing.

Chapter 4 focuses on Specific Aim # 3: identifying intra-individual temporal neighborhood importance to explain injury forecasts. If an athlete's graph snapshot is issued a high injury risk, a practitioner will likely want to know why. Prior work in edge importance has been restricted to static graphs, so in vein with Chapter 3, adding a temporal element is an experimental enterprise with minimal research backbone.

The purpose of Chapter 5 is to thread Chapters 2-4 into an end-to-end application for sport epidemiology. An authentic athlete monitoring dataset was retrospectively extracted from University of Iowa Department of Intercollegiate Athletics. Intra-individual dynamic graphs were constructed from longitudinal force plate data using the methodology from Chapter 2. The longitudinal time-to-event model from Chapter 3 was trained and tested to forecast musculoskeletal injury from the dynamic graph dataset. Athlete-specific injury forecasts are then explained using the architecture from Chapter 4, where novel idiographic inferences into the temporal interactions leading to injury are unveiled.

Chapter 6 summarizes the overarching contributions, addresses limitations, and recommends future research directions.

CHAPTER 2

INTRA-INDIVIDUAL DYNAMIC GRAPH CONSTRUCTION

2.1 Introduction

There are over seven billion human organisms in the world. Each person is governed by their unique set of high-dimensional, interdependent physiological systems which continuously adapt to external conditions to survive and reproduce. Consequently, an athlete's adaptability to rigorous training is complex. The general hypothesis of the complex systems theory discussed above, is that observed macroscopic behavior is not solely due to common nomothetic influences, but rather the emergent idiographic properties of time-varying interactions that take the form of a graph. Thus, I emphasize the importance of intra-individual graph construction to help parse heterogeneous patterns and to preserve an athlete-centered philosophy to health monitoring.

While there are many datasets that describe physical graphs – protein-protein interactions, social interactions, brain connectomics – the graphs in the present context are extracted from raw observations where interactions are not directly ascertained, and thus estimated. Each covariate (internal load marker) is represented by a node in such graph. Each node is sequentially ordered and time stamped when the observations were recorded and is comprised as the functional form of the respective internal load measure. By defining a distance measure for any two measures, it is possible to construct a functional graph, where dependencies are conveyed by the estimated connectivity pattern; presence or lack of an edge indicates dependence or independence, respectively. The assigned edge weight is conveyed by thickness and is proportional to estimated dependence quantity. I used the

term functional to help distinguish from other common graph types, such as neural networks or social networks.

Configurational shifts in such graphs, due to changes in the covariance of raw observations over time, may indicate that the data generating mechanism – the athlete – has changed. Capturing these meaningful adaptive signatures is the ultimate goal of athlete monitoring. But as of yet, the field has not formulated the problem under complex systems theory and has neglected the dynamic graph as an inferential apparatus. The purpose of this chapter is to survey the literature for tenable dynamic graph estimation techniques for representing an athlete's time-evolving signature through the lens of complexity theory. I will select numerous techniques on the basis of reasonable assumptions, which will be evaluated on the premise of edge reconstruction performance on realistically sized datasets.

2.2 Background

This section is dedicated to review the literature of methods for estimating graph topologies from raw Euclidean observations. I will begin by defining graph-theoretic preliminaries to equip the reader with the necessary notation. The logical flow of the review will remain chronologically attuned when possible, but the primary organization is meant to progress from static to dynamic, differentiate between node-wise versus covariance selection approaches, then touch on maximum likelihood estimation versus recent Bayesian solutions.

2.2.1 Preliminaries

The general philosophy behind the statistical perspective is that there exists a graph $\mathcal{G}$ whose structure governs the joint probability distribution of the observations on the data entities; columns of the data matrix $\boldsymbol{X}$. While there are many possible models of vary-

ing complexity, a general proximity relationship between columns is not only a plausible framework but a ubiquitous tool for study complex systems.

Consider random variables whose values are associated with nodes of a weighted, undirected graph. We let $\boldsymbol{x} = [x_i : x_P] \in \mathbb{R}^P$ and an unknown weighted graph $G(V, E, W)$, where $V = [1 : P]$ is the set of P nodes, $E \subset V \times V$ is the set of edges, and $W : V \times V \to \mathbb{R}$ denotes the set of unordered pairs of nodes to a scalar which associates an edge $(i, j) \in E$ if $W_{ij} \neq 0$. The symmetric pairs $W_{ij} = W_{ji}$ represent the strength of connection between nodes i and j (i.e., similarity or influence). Thus, when W_{ij} is large, the statistical signal between x_i and x_j tend to be associated, whereas $W_{ij} = 0$ indicates independence. Such an interpretation of edge presence establishes a connection between the raw $\boldsymbol{x}$ values and the graph connectivity structure (topology). Therefore, given an observed data matrix $\boldsymbol{X} = (\boldsymbol{x}_n)_{n=1}^N$, the goal is to estimate the graph topology $G(V, E, W)$ with $\mid V \mid = P$ which best encodes the entities of $\boldsymbol{X}$.

2.2.2 Static Graph Topology Inference

As alluded to above, a natural starting point toward constructing a graphical representation of a data matrix is to associate edge weights with non-trivial association measures between nodes. Informal scoring methods could rely on ad-hoc thresholding of user-defined edge-wise score functions. Examples among diverse domains could include the Pearson product-moment correlation used to quantify gene-regulatory interactions [73], the Jaccard coefficient for scientific citation networks [160], or mutual information to capture non-linear interactions [145]. Often thresholds are manually tuned so that the resulting graph is deemed to accurately capture the relational structure in the data; a choice possibly informed by hu-

man domain experts.

Such informal approaches fail when it comes to deciphering whether an estimated graph is accurate in some application-dependent sense as they lack a framework for objective validation. A different paradigm is to frame the graph-learning problem as one of selecting the best representation from a family of candidate graph structures by applying statistical modeling and inference. This adoption is advantageous to formally examine issues of identifiability or measurement error and sampling robustness.

For example, the most widely adopted linear measure of similarity between random variables x_i and x_j is the Pearson correlation coefficient (r). One could define a correlation graph $G(V, E, W)$ where $E = \{(ij) \in V \times V : r_{ij} \neq 0\}$. The resultant graph topology is then identified as the edge set with non-zero correlation entries via hypothesis testing:

$$H_0 : r_{ij} = 0$$

$$H_1 : r_{ij} \neq 0$$

for each $\binom{P}{2} = P(P-1)/2$ candidate edges in $\mathcal{G}$. However, this testing requires $\mathcal{O}(P^2)$ procedures and may not scale to large graphs – not to mention tests to control for false discovery rate.

Another note on the scope of correlation graphs, they can only capture linear, pairwise dependencies among random variables. Most importantly, such marginal correlations can be due to mediators rather than from strong direct influence among node pairs. For instance, a suspected regulatory interaction among biomechanical variables (i, j), inferred from their highly correlated kinteic profiles, could be an artifact from the variance of a third variable k that is actually governing the expression of both i and j. When seeking a graph reflective

of direct influences between pairwise nodal elements, correlation graphs may be unsuitable. This issue of confounding can be addressed by considering partial correlations

$$\rho_{ij|V\backslash ij} = \frac{\mathrm{cov}(x_i, x_j \mid V\backslash ij)}{\sqrt{\mathrm{var}(x_i \mid V\backslash ij)\mathrm{var}(x_j \mid V\backslash ij)}},$$

where $V\backslash ij$ denotes the collection of all $V-2$ random variables $\{x_k\}$ after excluding those indexed by nodes i and j. A partial correlation graph can be defined analogous to the marginal correlation graph counterpart, but instead with an edge set of $\mathcal{E} = \{(i,j) \in V \times V : \rho_{ij|V\backslash ij} \neq 0\}$. The problem of inferring non-zero partial correlations from a data matrix $\mathcal{X} = \{\boldsymbol{x}_n\}_{n=1}^N$ can be also be framed as one of hypothesis testing [111].

In lieu of confronting the multiple-testing problem, an alternative view of graph construction is through the lens of covariance selection. Suppose now that the data $\boldsymbol{x} = [x_1 : x_V]$ is a Gaussian random vector, meaning the random variables are jointly Gaussian. Under such a distributional assumption, $\rho_{ij|V\backslash ij} = 0$ is equivalent to x_i and x_j being conditionally independent given all of the other variables in $V\backslash ij$. Consequently, the partial correlation network with edges $\mathcal{E} = \{(i,j) \in V \times V : \rho_{ij|V\backslash ij} \neq 0\}$ specified conditional independence relations among the entries of $\boldsymbol{x}$. This is known as a Gaussian graphical model (GGM). A host of opportunities for inferring GGMs emerged after [118] recognized that the partial correlation coefficients can be expressed as

$$\rho_{ij|V\backslash ij} = -\frac{\theta_{ij}}{\sqrt{\theta_{ii}\theta_{jj}}} \tag{2.1}$$

where θ_{ij} is the $(i,j)^{\text{th}}$ entry of the precision matrix $\boldsymbol{\Theta} = \boldsymbol{\Sigma}^{-1}$; the inverse of the covariance matrix $\boldsymbol{\Sigma}$ of $\boldsymbol{x}$. The result of Equation 2.1 is that it reveals a bijection between the set of

non-zero partial correlations (edge set) and the sparsity pattern of the precision matrix $\boldsymbol{\Theta}$. Thus, this family of graphical model selection by identifying conditional independencies in $\mathcal{G}$ given *iid* realizations $\mathcal{X} = \{\boldsymbol{x}_n\}_{n=1}^N$ from a multivariate Gaussian distribution is termed covariance selection herein.

The term covariance selection was coined by [40] who explored the role of sparsity in estimating the elements of $\boldsymbol{\Theta}$ via an iterative likelihood-based thresholding on $\hat{\boldsymbol{\Theta}} = \hat{\boldsymbol{\Sigma}}^{-1}$. But this classic algorithm does not scale well to large-scale networks when $N \ll P$; this method breaks down since the sample covariance matrix $\hat{\boldsymbol{\Sigma}}$ is rank deficient. Such a predicament calls for regularization, and next we describe graphical model selection approaches which penalize the likelihood based on the ℓ_1-norm.

Henceforth, lets assume a zero-mean $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ since the focus is on estimating a graph structure encoded by the precision matrix entries $\boldsymbol{\Theta} = \boldsymbol{\Sigma}^{-1}$. Under such model, the maximum likelihood estimate of the multivariate normal precision matrix is given by a concave log-determinant operation

$$\hat{\boldsymbol{\Theta}}_{\text{ML}} = \min_{\boldsymbol{\Theta} \succeq 0} \left(\log(\det(\boldsymbol{\Theta})) - \text{tr}(\hat{\boldsymbol{\Sigma}}\boldsymbol{\Theta}) \right)$$

where $\boldsymbol{\Theta} \succeq 0$ requires the matrix to be positive semi-definite and $\hat{\boldsymbol{\Sigma}} = (1/N) \sum_{n=1}^N \boldsymbol{x}_n \boldsymbol{x}_n^\top$ is the empirical covariance matrix obtained from $\boldsymbol{X}$. But the maximum likelihood does not exist if $\hat{\boldsymbol{\Sigma}}^{-1}$ is singular, e.g., in high-dimensional settings. To overcome this challenge and encourage parsimonious graphs, the graphical LASSO penalizes the maximum likelihood

estimator with the ℓ_1-norm of $\boldsymbol{\Theta}$ as follows [197]:

$$\hat{\boldsymbol{\Theta}} = \min_{\boldsymbol{\Theta} \succeq 0} (\log(\det(\boldsymbol{\Theta})) - \operatorname{tr}(\hat{\boldsymbol{\Sigma}}\boldsymbol{\Theta}) - \lambda \left\| \boldsymbol{\Theta} \right\|_1). \tag{2.2}$$

Although Equation 2.2 is convex, the objective is non-smooth with unbounded constraint set [12]. Therefore, [61] offered an efficient first-order cyclic descent algorithm to address the space and time complexity issues identified in [12]. However, to bypass the consistency issues of cross-validation [158], [186] developed a full Bayesian treatment of the graphical LASSO estimator. Their block Gibbs sampler showed how representative the mode is of the full posterior in relation to the ℓ_1-penalty.

Yet another way to estimate a graphical model is to search for the set of neighbors $\{j : (i,j) \in E\}$ of each node $i \in V$ in the graph by regressing x_i against all other variables $\boldsymbol{x}_{\setminus i} \in \mathbb{R}^{P-1}$. To illustrate this idea, let $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}^{-1})$ where the conditional distribution $(x_i \mid \boldsymbol{x}_{\setminus i})$ is also Gaussian. The minimum mean squared error predictor of x_i based on $\boldsymbol{x}_{\setminus i}$ is $\mathbb{E}[x_i \mid \boldsymbol{x}_{\setminus i}] = \boldsymbol{x}_{\setminus i}^\top \boldsymbol{\beta}^{(i)}$, which is linear in $\boldsymbol{x}_{\setminus i}$ and is decomposed as

$$x_i = \boldsymbol{x}_{\setminus i}^\top \boldsymbol{\beta}^{(i)} + \epsilon_{\setminus i},$$

where $\epsilon_{\setminus i}$ is the zero-mean Gaussian prediction error uncorrelated with $\boldsymbol{x}_{\setminus i}$. The dependence between x_i and $\boldsymbol{x}_{\setminus i}$ is thus exclusively captured by the regression coefficients $\boldsymbol{\beta}^{(i)} \in \mathbb{R}^{N-1}$ and specified the incident edges to $i \in V$ in $\mathcal{G}$. In terms of the $\boldsymbol{\Theta}$ entries, this is expressed as

$$\beta_j^{(i)} = -\frac{\theta_{ij}}{\theta_{ii}}. \tag{2.3}$$

Importantly, Equation 2.3 together with Equation 2.1 reveals that a candidate edge (i,j)

belongs to $\mathcal{E} = \{(i,j) \in V \times V : \rho_{ij|V\setminus ij} \neq 0\} \Leftrightarrow \beta_j^{(i)} \neq 0$, and $\beta_i^{(j)} \neq 0$. Such property casts the problem of GGM selection as one of sparse linear regression using observations $\mathcal{X} = \{\boldsymbol{x}_n\}_{n=1}^N$. This neighborhood-based (node-wise) LASSO method proposed by [137] cycles over all nodes $i = [1:P]$ to estimate

$$\hat{\boldsymbol{\beta}}^{(i)} \in \min_{\boldsymbol{\beta}\in\mathbb{R}^{P-1}} \Big\{ \sum_{n=1}^{N} (x_{ni} - \boldsymbol{x}_{n\setminus i}^\top \boldsymbol{\beta})^2 + \lambda \left\|\boldsymbol{\beta}\right\|_1 \Big\}. \tag{2.4}$$

[137] enforces one of two rules to declare an edge $(i,j) \in E$. The first (OR) rule, requires that either $\hat{\beta}_j^{(i)}$ or $\hat{\beta}_i^{(j)}$ are non-zero, whereas the second (AND) rule requires that both coefficients are non-zero. The information in i and j's neighborhood are combined to enforce symmetry. For a sensible choice of λ in Equation 2.4, and under suitable conditions of $N \gg P$, [137] showed that a resultant graph can be consistently identified using either edge selection rule.

2.2.3 Dynamic Graph Topology Inference

The methods listed thus far learn a single graph topology. When data are observed over a temporal grid, the underlying connectivity patterns could evolve over time. When the covariance matrix evolves over time, the observations are not *iid*, and the results from graph inference approaches listed above would be misleading. This necessitates the study of temporal graphical models to unveil the structure of intra-individual covariation dynamics over time. Here we extend the problem formulations from Section 2.2.2 to accommodate the dynamic (temporal) graph topology inference problem.

Consider a sequence of $t = [1:T]$ graphs $\mathcal{G}_t(V, E_t, W_t)$ where each graph shares a common node set $|V| = P$ while being flexible to rearrange the corresponding edge sets E_t and weights W_t through time t. This implies there are potentially T different graphs to

recover each represented by $P \times P$ matrices, whose sparsity pattern may change across t. The main question at this stage is how to treat time. There seems to be two strands of beliefs regarding topological smoothness over time. In the maximum likelihood perspective, this primarily differs by the selected penalty.

One such assumption is to estimate graph topologies that are piece-wise constant. This is particularly convenient when we assume local stationarity and are useful for detecting jump points or abrupt changes in the graph signal. The common formulation is to apply some variant of the joint graphical LASSO penalty [38], such as:

$$\begin{aligned}\{\boldsymbol{\Theta}_t\}_{t=1}^T = \min_{\boldsymbol{\Theta}_t} \sum_{t=1}^T \Big[\log(\det(\boldsymbol{\Theta}_t)) - \operatorname{tr}(\hat{\boldsymbol{\Sigma}}_t \boldsymbol{\Theta}_t)\Big] \\ + \lambda_1 \sum_{t=2}^T \left\|\boldsymbol{\Theta}_{t\backslash ii}\right\|_1 \\ + \lambda_2 \sum_{t=2}^T \left\|\boldsymbol{\Theta}_{t\backslash ii} - \boldsymbol{\Theta}_{t-1\backslash ii}\right\|_F ,\end{aligned} \tag{2.5}$$

where $\|\cdot\|_F$ is the Frobenius norm and $\left\|\boldsymbol{\Theta}_{t\backslash ii}\right\|_1$ is the matrix ℓ_1-norm with the diagonal entries omitted. The first term in Equation 2.5 is the standard multivariate normal likelihood, the second term is the shrinkage of the sparse dependency structure $\boldsymbol{\Theta}_t$, and the third term carries the smoothness assumption of piece-wise constant [71]. Another popular approach to collectively inferring dynamic graphs is to assign a notion of similarity between consecutive graphs $\mathcal{G}_t$ and $\mathcal{G}_{t-1}$ encoded by a matrix distance $r(\boldsymbol{\Theta}_t, \boldsymbol{\Theta}_{t-1})$. [82] proposed this sequential graph inference problem as a time-varying graphical LASSO by solving

$$\{\boldsymbol{\Theta}_t\}_{t=1}^T = \min_{\boldsymbol{\Theta}_t} \Big\{ \sum_{t=1}^T \boldsymbol{\Phi}_t(\boldsymbol{\Theta}_t \boldsymbol{x}_t) + \alpha \sum_{t=2}^T r(\boldsymbol{\Theta}_t, \boldsymbol{\Theta}_{t-1}) \Big\}, \tag{2.6}$$

where $\alpha > 0$ is a regularization parameter and $\mathbf{\Phi}_t$ corresponds to the penalized global likelihood of the graphical LASSO from Equation 2.2, per time stamp.

However, interpreting and inferring insight about the complex system dynamics under consideration are greatly influenced by the imposition of structural assumptions. The following formulations were developed under the assumption that the graph gradually evolves over time. [201] developed a weighted negative log-likelihood function with a lasso penalty:

$$\min_{\mathbf{\Theta}_t \succeq 0} \log(\det(\mathbf{\Theta}_t)) - \mathrm{tr}(\hat{\mathbf{\Sigma}}_t \mathbf{\Theta}_t) + \lambda \sum_{u \neq v} \det(\mathbf{\Theta}_{t(uv)}), \tag{2.7}$$

This method utilizes the smoothness of the covariance matrix by introducing a kernel estimate $\hat{\mathbf{\Sigma}}_t$ into the likelihood function. $\hat{\mathbf{\Sigma}}_t = \sum_{j=1}^{N} w_h^{t_j}(t) \boldsymbol{x}_j \boldsymbol{x}_j^\top / \sum_{j=1}^{N} w_h^{t_j}(t)$ is the kernel estimate of the covariance matrix at time t, where the weights $w_h^{t_j}(t) = K_h(t_i - t/h)$ are derived from a symmetric non-negative unbounded smoothing kernel function $K(\cdot)$ of bandwidth $h > 0$. The use of this kernel estimate $\hat{\mathbf{\Sigma}}_t$ is to apply a time-sensitive smoothing factor across t. Similarly, [140] approached temporal smoothness by inserting the joint graphical LASSO formulation from Equation 2.5 into a kernel smoothed sliding window. This allows one to borrow information between more time points within the window, but this formulation leverages the joint graphical LASSO and inherits the assumptions above. Rather, [192] inserted the group-LASSO into a bounded kernel smoothed sliding window as:

$$L(\Theta_k) = \sum_{i \in \mathcal{M}_{gq}} \Big[\log(\det(\mathbf{\Theta}_{t_i})) - \mathrm{tr}(\hat{\mathbf{\Sigma}}_{t_i} \mathbf{\Theta}_{t_i}) \Big] + \lambda \sum_{u \neq v} \sqrt{\sum_{i \in \mathcal{M}_{gq}} \Theta^2_{t_i(uv)}}, \tag{2.8}$$

where $\mathcal{M}_{gq} = \{i \in \mathcal{I} : |t_i - t_g| \leq q\}$ represents the indices of time stamps centered around t_g with window width q and $|\mathcal{M}_{gq}|$ is the number of elements in $\mathcal{M}_{gq}$; $\Theta_k = \{\mathbf{\Theta}_{t_i}\}_{i \in \mathcal{M}_{gq}}$ is the

set of precision matrices within the temporal window and $\Theta_{t_i(uv)}$ is the $(u,v)^{\text{th}}$ element of $\mathbf{\Theta}_{t_i}$. Rolling the group-LASSO penalty $\lambda \sum_{u \neq v} \sqrt{\sum_{i \in \mathcal{M}_{gq}} \Theta^2_{t_i(uv)}}$ [38] over time promotes a sparse topology with smooth temporal evolution. The degree of smoothness is governed by the parameter q; the wider the temporal window, the more gradual the topology will evolve. The overall sparsity pattern of each time-dependent graph is conventionally governed by the λ parameter; increasing λ increases sparsity.

There are enticing properties in Equation 2.8 for the present context. First, the temporal window fosters locally smoothed estimates through time. The frequency of athlete monitoring occurs at relatively low fidelity, and therefore, it is advantageous to assume smooth evolutions instead of choppy stationarity. Second, enabling the temporal window size q to roll with the group-LASSO penalty over time implies that the graph's smoothness and sparsity behavior temporally adapts to an individual's incoming data. These properties successfully facilitate idiographic inference under the complex systems theory.

Unfortunately, the dynamic graphs inferred by the aforementioned categories are sensitive to the tuning parameters, including the kernel bandwidth and the penalty parameters that control the sparsity and smoothness. Classical brute-force grid search approaches for selecting these parameters via cross validation are fraught with difficulties of daunting computational cost; the learning algorithm needs to be run once for every combination of all possible values of the tuning parameters in a pre-defined set before the one associated with the largest score is chosen. Moreover, it has been demonstrated in [127] that such parameter selection approaches may yield unsatisfactory results for graphical model selection in high dimensions. Further, the time complexity for penalized likelihood-based categories above

require $\mathcal{O}(NP^3)$ [22] due to the alternating directions method of multipliers (ADMM) algorithm involving t eigen-decompositions of $V \times V$ matrices in every optimization iteration. [192] implemented an efficient cross-validation scheme to avert this important scalability issue by including early stopping and coarse search followed by refined search to hasten computation. [192] also developed a fast blockwise computation and pseudo-likelihood approximation, which reduced the time complexity to $\mathcal{O}(NP_b^3)$, where P_b serve as the block sizes $b \ll P$.

Rather, [195] recently proposed a Bayesian formulation for estimating dynamic graphs and reduced the time complexity to $\mathcal{O}(NP^2)$. Their proposition imposed a temporally dependent spike-and-slab prior [5] on the off-diagonal entries (j, k) of the precision matrix Θ at each time point t:

$$\Theta_t^{jk} = s_t^{jk} J_t^{jk}$$

$$\boldsymbol{J}_{1:T}^{jk} \mid \alpha^{jk} \sim \mathcal{N}(\boldsymbol{\mu}_{1:T}^{jk}\,,\ (\Theta^J)^{-1})$$

$$\Theta^J = (\Sigma^J)^{-1} = \alpha^{jk}\Theta_{\text{TM}}$$

$$s_t^{jk} \mid \pi_1, A_{00}, A_{11} \sim Bernoulli(\pi_t^{jk})$$

$$\pi_t^{jk} \begin{cases} \pi_1 = P(s_1^{jk} = 1) \\ A_{00} = P(s_{t-1}^{jk} = 1, s_t^{jk} = 1) \\ A_{11} = P(s_{t-1}^{jk} = 0, s_t^{jk} = 0) \end{cases}$$

where s_t^{jk} and $\boldsymbol{J}_t^{jk}$ are the spike-and-slab, respectively. The temporal smoothness for s_t^{jk} is modeled as binary Markov chain where A_{00} and A_{11} are the diagonal elements and π_1 is the Bernoulli success probability. π_1, A_{00}, and A_{11}, assume to follow a Uniform hyperprior. The temporal smoothness for $\boldsymbol{J}_t^{jk}$ is modeled as a Gaussian Markov chain where $\alpha^{jk}\Theta_{\text{TM}}$ is the precision matrix of the thin membrane model [31]. The authors also impose a hyperprior on

α^{jk} as a vague Jeffrey's prior. Temporal smoothness of the diagonal elements $\Theta_t^{jj} = \exp(\kappa_t^j)$ were also modeled as a Gaussian Markov chain:

$$\boldsymbol{\kappa}_{1:T}^j \mid \beta \sim \mathcal{N}(\boldsymbol{\mu}_{1:T}^j \,,\, (\Theta^\kappa)^{-1})$$

$$\Theta^\kappa = (\Sigma^\kappa)^{-1} = \beta\Theta_{\text{TM}}$$

where the $\beta\Theta_{\text{TM}}$ is the precision matrix of the thin membrane model (they also impose a Jeffrey's prior on β). Their reduction in time complexity stems from their variational inference algorithm, which replaces the exact likelihood of Θ_t by the pseudo-likelihood consisting of the conditional distributions of one variable conditioned on the remainder variables:

$$x_t^j \mid \mathbf{x}_t^{-j}, \Theta_t^{jj}, \Theta_t^{j,-j} \sim \mathcal{N}((\Theta_t^{jj})^{-1}\Theta_t^{j,-j}\boldsymbol{x}_t^{-j} \,,\, (\Theta_t^{jj})^{-1}).$$

The main reason they utilized the pseudo-likelihood is to simplify the determinant term in the exact likelihood, $\det(\Theta_t)$, like the penalized maximum-likelihood counterparts described above. They also resorted to simulated annealing [79] to combat local maxima problems during the approximate variational inference.

Lastly, the node-based regression formulation in Equation 2.4, can easily extend to the dynamic setting by applying a time-sensitive kernel smoothing parameter, as seen in Equation 2.7 [201]. Specifically, an unbounded Gaussian kernel is introduced into the likelihood function such as:

$$\min_{\boldsymbol{\beta}\in\mathbb{R}^{P-1}} \left\{ \frac{1}{\sum_{t=1}^T w_t} \sum_{t=1}^T w_t (x_{ti} - \boldsymbol{x}_{t\setminus i}^\top \boldsymbol{\beta})^2 + \lambda \left\|\boldsymbol{\beta}\right\|_1 \right\} \tag{2.9}$$

where w_t is the time-sensitive normalized kernel centered at t. This will serve as the maximum-likelihood node-based regression candidate for the present dynamic setting.

2.3 Problem Formulation

Given the slew of options for estimating graph structures, it is important to pair candidate methods which best conform to the assumptions and philosophical pillars set aside in Section 1.4.

By virtue of estimating time-evolving graphs as the inferential apparatus innately obeys the theory of complex systems conceptual framework. Considering the intended application is for idiographic exploration, and practitioners are likely to be responsible for many athletes, runtime scalability shall be scrutinize to remain pragmatic. It is assumed that internal load parameters are collected throughout the natural progression of a season, and thus, irregular measurements are often the rule rather than the exception. However, the granularity is assumed to be no more frequently than once per day; streaming real-time data, such as signal processing or waveform data is outside the present scope of temporal measurements.

Formally, the present interest is estimating undirected graphical models $G = (V, E)$ through time, where V denotes nodes relating to internal load variables and E as the corresponding edge set that encodes interdependencies; each node $i \in V$ is associated with a random variable x_i. An edge $(i, j) \in E$ is absent if the respective two variables x_i and x_j are conditionally independent: $p(x_i, x_i \mid x_{-ij}) = p(x_i \mid x_{-ij})p(x_j \mid x_{-ij})$, where $-ij$ represents all nodes in V except i and j. A dynamic (time-varying) weighted graph $\mathcal{G}$ is a sequence of T graphs, i.e., $\mathcal{G} = [G_1 : G_T]$, such that the graph at time stamp t is $G_t = (V, L, E_t, W_t)$, where V is the set of nodes, L is the set of node labels, E_t is the set of time-dependent edges $\subseteq V \times V$, and W_t is the set of associated edge weights.

Let $\boldsymbol{X}_t = [X_t^1 : X_t^p]^T \sim \mathcal{N}_p(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ be a p-variate Gaussian random vector indexed by $t \in [0, 1]$. It is implausible, and arguably unimportant, to estimate a graph topology for every time stamp t. Therefore, graph estimation time points are denoted as $\{\boldsymbol{x}_g\}_{g \in \mathcal{T}}$ where $\mathcal{T} = [1 : T]$, $\boldsymbol{x}_g$ is a realization of $\boldsymbol{X}_{t_g}$ $(g \in \mathcal{T})$. The present aim is to estimate the precision matrix $\boldsymbol{\Theta}_t = \boldsymbol{\Sigma}_t^{-1}$ based on the raw observations $\{\boldsymbol{x}_g\}_{g \in \mathcal{T}}$ and construct the edge set E_t according to the sparsity pattern of the estimated precision matrix $\hat{\boldsymbol{\Theta}}_t$. It is further assumed that the topology evolves gradually over time.

2.4 Experiments

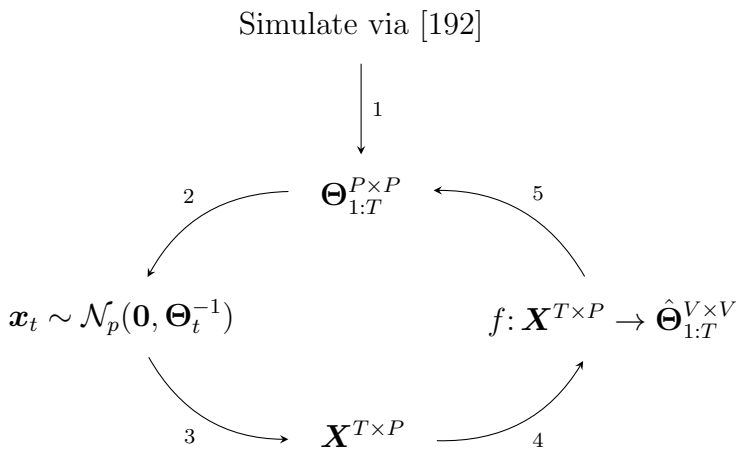
This section details the experiments to empirically evaluate the performance of the numerous dynamic graph estimation techniques. The experimental design is clarified first. The range of synthetic datasets are then described before justifying the candidate algorithms under scrutiny. The section ends by notating three quantitative criteria for validating temporal edge reconstruction performance.

2.4.1 Experimental Design

Figure 2.1 illustrates the present experimental design. Synthetic time-evolving precision matrices are simulated under the same mechanism as [192]. This data object serves as the ground truth. The next step is to sample time-dependent, normally distributed data from the time indexed precision matrices. This resultant $\boldsymbol{X}$ matrix emulates the raw data structure observed in the wild. The sampled $\boldsymbol{X}$ matrix is then fed to each candidate dynamic GGM algorithm to estimate time-evolving precision matrices $\hat{\boldsymbol{\Theta}}_{1:T}^{V \times V}$. The final step is to compare the time-evolving precision matrices generated from the candidate algorithms to the original ground truth object. Quantitative performance indices are then computed for

objective decision-making.

Figure 2.1. Experimental Design



2.4.2 Synthetic Datasets

The data generating mechanism was followed by [192], but with alterations to fit the present objectives. Below are the steps in which [192] simulates ground truth complex dynamic graphs:

1) Generate four lower triangle matrices $\boldsymbol{B}_1, \boldsymbol{B}_2, \boldsymbol{B}_3, \boldsymbol{B}_4 \in \mathbb{R}^{V\times V}$ with elements independently drawn from $\mathcal{N}(0, 0.5)$.

2) Set $G_t = (\boldsymbol{B}_1\gamma_1(t) + \boldsymbol{B}_2\gamma_2(t) + \boldsymbol{B}_3\gamma_3(t) + \boldsymbol{B}_4\gamma_4(t))/2$, where $\gamma_1(t) = \sin(\pi t/2)$, $\gamma_2(t) = \cos(\pi t/2)$, $\gamma_3(t) = \sin(\pi t/4)$ and $\gamma_4(t) = \cos(\pi t/4), t \in [0, 1]$.

3) Define $\boldsymbol{\Theta}_t^0 = G_t G_t^\top$.

4) Soft threshold the off-diagonal elements to obtain $\mathbf{\Theta}_t$:

$$\Theta_{uv}(t) = \text{sign}(1 - \frac{\alpha}{|\Theta^0_{uv}(t)|}) \cdot (1 - \frac{\alpha/2}{|\Theta^0_{uv}(t)|})_+ \Theta^0_{uv}(t),$$

where $(x)_+ = \max\{x, 0\}$, $\alpha = 0.22$.

5) Add $\log_{10}(p)/4$ to the diagonal elements of $\mathbf{\Theta}_t$ to ensure positive definiteness.

Various models were then constructed to sample from the ground truth dynamic graphs. Specifically, each model was generated by $\boldsymbol{x}_g \sim \mathcal{N}_p(\mathbf{0}, \mathbf{\Theta}_{t_g}^{-1})$, with $t_g = \frac{g-1}{T}(g = 1 : T + 1)$. The bounded parabolic kernel $K_h(x) = \frac{3}{4}(1 - (x/h)^2)I_{\{|x| \leq h\}}$ is applied to obtain smoothed estimates of the covariance matrices. Although different kernel functions vary the local weights, theoretical and practical evidence suggests that kernel estimators are influenced by the bandwidth parameter more than the kernel shape [86]. Therefore, the parabolic kernel was chosen for each dataset due to its optimality property of minimizing the asymptotic mean squared error under mild assumptions [86]. Prior studies of [192, 195] generated time series with both V and T well into the hundreds. This is implausible for practical athlete monitoring settings. Rather, the goal of this simulation was to test numerous dynamic graph estimation performances while scaling to realistic ranges of V and T.

The first round was a very small dataset to test the capability of edge selection in extremely sparse conditions $\boldsymbol{X}^{25\times25}$. The second round increase the time dimension to a more reasonable number for the graph estimation procedures ($\boldsymbol{X}^{100\times25}$). The third simulation then increased the dimensionality ($\boldsymbol{X}^{100\times50}$) to identify the source of gain in terms of both node dimensions and runtime scalability.

2.4.3 Benchmark Algorithms

The maximum likelihood-based covariance selection strategies of [201, 140, 192], as well as the Bayesian covariance selection strategy [195] were used as benchmark algorithms due to their shared assumptions of temporal smoothness. The penalty terms in piece-wise constant covariance selection strategies [82, 71] were excluded from the present simulations as they are better suited to identify changepoints, assume piece-wise stationarity, or intended for streaming applications. Such assumptions disqualify these methods under the pre-specified assumptions described in Section 2.3. However, the maximum likelihood node-wise regression from Equation 2.9 [137] was also included due to the time-sensitive unbounded kernel smoothing, which qualifies this method for fair comparison and assumption agreeance.

2.4.4 Performance Criteria

The performance evaluation metrics include precision (Prc): $\frac{1}{S}\sum_{g=1}^{S}|E_g \bigcap \hat{E}_g| \ / \ |\hat{E}_g|$ and recall (Rec): $\frac{1}{S}\sum_{g=1}^{S}|E_g \bigcap \hat{E}_g| \ / \ |E_g|$ for edge detection (averaged over the S time points where graphs are estimated), where E_g and $\hat{E}_g$ are the true edge set and the estimated edge set at time point $\widetilde{t}_g$, respectively. Precision in the present context translates to erroneous edge detection, whereas recall translates to missing truly important edges. Hereafter, overfitting serves as a loose description of error within precision (proportion of correct edges out of all *predicted* edges), whereas bias serves as a loose description of error within recall (proportion of correct edges out of all *true* edges). An overall metric for edge selection performance is $F_1 = 2 \cdot \frac{\text{Prc}\cdot\text{Rec}}{\text{Prc}+\text{Rec}}$ which strikes a balance between precision and recall; larger F_1 indicates superior edge selection performance.

2.5 Results and Discussion

Table 2.1 displays the temporal edge reconstruction performance from the simulation experiments. The first noteworthy phenomenon is that the three maximum likelihood-based covariance selection strategies failed to provide inference for the small (25×25) dataset; program crashed. Perhaps this is due to the data generating mechanism from [192] which could have produced a singular X matrix with such low T. This dataset dimension was chosen to represent pragmatic data collection scenarios found in sport (e.g., early phase of a season). Rather, the fully Bayesian covariance selection method, BADGE, was able to estimate a time-evolving precision matrix given the small samples. The maximum likelihood node-based regression formulation was also able to run without crashing, but struggled to detect any signal.

Table 2.1. Temporal Edge Reconstruction Performance
Mean (± standard deviation).

Algorithm	$(T \times V)$	**Precision**	**Recall**	$\mathbf{F}_1$	**Runtime**
Kernel		0.00 (± 0.00)	0.00 (± 0.00)	0.00 (± 0.00)	N/A
SINGLE		0.00 (± 0.00)	0.00 (± 0.00)	0.00 (± 0.00)	N/A
LOGGLE	25×25	0.00 (± 0.00)	0.00 (± 0.00)	0.00 (± 0.00)	N/A
BADGE		**0.65 (± 0.01)**	**0.29 (± 0.01)**	**0.40 (± 0.01)**	**0.1 m**
Node-wise		0.59 (± 0.01)	0.19 (± 0.01)	0.29 (± 0.01)	16.5 m
Kernel		0.60 (± 0.02)	**0.47 (± 0.01)**	0.36 (± 0.01)	2.3 m
SINGLE		0.48 (± 0.03)	**0.47 (± 0.03)**	0.47 (± 0.03)	23.3 m
LOGGLE	100×25	0.47 (± 0.02)	0.43 (± 0.02)	0.45 (± 0.02)	2.3 m
BADGE		**0.85 (± 0.01)**	0.38 (± 0.02)	**0.53 (± 0.02)**	**1.6 m**
Node-wise		0.60 (± 0.02)	**0.47 (± 0.02)**	0.52 (± 0.02)	17.2 m
Kernel		**0.74 (± 0.02)**	0.28 (± 0.03)	0.39 (± 0.02)	15.8 m
SINGLE		0.28 (± 0.02)	**0.52 (± 0.03)**	0.36 (± 0.02)	67.8 m
LOGGLE	100×50	0.53 (± 0.02)	0.35 (± 0.02)	**0.42 (± 0.02)**	15.9 m
BADGE		0.69 (± 0.02)	0.29 (± 0.02)	0.41 (± 0.02)	**5.9 m**
Node-wise		0.73 (± 0.02)	0.28 (± 0.02)	0.41 (± 0.02)	39.8 m

The second dataset enabled all candidate algorithms to estimate time-evolving precision matrices with the enlarged time dimension. This dataset begins to identify the sources of gain with respect to runtime; the time complexity of $\mathcal{O}(NP^2)$ and $\mathcal{O}(NP_b^3)$ for [195] and [192], respectively, are empirically validated against the $\mathcal{O}(NP^3)$ time complexity of [140] and [137]. Scaling T also boosted edge reconstruction performance, where BADGE [195] greatly outperformed all other candidates with precision, but suffered with recall. This means that while [195] generated graphs that were too sparse, many of the estimated edges matched the ground truth.

The third dataset challenged the candidate algorithm's flexibility in higher dimensions. The differing performance between Kernel [201] and SINGLE [140] is an interesting observation. [201] produced the least amount of erroneous edges but suffered from poor recall, whereas [140] exhibited superior recall but at the cost of many erroneous edges. These two methods performed well in their own right, but their overall performances fell behind the other contending algorithms. While LOGGLE produced the most (statistically) efficient overall performance (F_1), the scalability of BADGE is more pragmatic to deploy in real-world scenarios.

Overall, the dynamic graph estimation procedures suffered from systematic bias; the methods were unable to reconstruct the edges, illustrated by poor recall. The Bayesian solution [195] time complexity was the most scalable in all three datasets and displayed superior, or matched, overall edge reconstruction performance. On these grounds, [195] is the selected dynamic GGM approach for the thesis hereafter, and is suggested for real-

world sport environments over the maximum likelihood-based covariance selection models and node-based regression models.

2.5.1 Limitations

The dimensionality of raw observations should be carefully considered before embarking on dynamic graph estimation. The results highlight that runtime can be sensitive to V for both node-wise and covariance selection methods. [22] showed the computational complexity for ADMM-based algorithms is $\mathcal{O}(NP^3)$. This is only exacerbated by the repetitive deployment of cross-validation per time stamp (per observational unit) in the present formulation. Future work into the mechanics, applications, or approximations of ADMM is warranted.

A major drawback of the kernel-based solutions tested above is that these estimators exhibit excessive bias when t is close to the support boundary [54]. Applying local polynomial estimators to remedy the potential boundary bias associated with the kernel estimators may be an avenue for future work in maximum likelihood-based dynamic GGM estimation.

Another limitation is that all simulations herein assumed complete data (no missing values) per time stamp. While handling irregular time measurements is of practical value, only one method ([195]) can tolerate missing measurements within a time stamp. This performance was not explicitly tested, though.

Finally, estimating graphs from small sample sizes with the Bayesian formulation could result in a lack of information for the likelihood and produce heavily influential priors. Imposing subjective or empirical priors onto the spike-and-slab or Markov chains in these situations may foster fruitful investigations.

2.6 Conclusion

This chapter formally surveyed the graph topology inference landscape before embarking on numerous simulations to evaluate temporal edge reconstruction performance in realistically sized datasets found in sport. The candidate algorithms were maximum likelihood-based formuations of covariance selection versus node-based regression, with one Bayesian covariance selection method. A maximum likelihood-based candidate solution was identified to successfully facilitate idiographic inference under the complex systems theory, which temporally adapts the smoothness and sparsity behavior to an individual's incoming data. The greatest sources of gain for the Bayesian formulation was runtime and precision. However, dynamic graph estimation procedures tested herein systematically suffered from severe bias and struggled to detect true signals, especially in small datasets.

CHAPTER 3

LONGITUDINAL TIME-TO-EVENT FORECASTING OF DYNAMIC GRAPHS

3.1 Introduction

This chapter presents a modeling framework to forecast longitudinal time-to-event data from dynamic graphs as inputs. Longitudinal (temporal) time-to-event data arises from periodic follow-up settings where each athlete is repeatedly measured at discrete time points and the event (failure) times of interest can be right-censored; beyond a specified value but unknown by how much [110]. Joint modeling [92] and landmarking [200] are the two most common analytical solutions for longitudinal time-to-event models when the measured data is in standard Euclidean space. But analyzing complex adaptive systems generate data in the form of non-Euclidean dynamic graphs [23], where the configuration of nodes (covariates) and edges (relations) evolve over time [120]. To my knowledge there is no framework for forecasting events from dynamic graphs given as covariate information, at the time of this writing.

Chapter 2 viewed dynamic graphs as an estimation problem from temporal Euclidean data. But dynamic graph-based data are ubiquitous to other domains which organically generate graphical structures. Some motivating examples of dynamic graph-based time-to-event data could be derived from time-varying brain connectomics monitoring the time to an epileptic seizure event [74], sampling vaginal microbiota to forecast time to birth [43], or using the stock market's evolving topology to predict when a market bubble may burst [169]. Each of these systems – an athlete's organism, the brain, a developing embryo, a

financial market – are characterized as complex adaptive systems; their behavior emerges as a result of non-linear temporal interactions among a large set of nodes which adapt to a changing environment [96]. Further, these examples are concerned with forecasting the time to a discrete event which warrants time-to-event modeling.

Prior work in time-to-event forecasting from graph inputs have thus far only considered the static setting. This would translate to collecting one graph measurement per athlete before recording injury. A common strategy is to encode the graphs into a kernel to estimate pairwise similarities. For example, [41] utilized the support vector machine algorithm to learn the time-to-event distribution from the graph kernel, whereas [108] estimated the kernel for a Gaussian Process prior of a Bayesian analysis. But these tools concentrate on the results of adaptation – rather than the process of adaptation – by dismissing the formation, continuation, or dissolution of edges over time. On the other hand, current methods for mining temporal graphs, such as dynamic node classification in citation networks [4] or dynamic link prediction in social networks [102], are ill-suited for comparing multiple graphs and do not incorporate event censorship. Such tasks are distinct from the present problem.

With Euclidean data, the closest solution is joint modeling, which estimates the temporal process and time-to-event data from two respective sub-models linked by random effects [92]. A typical formulation involves a repeated measures mixed-effect model with the semi-parametric Cox regression. But optimizing this joint likelihood produces a model sensitive to mis-specifications and imposes computational challenges when implemented in high-dimensional domains [101]. Another framework for analyzing longitudinal time-to-event data is via landmarking [200], which incorporates only the most recent available measure-

ment when making time-to-event predictions. While landmarking has shown competitive performance to joint modeling in practice [174], such method neglects the temporal trajectory and suffers severe loss of information when making forecasts. A naïve solution to the present problem is to inject time-varying graph-level indices into a joint model or a landmarked model, but the aforementioned limitations would nevertheless apply.

The purpose of the present chapter is to propose a unified framework for longitudinal time-to-event forecasting from dynamic graphs, guided by complexity theory [97, 134]. The properties of complex adaptive systems necessitate learning non-linear temporal patterns from macro-structural topologies, where forecasts should update in lockstep with time as new graph snapshots are accrued. Specifically, an embedding perspective is implemented to learn latent nodal features from the non-Euclidean graph domain. Deep learning autoencoders are arranged sequentially in a Markovian design to learn the latent temporal trajectories. That is, the autoencoder embeddings are initialized from $t-1$ before performing gradient training on time t in order to capture non-linear nodal interactions within time points and across time points. A peephole long short-term memory (pLSTM) neural network [70, 151] is then trained from the latent trajectories to learn long-range graph dynamics preceding the event. The loss function of the pLSTM is customized with the log-likelihood to incorporate right-censorship. Finally, the flexibility of the trained architecture enables event forecasting at any time of interest in an online fashion. The resultant model performance is evaluated against rigorous synthetic graph data, which expresses promising results for a generalized modeling framework for complex system time-to-event forecasting.

The rest of the chapter is organized as follows. Section 3.2 begins by introducing

preliminary notation to sufficiently formalize the problem statement in Section 3.3. Section 3.4 establishes the full modeling architecture. Section 3.5 details the validation experiments, where the performance results are presented in Section 3.6. Some concluding remarks and discussion are provided in Section 3.7.

3.2 Preliminaries

3.2.1 Dynamic Graphs

This chapter focuses on analyzing dynamic graphs as a series of individual snapshots of a complex system through time. The chronology of these time-specific snapshots reveal the temporal evolution of interactions within the complex adaptive system under consideration. A dynamic graph $\mathcal{G}$ is a sequence of T graphs, i.e., $\mathcal{G} = [G_1 : G_T]$, such that the graph at time stamp t is $G_t = (V, L, E_t, W_t)$, where V is the set of nodes, L is the set of node labels, E_t is the set of time-dependent edges $\subseteq V \times V$, and W_t is the set of associated edge weights $\in \mathbb{R}^E$. However, the present methodology can easily generalize to directed or unweighted graphs. Since V and L are held constant over time, evolutionary characteristics are depicted by time-dependent edge (and weight) transformations. Edge transformations appertain to three rules defined as follows. Edge insertions over time follow a discrete operation defined as:

$$\begin{aligned} e+ \coloneqq\ & \neg\exists(v_i, v_j) \in E(G_t), \exists v_i \in V(G_t), \exists v_j \in V(G_t) \\ & \text{s.t. } (L(v_i) = u_i) \wedge (L(v_j) = u_j) \text{ and } \exists(v_i', v_j') \in E(G_{t+1}) \\ & \text{s.t. } (L(v_i') = u_i) \wedge (L(v_j') = u_j) \end{aligned}$$

which implies the formation of an interaction between nodes i and j from time t to $t + 1$. Edge deletions over time follow a discrete operation defined as:

$$
\begin{aligned}
e- \coloneqq\ & \exists (v_i, v_j) \in E(G_t) \ \text{ s.t. } \ (L(v_i) = u_i) \wedge (L(v_j) = u_j) \\
& \text{and } \neg\exists (v'_i, v'_j) \in E(G_{t+1}), \exists v'_i \in V(G_{t+1}), \exists v'_j \in V(G_{t+1}) \\
& \text{s.t. } \ (L(v'_i) = u_i) \wedge (L(v'_j) = u_j).
\end{aligned}
$$

which implies the dissipation of an interaction between nodes i and j from time t to $t+1$. Edge continuation is simply $\exists (v_i, v_j) \in E(G_t)$ and $\exists (v'_i, v'_j) \in E(G_{t+1})$, which implies the preservation of an interaction between nodes i and j from time t to $t+1$. If $(v_i, v_j) \in E_t$, then $w(v_i, v_j) \neq 0$ denotes the corresponding edge weight $\in W_t$. Edge weights are considered continuous numeric values which can fluctuate over time so long as the edge continues from t to $t+1$ according to the edge continuation rule just specified.

Each observational unit $\mathcal{G}^k$ is stored as an adjacency tensor denoted as $\mathcal{A}^k = \{a^k_{ijt}\}$, where $k \in [1:n]$ is the index of n sampled units, i and $j \in [1:V]$ are the node indices in the tth adjacency matrix, where $t \in [1:T]$ denotes time indices. Therefore, $\mathcal{A}^k \in \mathbb{R}^{V \times V \times T}$ denotes an entire longitudinal adjacency tensor available for unit k until the last recorded measurement time T for that unit. Let $A^k_t \in \mathcal{A}^k$ denote the tth adjacency matrix for unit k, which represents a time-specific graph topology snapshot. Lastly, let $\mathbf{a}^k_{i-t}$ denote the ith row for the tth time stamp of unit k, which represents the neighborhood for node i in time t.

3.2.2 Longitudinal Time-to-Event Data

Time-to-event modeling analyzes the time leading to a discrete event as the dependent variable to estimate. The dependent variable is commonly incomplete for a subset of the sample, which is termed censored. To be clear, an athlete k is recorded as a three-piece tuple: 1) recorded covariates, 2) time-to-event under consideration, and 3) a label indicating event occurrence or censorship. In the present setting, covariate information takes the form

of dynamic graphs (defined above), and is thus longitudinal. It is assumed that the start of the season (or study) and last follow-up time are based on a synchronous interval for all athletes. However, irregular repeated measurement intervals between athletes is allowed. An anatomically homogeneous injury is also assumed, where each dynamic graph is at risk for one injury type (defined by the practitioner). The event label currently considered is Type I right-censoring, where a pre-specified follow-up time arrives before an event occurs.

Formally, consider a dataset $\mathcal{D} = \{\mathcal{A}^k, \tau^k, \delta^k\}_{k=1}^n$ comprising the time-to-event data for n athletes who have been followed up for a certain amount of time. Time is considered as discrete intervals with a fixed temporal resolution (i.e., minute, day, week); $T = \tau$ indicates contiguous pairs of intervals $T \in [\tau, \tau + \Delta t)$ where Δt is the temporal resolution. It is reasonable to assume a finite time horizon and hence the range of possible times is thereby indicated as $\mathcal{T} = [0 : T_{\text{end}}]$ as T_{end} is the pre-determined endpoint (e.g., end of season). In the resultant dataset $\mathcal{D}$, $\tau^k = \min(T^k, C^k)$ is the time-to-event with $T^k \in \mathcal{T}$ indicating event occurrence and $C^k \in \mathcal{T}$ indicating censorship. The event indicator is labeled as $\delta^k \in \{\varnothing, 1\}$ where $\varnothing$ denotes right-censorship. Note that τ is either the time at which an event occurred or the time at which the observational unit was censored; in both instances, the unit was known to be event-free at all times preceding τ.

3.3 Problem Formulation

Time-to-event forecasting is concerned with estimating the risk of an event given the history of observations over time and to provision risk predictions when new data become available. The present aim is to capture how time-varying topological patterns lead to injury, with the ultimate goal of mitigating injury occurrence in future graphs. To quantify

the strength of association between dynamic graph structures and the risk for an event, a direct approach is to postulate a cumulative hazard model of the form:

$$\begin{aligned} F(\tau^* \mid \mathcal{A}^k) &= \mathbb{P}(T \leq \tau^* \mid \mathcal{A}^k) \\ &= \sum_{\tau \leq \tau^*} \mathbb{P}(T = \tau \mid \mathcal{A}^k) \end{aligned} \tag{3.1}$$

where τ^* is a future unobserved time point ($t > \tau^k$) for athlete k, and $\mathcal{A}^k$ is the entire history of measured graphs for the kth unit. Whenever a new graph measurement is recorded for athlete k, Equation 3.1 is updated accounting for the new covariate information in an online fashion. Accordingly, the present problem statement is as follows:

Problem 1. Given a dataset $\mathcal{D} = \{\mathcal{A}^k, \tau^k, \delta^k\}_{k=1}^n$, forecast injury occurrence δ^k at a future time τ^* via $\hat{F}(\tau^* \mid \mathcal{A}^k)$.

The specifics for how $\hat{F}(\cdot)$ is estimated to perform online risk forecasts are detailed in the following section.

3.4 Methods

3.4.1 Dynamic Neighborhood Embeddings

The first segment of the overall model applies the embedding perspective to dynamic graphs. The goal of this segment is to project the non-Euclidean dynamic graphs into a latent Euclidean vector space, where geometric relations in the latent space resemble relationships in the original graph topology [78]. Given a dynamic graph $\mathcal{A}^k$ (defined in Section 3.2.1), a dynamic graph embedding is a sequence of mappings $\mathcal{F}^k = [f_1^k : f_T^k]$ such that mapping f_t^k is a graph embedding for snapshot A_t^k. Specifically, given a time slice $A_t^k \in \mathcal{A}^k$, the graph embedding optimizes a mapping function $f : V \to \mathbb{R}^d$, where the function f preserves the proximity of nodes in A_t^k, and that $d \ll V$, namely $f(A_t^k) \in \mathbb{R}^{V \times d}$. Let $\mathbf{z}_v = f(v)\ \forall_v \in V$,

such that if two nodes v_i and v_j are structurally similar in A_t^k, then $\mathbf{z}_{v_i}$ and $\mathbf{z}_{v_j}$ should be in direct proximity within the latent Euclidean space.

3.4.1.1 Sequential Autoencoders

Extensive literature has shown that deep autoencoders can effectively learn complex low-dimensional representations of data for numerous unsupervised learning tasks [15]. The present solution employs deep autoencoders to map the input graph data to a latent space to extract the complex connectivity patterns in any time-specific graph snapshot.

Recall that the adjacency tensor $\mathcal{A}^k$ stores the entire dynamic graph for athlete k, and that A_t^k is the adjacency matrix storing the tth graph snapshot. Then $\mathbf{a}_{i-t}^k$ is a $1 \times V$ vector corresponding to the ith row of the adjacency matrix as the neighborhood for node v_i^k. For any pair of nodes v_i^k and v_j^k from graph A_t^k, the model takes their neighborhoods $\mathbf{x}_i^k = \mathbf{a}_{i-t}^k$ and $\mathbf{x}_j^k = \mathbf{a}_{-jt}^k$ as inputs into the autoencoder to generate d-dimensional latent embeddings $\mathbf{z}_i^k = \mathbf{z}_i^{(H)}$ and $\mathbf{z}_j^k = \mathbf{z}_j^{(H)}$ as the outputs of the encoder. Here, H defines the total number of neural network layers. The decoder then reconstructs the neighborhoods $\hat{\mathbf{x}}_i^k$ and $\hat{\mathbf{x}}_j^k$ from embeddings $\mathbf{z}_i^k$ and $\mathbf{z}_j^k$ during training. Lastly, each $1 \times d$ neighborhood embedding vector $\mathbf{z}_v^k \in V$ is averaged $\bar{z}_v^k = \frac{1}{d}\sum_{m=1}^{d} \mathbf{z}_{v(m)}^k$ to retrieve $\mathbf{e}_t^k = [\bar{z}_1^k : \bar{z}_V^k]$ as the latent graph embedding of time t for unit k. The cumulative dynamic graph embedding for an entire $\mathcal{A}^k$ corresponds to $\mathcal{E}^k = [\mathbf{e}_1^k : \mathbf{e}_T^k] \in \mathbb{R}^{T \times V}$.

The first deep autoencoder model is trained fully on A_1^k using random initialization parameters θ_1^k. For each subsequent time stamp $[2; T]$ the model parameters θ_t^k are initialized from the previously trained time stamp parameters θ_{t-1}^k in line with the Markov process

formalism [149]; θ_{t+1} only depends on the current state of the system θ_t. In terms of neural network training, the autoencoder at time t is initialized from the $t-1$ weights before performing gradient training on time t, which converges in only a few iterations. This design was chosen for two primary reasons. The first reason is to avert overfitting the temporal dynamics; embedding the entire sequence of graphs via deep learning has potential for overfitting the latent graph structures which would significantly inhibit downstream learning tasks. The second reason is for the utility of online predictions; the model only needs to learn about the evolution between A^k_{t-1} and A^k_t when provisioning risk predictions, whereas retraining the entire sequence is counterintuitive. Another byproduct of the Markov design is that $\mathcal{F}^k(\mathcal{A}^k)$ should generate temporally stable embeddings without the need for extra regularization parameters. That is, the magnitude of change between $\mathbf{e}^k_t = f^k_t(A^k_t)$ and $\mathbf{e}^k_{t+1} = f^k_{t+1}(A^k_{t+1})$ should be proportional to the magnitude of change between the consecutive graph snapshots A^k_t and A^k_{t+1}.

Each autoencoder model is semi-supervised and minimizes a combination of objective functions including both the second-order proximity (unsupervised) and first-order proximity (supervised), respectively. The model parameters are learned by minimizing a weighted summation of three objective functions at each time stamp, defined by a loss function similar to [185]:

$$\mathcal{L}_{emb} = \sum_{i=1}^{V} \|(\mathbf{x}_i - \hat{\mathbf{x}}_i) \odot \mathbf{b}_i\|_2^2 + \phi \sum_{i,j}^{V} a_{ij} \|\mathbf{z}_i - \mathbf{z}_j\|_2^2 + \lambda \sum_{h=1}^{H} (\left\|\mathcal{W}^{(h)}\right\|_1 + \left\|\hat{\mathcal{W}}^{(h)}\right\|_1), \quad (3.2)$$

where ϕ and λ are hyperparameters acting as relative weights, h are the number of neural network hidden layers $\in [1 : H]$, and $\odot$ is the element-wise product. Note that the depen-

dence on the kth athlete was omitted for ease of notation. The first term in $\mathcal{L}_{emb}$ is the unsupervised portion, which learns the context of a node's neighborhood within the global graph topology by minimizing the decoder's reconstruction error $\hat{\mathbf{x}}_i$. The second term is the supervised portion, which learns the pairwise similarity between nodes i and j in the observed topology. The third term corresponds to the ℓ_1-norm penalty to promote a sparse solution whereas [185] employs the ℓ_2-norm. Here, $\mathbf{b}_i$ is a vector with $b_{ij} = 1$ if $a_{ij} = 0$ else $b_{ij} > 1$ to penalizes inaccurate reconstructions of true edges a_{ij} more than inaccurate reconstructions of missing edges. To be clear, the resultant parameters θ_t of the deep autoencoder at each time step t and are used as the embedding output $\mathbf{e}_t^k = f_t^k(A_t^k)$ for graph snapshot A_t^k.

3.4.1.2 Network Training

Each autoencoder layer uses the rectified linear activation function (ReLU) [76] in order to support unweighted and weighted graphs. ReLU also accelerates training with the straightforward derivatives compared to other activation functions [76]. ReLU can also propagate the gradients from both first- and second-order encoder objectives and helps evade the vanishing gradient problem [76]. The Nesterov-accelerated adaptive moment estimation (Nadam) [45] optimizer was used with carefully tuned hyperparameters to promote convergence speed. Weights for first network θ_1^k are initialized with the Glorot normal distribution [75].

3.4.2 Longitudinal Time-to-Event Model

This second segment of the overall model learns the longitudinal time-to-event patterns from the dynamic neighborhood embeddings described in Section 3.1. The architecture consists of two neural networks: 1) a recurrent neural network that captures the latent temporal trajectories and predicts the subsequent measurements of dynamic neighborhood embeddings, and 2) a fully-connected neural network for estimating the event occurrence. Both network architectures are jointly optimized to help extract associations between event times and the underlying process governing the time-evolving graphs.

3.4.2.1 Recurrent Neural Network

The recurrent neural network consists of a peephole long short-term memory (pLSTM) structure to flexibly capture the latent temporal patterns with each observational unit having different numbers of measurements collected at uneven time intervals. For each observed time stamp t, the pLSTM architecture takes a two-tuple as input $(\mathbf{e}_t, \tau)$ and outputs $(\mathbf{y}_t, \mathbf{h}_t)$ where $\mathbf{y}_t$ is the estimate of the dynamic neighborhood embeddings after time τ, (i.e., $\mathbf{e}_{t+1}$). Here, $\mathbf{h}_t$ is the neural network hidden state at time stamp t, which is derived utilizing the pLSTM structure as follows:

$$\mathbf{r}_t = \sigma(\mathcal{W}_r[\mathbf{e}_t, \tau] + \mathcal{U}_r h_{t-1} + \mathbf{b}_r)$$

$$\mathbf{i}_t = \sigma(\mathcal{W}_i[\mathbf{e}_t, \tau] + \mathcal{U}_i h_{t-1} + \mathbf{b}_i)$$

$$\mathbf{o}_t = \sigma(\mathcal{W}_o[\mathbf{e}_t, \tau] + \mathcal{U}_o h_{t-1} + \mathbf{b}_o)$$

$$\mathbf{c}_t = \mathbf{r}_t \circ \mathbf{c}_{t-1} + \mathbf{i}_t \circ \sigma(\mathcal{W}_c[\mathbf{e}_t, \tau] + \mathbf{b}_c)$$

$$\mathbf{h}_t = \mathbf{o}_t \circ \sigma(\mathbf{c}_t)$$

where $\mathcal{W}$, $\mathcal{U}$, and $\mathbf{b}$ are weight matrices and vectors parameterizing the neural network, and $\sigma(\cdot)$ is the sigmoid function. Again, the dependence on the kth unit was omitted for notational clarity.

Training the pLSTM resorts to minimizing the prediction error of latent temporal trajectories. The pLSTM loss function $\mathcal{L}_{rnn}$ serves as an auxiliary task to extract the hidden representations of historical time-varying embeddings. Since the pLSTM makes predictions $\mathbf{y}_t$ on the step-ahead covariate information $\mathbf{e}_{t+1}$ to preserve the hidden representations, the prediction loss is defined as:

$$\mathcal{L}_{rnn} = \alpha \sum_{k=1}^{N} \sum_{t=0}^{T^i-1} (\mathbf{e}_{t+1}^k - \mathbf{y}_t^k),$$

where $\alpha \geq 0$ is a hyperparameter.

3.4.2.2 Event Network

The aim of the second deep learning structure is to learn the time-to-event distribution. But in order to provide context of the longitudinal process, the following vector is derived from the pLSTM hidden states in Section 3.2.1:

$$\mathbf{p} = \sum_{t=1}^{T} \frac{e^{f(\mathbf{h}_t, \mathbf{e}_t)}}{\sum_{l=1}^{T-1} e^{f(\mathbf{h}_l, \mathbf{e}_l)}} \mathbf{h}_t$$

where $f(\cdot)$ is a fully-connected feed-forward neural network and $\mathbf{h}_t$ is the pLSTM hidden state vector. Once optimized, this $\mathbf{p}$ vector enables the time-to-event network access to the learned representation of the latent temporal trajectories.

The final structure is another fully-connected feed-forward network to jointly capture

relations between the latent temporal trajectories and event occurrence. This neural network directly learns the latent temporal patterns that distinctly lead to event occurrence. Formally, the time-to-event network takes as input the context vector $\mathbf{p}^k$ and the last embedded measurement ($\mathbf{e}_T^k$) and outputs a vector $f(\mathbf{p}^k, \mathbf{e}_T^k)$, where the final layer employs the conventional soft-max to summarize the estimated failure outcome $f(\cdot)$ into a probabilistic quantity. In particular, given an observational unit $\mathcal{E}^k$, the output node represents the event probability at time τ: $\hat{P}(T = \tau \mid \mathcal{E}^k)$. Therefore, the estimated cumulative hazard at a future time τ^*, given the entire dynamic graph embedding, is defined as:

$$\hat{F}(\tau^* \mid \mathcal{E}^k) = \sum_{t<\tau\leq\tau^*} \mathbb{P}(T = \tau \mid \mathcal{E}^k). \tag{3.3}$$

Note that this equation assumes the athlete is injury-free up to the last measured time stamp.

To accommodate injury censorship into the neural network optimization, a log-likelihood loss function $\mathcal{L}_{evt}$ is customized as follows. For an athlete who experienced an injury, $\mathcal{L}_{evt}$ integrates both the injury and the time at which the injury occurred, conditional on the dynamic graph embeddings leading to the injury. For an athlete who is censored, it integrates the time of censorship conditional on the dynamic graph embeddings until the last follow-up time:

$$\mathcal{L}_{evt} = -\sum_{k=1}^{N} \Big[\mathbb{1}(\delta^k = 1) \cdot \log(\hat{F}(\tau^k \mid \mathcal{E}^k)) + \mathbb{1}(\delta^k = \varnothing) \cdot \log(1 - \hat{F}(\tau^k \mid \mathcal{E}^k))\Big],$$

where $\mathbb{1}$ is the indicator function. The first term in $\mathcal{L}_{evt}$ corresponds to uncensored graphs, where the second term assumes the graph is event-free, and therefore, the event may occur beyond the time of censorship.

3.4.2.3 Network Training

The overall neural network architecture is optimized by minimizing a total loss function $\mathcal{L}_{total} = \mathcal{L}_{rnn} + \mathcal{L}_{evt}$ which is tailored to incorporate dynamic graph embeddings and Type I right-censoring. Parametric rectified linear activation function (PReLU) [89] is used for the hidden layers followed by stochastic dropout to improve generalization error. The networks are optimized with adaptive moment estimation (Adam) [112] and initialized with the Glorot uniform distribution [75].

3.5 Validation

This section details the validation experiments to evaluate the performance of the proposed solution. The practicality of a time-to-event model shall be judged by how well the forecasts discriminate individual risks. The experiments below rely on complex synthetic data to simulate plausible problem sets. The sources of gain are identified by comparing the present solution to existing longitudinal time-to-event models. Since extant models assume Euclidean input data, all respective model performances are based upon the latent embeddings described in Section 3.4.1 for fair comparison. More details are found below.

3.5.1 Synthetic Datasets

Data simulation is an important tool for examining model performance under pre-specified conditions. It builds trust for the human analyst when a model can reproduce or recover an *a priori* data generating process. The goal of these experiments is to test the proposed solution with plausible data that would be collected in the field to augment the simulation's generalizability for applied practitioners.

The synthetic datasets were generated under a longitudinal, multivariate mixed-effect

model framework with a time-to-event outcome. Such data is meant to emulate the natural athlete monitoring setting where repeated internal load measurements are collected while monitoring time-to-injury, for example. The first version of the synthetic data was generated in relatively small dimensions, where the second version simulated a high-dimensional covariate setting. The first simulation (Syn1) generated 500 observational units sampled from a random 25×25 variance-covariance matrix, whereas the second version (Syn2) sampled from a random 100×100 variance-covariance matrix. All eigenvalues were positive to ensure valid positive definite variance-covariance. The length of repeated measurements were within 10 to 50, i.e., the total number of time stamps T for an observational unit was between 10 and 50 measurements. For each unit $k = 1, \ldots, n$, $\mathbf{y}_k = (\mathbf{y}_{k1}^\top, \ldots, \mathbf{y}_{kP}^\top)$ denotes a $p-$variate continuous outcome vector where $\mathbf{y}_{kp}$ is an $(n_{kp} \times 1)-$vector of observed serial measurements for the pth outcome. The pth longitudinal model is given as:

$$y_{kp}(t) = \mu_{kp}(t) + W_{1k}^p(t) + \epsilon_{kp}(t)$$

where $\mu_{kp}(t)$ is the mean response and $\epsilon_{kp}(t)$ is the error term. The mean response is formulated as $\mu_{kp}(t) = \mathbf{x}_{kp}^\top(t)\beta_p$ where $\mathbf{x}_{kp}^\top(t)$ is the $p-$vector of time-varying covariates with β_p as the fixed effect term. $Z_{1k}^p(t)$ is simply $Z_{1k}^p(t) = \mathbf{z}_{kp}^\top(t)\mathbf{b}_{kp}$ as the $\mathbf{z}_{kp}^\top(t)$ term are the time-varying random effects. Finally, the time-to-event outcome is given by the hazard model as:

$$h_k(t) = h_0(t)e^{\mu_{kp}(t)+W_{1k}^p(t)}$$

where $h_0(t)$ is the baseline hazard function. Dynamic graphs for each k athlete were estimated

using the Bayesian dynamic GGM approach [195] selected from Chapter 2. To be clear, the Syn1 dataset generated $n = 500$ individual dynamic graphs $\in \mathcal{A}^{25\times25\times T}$, whereas the Syn2 dataset generated $n = 500$ individual dynamic graphs $\in \mathcal{A}^{100\times100\times T}$, where $[T : 10 \leq t \leq 50]$.

3.5.2 Benchmark Algorithms

Since there are currently no benchmark algorithms for dynamic graph-based time-to-event forecasting due to the novelty of the problem, the proposed solution was compared against the two benchmark algorithms discussed in Section 3.1. Recall that both joint modeling and landmarking assume Euclidean data and cannot accept dynamic graphs as inputs. Thus, for fair comparison, the synthetic dynamic graphs generated from Section 3.5.1 were analyzed according to the latent embedding procedures described in Section 3.4.1 prior to training the benchmark models.

Specifically, the joint model was implemented in a maximum likelihood framework. The longitudinal process was modeled by a multivariate repeated measures linear mixed-effect model. The time-to-event was modeled using the conventional Cox proportional hazard regression [35]. Joint modeling is the closest to the present solution on the premise of jointly optimizing the longitudinal process and event time together. Such formulation enables risk provisions at any time of interest. However, the joint model was only evaluated against the Syn1 dataset due to the computational limitations in high-dimensions.

On the other hand, the landmarking methodology was implemented using the survival support vector machine algorithm [175]. Landmarking requires pre-specified evaluation times in order to provision risks, which prohibits the utility of dynamic forecasting. Therefore, landmarking prediction times were preset to 20, 30, and 40, where only observational

units who were still at risk upon the landmarked times were examined when fitting the survival support vector machine. Forecasting performance was evaluated +5 and +10 from each landmarked time. All experiments used these prediction and evaluation times for fair comparison.

Each experiment entailed 10 trials with randomly split training data (75%) and testing data (25%). 10% of each training dataset was further partitioned as a validation set for optimal tuning. The neural network hyper-parameters, along with size (number of nodes) and depth (number of layers) were chosen via grid search [113]. Table 3.1 lists the permissible values of the hyper-parameter search for each architecture, respectively.

Table 3.1. Permissible Hyper-parameters

Hyper-parameter	Set
Optimization	{SGD, Adam, Nadam}
Initialization	{Glorot Uniform, Glorot Normal}
Learning rate	$\{10^{-3}, 10^{-4}\}$
Mini-batch	{25, 50, 75}
Architecture*	{Elman, pLSTM}
Activation*	{eLU, ReLU, PReLU}
Nodes*	{50, 100, 150}
Hidden layers*	{1, 2, 3}
Dropout*	{0.25, 0.50, 0.75}
Activation†	{tanh, ReLU, GELU}
Nodes†	{50, 100}
Hidden layers†	{1, 2}
Dropout†	{0.25, 0.50, 0.75}

* Section 3.4.2.1
† Section 3.4.2.2

3.5.3 Performance Criteria

This subsection presents the performance metric extended for the current longitudinal time-to-event setting, which requires a small variation to the conventional concordance index to account for repeated measures [69]. The time-dependent concordance index, $\mathrm{CI}_t(t, \Delta t)$, is used as the performance criteria for evaluating the present algorithm. CI_t properly accounts for both evaluation and prediction times to reflect temporal risk fluctuations. Given $\hat{F}(\cdot)$ for a pair of athletes q and u, CI_t is defined as:

$$\mathrm{CI}_t(t, \Delta t) = \mathbb{P}(\hat{F}(t + \Delta t \mid \mathcal{E}^q) > \hat{F}(t + \Delta t \mid \mathcal{E}^u) \mid \tau^q < \tau^u, \tau^q < t + \Delta t). \tag{3.4}$$

where t indicates prediction time and Δt denotes evaluation time. $\hat{F}(t + \Delta t \mid \mathcal{E})$ implies the injury time risk in Δt time, which is predicted at time t, given the dynamic graph embeddings until t.

For the Syn1 experiment, a one-way analysis of variance with Tukey's Honest Significant Difference post-hoc pairwise comparisons test was used to statistically judge whether discriminative performances differ between the three modeling frameworks. Only positive significant improvements are tabulated for meaningful inference. Family-wise error rate was set *a priori* to 0.05. As for the Syn2 experiment comparing the proposed solution only to landmarking, discriminative performance was deemed statistically significant by comparing the mean and standard deviation of CI_t using the t-statistic with Type I error rate set to 0.05.

3.6 Results and Discussion

The discriminative performance of the present algorithm is reported in Table 3.2. Mean and standard deviations of CI_t were obtained by 10 random trials as described in Section 3.5. The prediction and evaluation times were held constant throughout the tests for equal comparisons. The current architecture statistically outperformed one or both of the benchmark algorithms in all evaluation and prediction times in terms of CI_t. However, due to a lack of dynamic graph-based time-to-event algorithms, the sources of gain are isolated to the custom deep learning architecture described Section 3.4.2. All the experimental results discussed assume the joint model and landmarked model were supplied the same dynamic graph embeddings from Section 3.4.1, since these benchmark algorithms require Euclidean input data. Recall that joint modeling suffers from linearity assumptions and is sensitive to model mis-specification, whereas landmarking neglects the temporal trajectory and suffers severe loss of information when making forecasts. In contrast, the present solution leverages the pLSTM architecture to learn the associations between the longitudinal measurements and the event time, and incorporates the entire history of measurements when making risk predictions.

In the low-dimensional Syn1 dataset, landmarking exhibited inverse performance to joint modeling in relation to prediction time. Specifically, landmarking outperformed joint modeling in the earlier time stamp $t = 20$ but gradually diminished in discriminative power, whereas the joint model moderately increased in accuracy over time. This is perhaps due to joint modeling exploiting the entire trajectory when provisioning risk compared to landmarking. On the contrary, the present solution exhibited superior performance in discriminating

future risks from the latent temporal trajectories.

Table 3.2. Discriminative Performance
Mean CI_t ($\pm$ standard deviation) $\times 100$.

Dataset	**Algorithm**	**Pred Time**	**Eval Time**	
			+5	+10
Syn1	Joint modeling	$t = 20$	88.6 ($\pm$ 2.9)	89.1 ($\pm$ 2.5)
	Landmarking		91.3 ($\pm$ 5.2)*	92.0 ($\pm$ 4.7)*
	Proposed solution		**95.7 ($\pm$ 4.2)***	**94.5 ($\pm$ 4.1)***
	Joint modeling	$t = 30$	91.3 ($\pm$ 2.2)	91.1 ($\pm$ 2.1)
	Landmarking		85.9 ($\pm$ 8.1)	85.1 ($\pm$ 7.6)
	Proposed solution		**95.5 ($\pm$ 3.8)**†	**95.0 ($\pm$ 4.0)**†
	Joint modeling	$t = 40$	91.6 ($\pm$ 2.0)†	90.8 ($\pm$ 2.2)
	Landmarking		84.7 ($\pm$ 8.3)	85.2 ($\pm$ 7.5)
	Proposed solution		**97.6 ($\pm$ 1.7)***†	**96.1 ($\pm$ 1.8)***†
Syn2	Landmarking	$t = 20$	83.9 ($\pm$ 6.1)	82.3 ($\pm$ 6.6)
	Proposed solution		**92.6 ($\pm$ 3.3)**†	**88.7 ($\pm$ 2.4)**†
	Landmarking	$t = 30$	80.3 ($\pm$ 5.8)	79.5 ($\pm$ 6.0)
	Proposed solution		**93.0 ($\pm$ 3.3)**†	**92.2 ($\pm$ 2.9)**†
	Landmarking	$t = 40$	81.8 ($\pm$ 6.3)	76.8 ($\pm$ 5.9)
	Proposed solution		**93.8 ($\pm$ 2.8)**†	**91.9 ($\pm$ 3.2)**†

* Significant from Joint modeling p-value < 0.05
† Significant from Landmarking p-value < 0.05

As for the high-dimensional setting in Syn2, landmarking was unable to match the performance to the present solution in earlier time stamps – as it did in Syn1 at $t = 20$. While the mechanics of support vector machine convex solution is robust to high-dimensions, discarding the temporality of adaptation substantially curtailed the effectiveness of forecasting a time-dependent phenomenon. Quadrupling the network size also increases the opportunity for perpetually novel evolutionary entanglements to emerge [95].

At run-time, the present algorithm provisions risk predictions as defined in Equation

3.3 for each observational unit which incorporates the entire historical trajectory. By virtue of the pLSTM neural network utilized in the first structure, whenever a new snapshot is accrued for a particular graph, the proposed method is easily able to integrate the information into the history of measurements and provision new risk predictions in a fully dynamic (online) fashion. Joint modeling is also able to update the event risks when incorporating new measurements, but bears the limitations of maximum likelihood estimation by imposing *a priori* distributional assumptions. Rather, landmarking methods can only provision risk assessments at the predefined landmarked times.

3.6.1 Limitations

In athlete monitoring setting where there may be small numbers of athletes available, or when data collection is minimal from early adoption, the present solution may be prone to overfitting and result in unsatisfactory performance. Injecting graph embeddings into benchmarks such as joint models or landmarked models may better suit these scenarios. Otherwise, future research is warranted to develop flexible solutions that are more tailored to dynamic graphs in smaller sample sizes.

3.7 Conclusion

This chapter presented a novel longitudinal time-to-event forecasting solution, which takes in dynamic graphs as covariate information. A dynamic neighborhood embedding perspective was employed to learn the time-varying latent nodal features from the non-Euclidean graph domain. Deep autoencoders were arranged in a Markovian design to estimate stable latent trajectories over time and hasten the incremental training process. A peephole long short-term memory neural network was trained from the latent trajectories to learn the as-

sociations between the longitudinal measurements and the risk of the event, such as injury. The present deep learning architecture learns the joint distribution of time-dependent events without imposing assumptions regarding the underlying stochastic process. The present solution was evaluated through simulation experiments and compared against the joint modeling and landmarking benchmark algorithms. The experiments conveyed superior performance in favor of the present solution in terms of discriminative forecasting ability. The primary utility of the present solution is the capability to integrate the entire time-evolving graph trajectory and to provision new risk predictions in a fully online fashion.

CHAPTER 4

INTRA-INDIVIDUAL DYNAMIC NEIGHBORHOOD IMPORTANCE

4.1 Introduction

We as practitioners have mental models of our domain expertise. Unexpected outcomes naturally invoke curiosity for us to update our mental model for future encounters. Given that we believe the outcome is not a fluke, the first step is to search for an explanation of the unexpected outcome. This hints at the psychological concept of trustworthiness, which is challenging to obtain when humans interface with unfamiliar inorganic matter [47], or rely on autonomous black-boxes [88, 114]. If users discredit a model's predictions, they will not use it. If decision-makers fail to understand a model's predictions, they cannot intervene. If researchers use complex models without explanations, scientific discoveries remain concealed. To earn human trust, augment decision-making, and facilitate learning, explanations are essential.

So what is required for a model to earn a human's trust? Accuracy alone may not suffice. For example, the deep learning architecture from Chapter 3 was effective at forecasting risk of complex systems, but the interpretation of such risk remains hidden. If (when) hazardous configurations are identified, it is natural to ask: which time-varying interactions are contributing to the provisioned risk?

Global feature selection algorithms reduce dimensionality by discovering the most pertinent subset of features to a model's outcome. But selecting features for a global model assumes the feature subset is optimal for every realization (data point). Such framework can be viewed as a nomothetic mechanism. This bears a severe limitation when forecast-

ing risk among complex adaptive systems because hazardous configurations may emerge from disparate connectivity patterns between persons. Local feature selection methods have received recent attention to combat this assumption, which attempt to discover relevant features for the outcome a given realization [27]. This nicely dovetails the philosophical goals of idiography, but has been far less explored in the temporal setting.

Motivated by the rationale above, this chapter seeks to contribute a local (idiographic) interpretability module to explain the forecasts provisioned to the dynamic graph embeddings from the previous chapter. The intended utility is to convey which time-varying edges are contributing to an athlete's risk. Is it important to distinguish between explaining a model versus explaining a model's forecast. I reckon the former is of less value to decision-makers in the field. If athletes are trained on a personal level then coaches may only want explanations for the forecast made by the model for a specific person. Selecting local (idiographic) temporally important edges contributing an athlete's provisioned risk is a novel inferential apparatus for sport epidemiology.

The following section critically examines the current local interpretability landscape for temporal data. Section 4.3 formulates the present problem to be addressed in Section 4.4. Section 4.5 establishes the validation experiments where the results are discussed in Section 4.6. Section 4.7 concludes the chapter.

4.2 Background

The opacity of machine learning models have been a major hindrance to wide-scale deployment in many fields [29]. This obliged methodologists to address the problem which blossomed explainable (interpretable) machine learning techniques [6]. Interpretable machine

learning is now a well-established sub-domain of data mining for conventional Euclidean data, but far less explored in graph mining. Further, research attention has thus far been predominated by static methods. The present problem is attempting to provide deep learning model transparency at the level of the observational unit (local) in a temporal setting. The main classes of local interpretability methods for Euclidean data will be discussed first. Methods which have contributed temporal variants will be acknowledged. The subsection will cover the interpretability methods developed specifically for graph mining.

4.2.1 Local Methods

Diverse classes have been proposed for explaining time series model behavior. A number of these approaches were explicitly designed for time series settings, while others are originally designed for static supervised tasks.

While the attention mechanism was initially crafted for machine translation [9, 179], it has also been utilized to reap insight into time series model behavior. Attention models are adapted to work with time series thanks to their memory mechanism and their ability to learn long-range dependencies across time [191, 109]. The parameters (i.e., attention weights) intend to explain model behavior through time. A first attention weights could not be attributed to observational units of time series due the complex mappings of recurrent latent spaces [81]. [30] addressed this by proposing to augment the attention mechanism using separate parameter sets to derive importance scores for features over time, while [80] modified the actual hidden states of an LSTM to derive a local attention framework. But it has recently been shown that model attention parameters can be modified without effecting the model's predictions [107]. This renders the association between local attention parameters

and feature importance rather blurred.

Attribution-based methods explain models by evaluating the importance of each feature on the output [162, 198]. This was coined the term saliency methods in image classification nomenclature [139, 1]. Attributions can be assigned using gradient-based methods to assess the sensitivity of a prediction to small changes in the input features [7, 170, 161]. Their proposition is the following: if a feature is indeed salient, it shall have a significant impact on the model's prediction when modified locally. Other methods linearly approximate local classifiers to explain model outcomes with respect to local references [131, 32, 148]. Most of these methods have been designed for static settings and must to be carefully extended to time series; the risk of vanishing gradients in the recurrent structures could severely comprise feature importance estimations [106].

Similar to attribution-based methods, perturbation-based methods assign importance to a feature based on changes to model output but by perturbing the feature itself [202]. Common approximations perturb the features with mean values or random noise [37]. But [193] has noted that such methods are vulnerable to out-of-domain explanations that may not reflect a model's true behavior. [24] then formulated the black-box interpretability as a multiple hypothesis testing problem and assessed significance of model's change by replacing features with counterfactuals [72].

4.2.2 Graph Mining

Interpretable graph mining is quite embryonic compared the Euclidean counterparts above. This subsection recognizes the accomplishments thus far, beginning with global (nomothetic) developments before zeroing in on local methods.

[63] applied contrast learning to graph analysis to reveal unique characteristics in one graph by contrasting with another. Contrast learning strives to identify meaningful differences between classes by reverse-engineering key features in a post-hoc fashion [10]. This work implemented contrastive principal component analysis to node-level statistics and attributes as a way to interpret differences between graph nodes. To my knowledge this is the first contrast learning attempt for graphs, but unfortunately their formulation is restricted to two graphs under consideration. Rather, the present problem would require a method to contrast an array of individual graphs. Lastly, [196] proposed to train a graph generator, instead of optimizing an input graph, so that the generated graph can maximize a target graph prediction. The generated graph is thus regarded as the explanation for the target prediction, which is expected to contain discriminative graph patterns.

[194] were the first to incorporate graph structure and feature information to explain predictions from complex graph neural networks. Given a graph instance, GNNExplainer identifies a compact subgraph structure and a small subset of node features which carry an important role in the graph neural network's classification via masking. They formulated an optimization task which maximizes the mutual information between the predicted class and distribution of possible subgraph structures. [184] continued this work with PGM-Explainer: given a predicted class to be explained, PGM-Explainer also identifies subgraph components but generates an explanation in form of a probabilistic graphical model.

Rather, [124] attempted to detect an importance probability for each node and edge in the graph instance through a single self-attention layer and sparsity regularization. [132] offered another architecture, PGExplainer: given an input graph, it obtains embeddings for

each edge, where the predictor uses these embeddings to predict the importance score as the probability of each edge being selected. Similar to PGExplainer, [152] trained a classifier to predict whether an edge can be removed without affecting the original predicted class, akin to perturbation-based designs discussed above.

4.3 Problem Formulation

Continuing from Chapter 3, let $\mathcal{E} \subset \mathbb{R}^{T \times V}$ be the dynamic graph embeddings for one athlete under consideration. A cumulative hazard forecast $\hat{F}(\cdot)$ is provisioned to the embedding matrix (Chapter 3). For notational ease, let $\hat{F}(\cdot)$ be labeled as $\mathcal{Y} \subset \mathbb{R}^{+}$ and serve as the output space. We can frame the dynamic graph embeddings as a time series $(\mathbf{e}_t)$, where the inputs $\mathbf{e}_t \in \mathcal{E}$ are indexed by time $t \in [1 : T]$ with $T \in \mathbb{N}$. We consider the problem of predicting a sequence $(\mathbf{y}_t)_{t \in [t_y : \tau]}$ of outputs $\mathbf{y}_t \in \mathcal{Y}$ indexed by the same time parameter but beyond the embedding time; $\tau \geq T$. This allows for many-to-one or many-to-many forecasting designs. Finally, $\mathbf{E} = (e_{t,i})$ denotes the matrix in $\mathbb{R}^{T \times V}$ where rows are time stamps and columns are neighborhood embeddings. Accordingly, $Y = (\mathbf{y}_t)$ represents the vector in $\mathbb{R}^{\tau \times 1}$.

The task is to explain the risk $Y = f(\mathbf{E})$ provisioned from the trained black-box f in Chapter 3. The utility is to identify the athlete-specific connectivity patterns $\mathbf{E}$ that are most contributory for f to reproduce the athlete's injury forecast Y. Resultant estimations serve to illustrate human-understandable adaptive trajectories to indicate how an athlete's risk may have formed throughout time.

4.4 Methods

Owing to prior graph interpretability literature, the formulated problem is approached through a perturbation-based perspective via masking. Inspirations for this chapter are rooted in the procedures from [143], much of which were reduced to fit the present context, but I will acknowledge their work when appropriate. The mask matrix is first defined in order to construct a perturbation mechanism before applying it to the dynamic setting.

A mask connected between an athlete's embedding matrix $\mathbf{E} \in \mathbb{R}^{T \times V}$ and black-box $f : \mathcal{E}^{T \times V} \to \mathcal{Y}^{\tau}$ is a matrix $\mathbf{M} = (m_{t,i}) \in [0,1]^{T \times V}$ of the same dimensions. Once trained, the elements of this matrix $m_{t,i}$ represent the magnitude of importance for neighborhood embedding i at time t for f to forecast $Y = f(\mathbf{E})$. Thus, a resultant mask coefficient near one implies the neighborhood embedding is importance, while a coefficient near zero implies the opposite.

[143] trains a convolutional neural network to map the mask to an input sequence. This is far too complex for the present context because there is only one dynamic graph instance under consideration at a time. Instead, an optimization problem is designed to learn a mask matrix by iteratively perturbing the dynamic embeddings.

4.4.1 Temporal Perturbations

A perturbation mechanism corresponding to a mask $\mathbf{M} \in [0,1]^{T \times V}$ is a linear operation to the embedding space $\Psi : \mathbb{R}^{T \times V}$. There are two assumptions for the perturbation to follow for any given $(t,i) \in T \times V$. First, the perturbation for $e_{t,i}$ is governed by $m_{t,i}$ as $[\Psi(\mathbf{E})]_{t,i} = \psi(\mathbf{E}, m_{t,i})$. The first assumption ensures the perturbations are independent amongst the embeddings. Second, given resultant mask coefficients imply importance, the

perturbation magnitude shall diminish for embeddings $e_{t,i}$ whose mask coefficient $m_{t,i}$ equals 1: $\psi(\mathbf{E}, 1) = e_{t,i}$. The second assumption ensures that embeddings of high importance shall remain unperturbed. ψ must also be differentiable for $m \in (0, 1)$ to update.

In order to appreciate the chronological trajectory, the perturbations are assembled from temporal neighbors of the embedding through time. The perturbation is thus temporal if a range exists $(B, t) \in \mathbb{N} \times \mathbb{N} \setminus (0, 0)$ such that for all $(t, i) \in T \times V$ there is:

$$\frac{\partial[\psi(\mathbf{E}, m_{t,i})]}{\partial e_{t',i}} \neq 0$$

for all $t' \in [t - B : t] \ \cap \ [1 : T]$. The parameter B is the width of a temporal window and controls how the perturbation behavior. $B > 0$ indicates that the perturbation includes previous time stamps with $t \geq 1$.

Such definition allows creativity for designing a perturbation mechanism. The present goal is to construct a mask while appreciating the temporality of an athlete's graph trajectory. This is imperative to capture influential configurational shifts that are ideally detected by the temporal perturbation mechanism.

The functional form of ψ is now defined. The temporal perturbation utilizes previous neighborhood embeddings to compute a rolling-average as:

$$\psi(\mathbf{E}, m_{t,i}) = \mu_{t,i} \cdot (1 - m_{t,i}) + e_{t,i} \cdot m_{t,i},$$
$$\mu_{t,i} = \frac{\sum_{t'=t-B}^{t} e_{t',i}}{B + 1}$$

with B as the window width. Hereafter, Ψ_B denotes the temporal perturbation mechanism corresponding to parameter B.

The complexity for computing a roster of k athletes requires $\mathcal{O}(T^k \cdot V^k \cdot B)$ opera-

tions; one must compute a perturbation for each $T \cdot V$ component of the kth dynamic graph embedding and each moving average involves $B + 1$ terms.

4.4.2 Objective Function

Designing an objective function for the optimization scheme should follow the intuition behind the intended target. It is evident that the perturbed and original dynamic embedding matrix shall be compared via the black-box forecasts. In theory, the resultant mask should conceal a large portion of $e_{t,i}$ which does impact $f(\mathbf{E})$ and thus inform the perturbation mechanism to preserve important elements of $e_{t,i}$. A straightforward formulation is to minimize the mean squared error between original and perturbed forecasts as:

$$\begin{aligned} \widehat{\mathbf{E}}_{t,i} &= \Psi_{\mathbf{M}}(\mathbf{E}_{t,i}) \\ \mathcal{L}_{mse}(\mathbf{M}) &= -\sum_{t=t_y}^{\tau} \Big([f(\widehat{\mathbf{E}})]_t - [f(\mathbf{E})]_t \Big)^2 \end{aligned} \tag{4.1}$$

to promote small deviations in $f(\widehat{\mathbf{E}})$ between iterations.

The next goal is to ensure proper identification of $e_{t,i}$ in the mask. Recall that $m_{t,i} = 0$ implies $e_{t,i}$ is insignificant toward f forecasts. This idea relates to the imposition of sparsity in $\mathbf{M}$. Previous work has applied the ℓ_1 penalty on the $\mathbf{M}$ coefficients [94]. However, [59] realized this tactic produces masks with unrelated regularization coefficients, and instead introduced a novel penalty term to impose sparsity on the $m_{t,i}$ elements:

$$\mathcal{L}_c(\mathbf{M}) = \|\text{vecsort}(\mathbf{M}) - \mathbf{r}_c\|_2 \tag{4.2}$$

where $\|\cdot\|$ denotes the ℓ_2-norm, and vecsort vectorizes $\mathbf{M}$ before sorting the elements in ascending order. In this case, $\mathbf{r}_c$ contains $(1 - c) \cdot T \cdot V$ zeros followed by $c \cdot T \cdot V$ ones,

where $c \in [0, 1]$. [59] declares this term invites $\mathbf{M}$ to only select a proportion c of $m_{t,i}$. This c hyper-parameter can be thought of as the mask constraint. I see both an advantage and disadvantage to this penalty term. The advantage is that this hyper-parameter can be tempered by practitioners to pinpoint progressively varying levels of importance. This is also a disadvantage: human-level tampering induces human generated bias as to what is deemed "important", enticing end-users down a road of confirmation bias.

For a fixed fraction a, the final optimization problem can therefore be formulated as:

$$\mathbf{M}_c^* = \min_{\mathbf{M} \in [0,1]^{T \times V}} \mathcal{L}_{mse}(\mathbf{M}) + \mathcal{L}_c(\mathbf{M})\lambda_c.$$

Note how c is held constant. To alleviate the aforementioned temptations, c is swept through a grid search of ten equidistant intervals between $[0.02 : 0.20]$. The final solution to this brute force search returns the smallest c with the smallest mean squared error between $[f(\widehat{\mathbf{E}})]_t$ and $[f(\mathbf{E})]_t$ written as: $c^* = \min\{c \in [0, 1] \mid \mathcal{L}_{mse}(\mathbf{M}_c^*)\}$. The intuition behind the heuristic is to quantitatively encourage the tightest constraint c which expressed the least forecast reproduction error.

4.4.3 Algorithm Design

The mask optimization algorithm is presented in Algorithm 4.1. Stochastic gradient descent is used to minimize the specified objective function. The notation $(0.5)^{T \times V}$ is a $T \times V$ matrix with all elements set to 0.5, which imposes zero *a prior* assumptions on the importance of each element. In practice this could be adjusted but this was not presently explored. Likewise, $(0)^{(1-c) \cdot T \cdot V}$ denotes a vector with $(1 - c) \cdot T \cdot V$ components set to zero and $(1)^{c \cdot T \cdot V}$ denotes a vector with $c \cdot T \cdot V$ components set to one. B, N, ω and α, denote

the temporal perturbation window width, number of epochs, learning rate, and momentum, respectively. The sparsity coefficient λ_c grows exponentially to reach a maximum value of $\delta \cdot \lambda_0$ at the end of the optimization.

The optimization scheme operates in two ways. The total loss is first influenced by $\mathcal{L}_{mse}(\mathbf{M})$ which initially increases the coefficients of important $m_{t,i}$ elements. The $\mathbf{M}$ coefficients are then polarized towards zero and one as λ_c enlarges over the epochs. After sufficient epochs the final mask coefficients are preferably near binary.

Algorithm 4.1 Dynamic Neighborhood Importance

Input: $\mathbf{E} \in \mathbb{R}^{T \times V}$, $B \in \mathbb{N}$, $N \in \mathbb{N}$, $\omega \in \mathbb{R}^+$, $\alpha \in \mathbb{R}^+$, $\delta \in \mathbb{R}_{\geq 1}$

$\mathbf{M} \leftarrow (0.5)^{T \times V}$; $\Delta\mathbf{M} \leftarrow 0$; $\lambda_c \leftarrow \lambda_0$; $\mathbf{r}_c \leftarrow (0)^{(1-c) \cdot T \cdot V} \oplus (1)^{c \cdot T \cdot V}$
for $i \in \{0.02, \ldots, 0.20\}$ **do**
  for $j \in \{1, \ldots, N\}$ **do**
    $\widehat{\mathbf{E}}_{t,i} \leftarrow \Psi_B(\mathbf{E}_{t,i})$
    $\mathcal{L}_{mse}(\mathbf{M}) \leftarrow -\sum_{t=t_y}^{\tau} \Big([f(\widehat{\mathbf{E}})]_t - [f(\mathbf{E})]_t\Big)^2$
    $\mathcal{L}_c(\mathbf{M}) \leftarrow \big\|\text{vecsort}(\mathbf{M}) - \mathbf{r}_{[i]}\big\|_2$
    $\Delta\mathbf{M} \leftarrow \alpha \cdot \Delta\mathbf{M} + \omega \cdot \nabla_{\mathbf{M}}[\mathcal{L}_{mse} + \mathcal{L}_c \lambda_c]$
    $\mathbf{M} \leftarrow \mathbf{M} + \Delta\mathbf{M}$
    $\lambda_c \leftarrow \lambda_c \cdot e^{\log \delta/N}$
  end for
end for
return $\mathbf{M}_{c^*} \in [0,1]^{T \times V}$

4.5 Validation

Evaluating the quality of explanations is challenging due to the absence of a ground truth. It further complicates the comparison when there is a lack of benchmarks in the same domain. Since explanations are reflective of model behavior, evaluations are intimately

tied to the performance of the model itself. Therefore, the functionality of the present optimization scheme is evaluated in a controlled simulated environment where the ground truth of important temporal observations are known *a priori*. The first synthetic dataset is simple and intended to identify the source of gain when adding temporality in subsequent experiments. The second dataset serves to judge the method in a elementary temporal environment, where the third dataset challenges the method under complex dynamics. A description of each synthetic dataset is presented below.

4.5.1 Synthetic Datasets

Dataset 1: Spike

The first simulation entails a time series problem where the outcome (y_t) erupts as soon as a spike is observed in the relevant neighborhood. This task is kept simple to serve as a sanity check before exploring more complex data generating processes. I anticipate the idiographic explanation to assign importance to the one *a priori* neighborhood, at the actual spike time, even in the presence of spikes in irrelevant neighborhood. In the present context, this would correspond to a node's degree to suddenly explode then vanish over time. Five independent sequences were generated from an auto-regressive moving average time series as:

$$e_{t,i} = 0.5 \cdot e_{t-1,i} + 0.2 \cdot e_{t-2,i} + 0.1 \cdot e_{t-3,i} + \epsilon_t \tag{4.3}$$

where $\epsilon_t \sim \mathcal{N}(0, 1)$. The spikes are uniformly added to each instance and every d neighborhood according to:

$$\text{ind}_d = Bern(0.5)$$
$$\gamma_d = \begin{cases} Pois(\lambda = 2) & \text{if ind}_d = 1 \\ 0 & \text{else} \end{cases}$$

where γ_d amount of spikes are sampled between $[1 : T]$. The corresponding output $y_t = (e_{t,i})^3 \; \forall \; t > t_1$ where, in this case, t_1 is the smallest sampled time stamp; the output jumps when a spike is encountered.

Recall that a trained model (f) is required to iteratively compare the temporal perturbations in Section 4.4.2. Since the data generating function of this dataset is known (Equation 4.3), training a separate model would be redundant. Therefore, the temporal perturbations are iteratively ingested into Equation 4.3 (instead of a trained model) for this spike experiment.

Dataset 2: ARMA

The second simulation generates a more realistic temporal phenomenon. Using Equation 4.3, the output (y_t) now depends on a known subset $F = F_T \times F_V \subset T \times V$ of important inputs, where F_T and F_V govern the important time stamps and neighborhoods, respectively:

$$y_t = \begin{cases} \sum_{i \in F_V} (e_{t,i})^2 & \text{if } t \in F_T \\ 0 & \text{else.} \end{cases}$$

Such sequence is generated with $t \in [1 : 50]$ for each $i \in [1 : 50]$. Only five neighborhoods were deemed as important, $F_V \sim \mathcal{U}([1, V])$, where the important time window was defined between $F_T = [10 : 30]$.

This simulation considers the scenario when the temporal function depends on a small portion of important neighborhood $|F_V| \ll V$. This task is deemed challenging for

global explainability methods [105], which is an interesting experiment for local methods herein.

Dataset 3: State-Space

The first two synthetic datasets did not necessarily evaluate temporal importance under complex dynamics. This dataset attempts to realistically simulate real-world complexity with known ground truth explanations. A non–stationary Hidden Markov Model (HMM) with two latent states, and a linear transition matrix, is used to simulate observations over time. Specifically, at each time step t, observations are sampled from the multivariate normal distribution determined by the latent state of the HMM. The state transition probabilities are modeled as a function of time to induce non-stationarity. The outcome y is a Normal random variable, which, in state one, is only governed neighborhood one and in state two, only governed by neighborhood two. The ground truth explanation for output at time t is the observation $e_{i,t}$ where i is the neighborhood that governs the output in the current state.

Formally, the two-state HMM with $\pi = [0.5, 0.5]$ and transition probability $C = \left[\begin{smallmatrix} 0.15 & 0.85 \\ 0.15 & 0.85 \end{smallmatrix}\right]$. The time series observations are sampled from the HMM emission probability for each state s_t formulated as a multivariate Gaussian: $\mathcal{N}(\boldsymbol{\mu}_{s_t}, \boldsymbol{\Sigma}_{s_t})$ where the mean vectors are governed by the state as $\boldsymbol{\mu}_1 = [0.1, 1.6, 0.5]$ and $\boldsymbol{\mu}_2 = [-0.1, -0.4, -1.5]$. Marginal variance for all features in each state is 0.8 with only neighborhood 1 and 2 being correlated ($\Sigma_{12} = \Sigma_{21} = 0.05$) in state 1 and only 0 and 2 in state 2 ($\Sigma_{02} = \Sigma_{20} = 0.05$).

Depending on the hidden state at time t, only one of the features contribute to the output y_t and is deemed influential. In state one, y only depends on neighborhood one and in state two, y only depends on neighborhood two as:

$$p_t = \begin{cases} \dfrac{1}{1+e^{-e_{t,1}}} & s_t = 0 \\ \dfrac{1}{1+e^{-e_{t,2}}} & s_t = 1 \end{cases}$$

where s_t is the state of the neighborhood at time t, which signals distinct sampling schemes through an indicator as:

$$\text{ind}_t = Bern(p_t)$$
$$y_t = \begin{cases} \mathcal{N}(0, 0.25) & \text{ind}_t = 0 \\ \mathcal{N}(1, 0.25) & \text{ind}_t = 1. \end{cases}$$

Recall that a trained model (f) is required to iteratively compare the temporal perturbations in Section 4.4.2. The first two datasets had the function available which did not warrant a separate f. However, a separate model is needed to produce a viable f for this experiment due to the stochastic complexity of this state-space process. A lightweight Elman recurrent neural network architecture [51] with one hidden layer and $d \cdot 2$ cells was trained to produce f for the optimization scheme. Batch normalization [104] and dropout (0.5) were used to combat over-fitting [165]. 1000 synthetic time series were generated for each experiment described above, 900 were used for training and 100 were held-out for testing.

4.5.2 Benchmark Algorithms

Evaluating the quality of the present explanations is complicated by the lack of benchmark methods in explaining local, high-dimensional, temporal forecasts; most of the available local benchmarks are designed for static settings. A handful of competitive candidates were selected and rolled through time to compare the performance of the Spike and ARMA experiments. Since these first two datasets have the regressive function readily available prohibits some of the more recent local explainability methods which rely on a black-box

function to approximate. However, some of these methods are considered for the State-Space experiment.

Integrated Gradient [170], Feature occlusion [171], and Feature Permutation [57] were selected to compare the performance of the Spike and ARMA explanations. These methods estimate the importance of each neighborhood at each time stamp for fair comparisons. Integrated Gradient utilizes the gradient of a model's prediction to produce a local importance score with respect to the neighborhood. Feature Occlusion assigns importance based on the difference in model prediction when each neighborhood e_i is replaced with a random sample from a Uniform distribution. Feature Permutation quantifies neighborhood importance by calculating the change in a model's prediction error after permuting the neighborhood; a similar variant to the perturbations in Section 4.4.1 but via permutations.

The State-Space experiment now qualifies black-box methods such as: Gradient-SHAP [131], RETAIN [30], and DeepLIFT [161]. Gradient-SHAP is a gradient-based method which computes Shapley values [159] used to assign local importance to neighborhoods and is a competitive baseline for static data. While RETAIN is a global importance method, it was chosen as it is a temporal attention-based method for recurrent neural networks. DeepLIFT is a gradient-based method which decomposes the model output by backpropagating local neighborhood contributions.

4.5.3 Performance Criteria

The definition for each experimental evaluation metric is listed in this subsection. Since the "true" important neighborhoods are explicitly known, we can measure the proportion of selected features that are truly important via precision. We can also measure the

portion of true important neighborhoods that were selected by the model via recall. For the sake of decision-making when such explanations may guide a practitioner's interventions, both sources of error are harmful: detecting erroneous edges (precision) and missing truly important edges (recall). I argue that recall is more critical in the present context and should judge the value of the explanatory mechanism. But this is subjective to an investigator's opinion and thus both metrics will be tabulated.

Formally, let $\mathbf{R} = (r_{t,i}) \in (0,1)^{T \times V}$ be a matrix whose elements indicate the true importance of the elements included in $\mathbf{E} \in \mathbb{R}^{T \times V}$. By definition, $r_{t,i} = 1$ if neighborhood $e_{t,i}$ is important and 0 otherwise. This allows us to frame the optimized mask as a binary estimator $\hat{\mathbf{R}}$ via:

$$\hat{r}_{t,i} = \begin{cases} 1 & \text{if } m_{t,i} \geq 0.5 \\ 0 & \text{else.} \end{cases}$$

Therefore, precision is defined as $|\mathbf{R} \cap \hat{\mathbf{R}}|$ / $|\hat{\mathbf{R}}|$, and recall is defined as: $|\mathbf{R} \cap \hat{\mathbf{R}}|$ / $|\mathbf{R}|$.

4.6 Results and Discussion

Evaluating local importance comes with two difficulties: the true importance is never known in real-world data, and that the performance of an importance method depends on the black-box performance. The first two experiments were under a regressive setting with known local temporal importance so that both difficulties were avoided. The third experiment trained a black-box model on a complex dataset with known local temporal importance and hence only the second difficulty was confronted.

Precision was not effective to differentiate between candidate methods in both the regressive Spike and ARMA experiments. The Spike data generation scheme may have been

too tailored for static methods as the present temporal perturbations performed inferior to the static recall performances, perhaps due to the spike location randomness. However, the source of gain for the temporal perturbations begin to diverge in the ARMA experiment.

Table 4.1. Temporal Importance Experiment
Mean (± standard deviation).

Sim.	$(T \times V)$	**Algorithm**	**Precision**	**Recall**
Spike	50×5	Integrated Gradient	0.99 (± 0.00)	0.92 (± 0.01)
		Feature Occlusion	0.99 (± 0.00)	**0.93 (± 0.01)**
		Feature Permutation	**1.00** (± 0.00)	**0.93 (± 0.01)**
		Present Solution:		
		$B = 3$	0.99 (± 0.01)	0.89 (± 0.02)
		$B = 9$	0.98 (± 0.01)	0.84 (± 0.03)
		$B = 15$	0.98 (± 0.01)	0.85 (± 0.02)
ARMA	50×50	Integrated Gradient	0.97 (± 0.01)	0.16 (± 0.04)
		Feature Occlusion	**0.98 (± 0.01)**	0.17 (± 0.03)
		Feature Permutation	**0.98 (± 0.01)**	0.16 (± 0.03)
		Present Solution:		
		$B = 3$	**0.98 (± 0.01)**	0.57 (± 0.03)
		$B = 9$	**0.98 (± 0.01)**	**0.59 (± 0.03)**
		$B = 15$	0.94 (± 0.02)	0.35 (± 0.03)
State-Space	50×5	Integrated Gradient	0.64 (± 0.03)	0.33 (± 0.03)
		Feature Occlusion	0.56 (± 0.06)	0.44 (± 0.06)
		Feature Permutation	0.54 (± 0.06)	0.40 (± 0.05)
		Gradient-SHAP	0.60 (± 0.06)	0.38 (± 0.03)
		RETAIN	**0.73 (± 0.06)**	0.15 (± 0.02)
		DeepLIFT	0.54 (± 0.06)	0.21 (± 0.02)
		Present Solution:		
		$B = 3$	0.63 (± 0.05)	0.60 (± 0.04)
		$B = 9$	0.70 (± 0.05)	**0.64 (± 0.05)**
		$B = 15$	0.69 (± 0.05)	0.63 (± 0.05)

The discrepancy between precision and recall between the local static methods in ARMA suggests that detecting erroneous edges was not a major source of error but that the

methods were consistently biased. The poor recall of Integrated Gradient, Feature Occlusion, and Feature Perturbation, exposes their inability to detect local (temporal) signals. Rather, the temporal perturbations were much better at discovering local ground-truth dynamics. However, performance in this higher dimensional experiment demonstrated sensitivity to the window width parameter B.

The complexity of the State-Space experiment warranted a black-box function to train and thus qualified other candidate solutions which also rely on a trained model. First to note is the systematic decrease in precision performance among all the candidate models from Spike and ARMA. This behavior is attributable to the HMM complexity; generating a less prominent signal produced more overfitting at the local level. Second to note is the performance behavior of RETAIN. RETAIN was the only model specifically designed for temporal importance but for global explanations. The expression of superior precision suggests that global temporal methods may mitigate overfitting to irrelevant edges, but suffer when asked to detect locally relevant edges. RETAIN's recall performance justifies the need for local temporal developments. Third to note is the superior recall performance of the present temporal perturbations. The iterative loss function was able to inform the mask matrix of temporally important edges in a more efficient manner than the candidate local methods. However, there is room for improvement as the State-Space experiment only tested the ability to detect edges between five nodes.

Another noteworthy aspect of the experiments was the query runtime. While speed was not the highest priority when developing the present solution, is it worth noting that the average run time for the Spike and State-Space was 6.8 seconds, whereas Integrated Gradient

and Feature Occlusion were < 1 second. This is specifically the runtime of generating a test instance after having trained f. The average run time for the temporal perturbations in the higher dimensional ARMA model was 40.1 seconds, compared to < 5 seconds of the candidate algorithms. This computational lag stems from the iterative nature of the optimization scheme and the brute force grid search for the optimal constraint hyper-parameter c.

4.6.1 Limitations

Machine learning is innately riddled with hyper-parameters, and optimization is no exception. Searching for the two hyper-parameters (window width and constraint penalty) exposed the Achilles' heel of the present solution: runtime. Computational speed was not fundamental goal during development, but the results were striking compared to other benchmarks. The sources of improvement are to address an alternative to grid search for the constraint parameter. Perhaps Bayesian optimization could be a promising exploration to stochastically tune the hyper-parameters in lieu of laborious grid search or human tampering.

Future work could also develop more sophisticated consistency tests for local explainability methods, like extant tests for image classification [173], but in a temporal setting. This could study the advantages or disadvantages of the present solution in greater detail.

4.7 Conclusion

This chapter proposed a local (idiographic) temporal importance method designed for estimating which dynamic graph embeddings most contribute to a provisioned forecast. The solution was formulated as an optimization problem where the forecasts from the original embedding matrix are iteratively compared to a temporally perturbed embedding matrix. The optimization objective thus searches for a reduced embedding matrix with the least

amount of error from the original forecasts. The present solution was evaluated through three simulation experiments and compared against multiple interpretable machine learning benchmark algorithms. The experiments conveyed promising performance, especially with respect to recall in complex temporal environments. However, runtime warrants future attention. The primary utility of the present solution is to estimate the intra-individual dynamic neighborhood embeddings most contributory towards the provisioned risk forecast. This is crucial when black-box forecasts require transparency for human decision-makers.

CHAPTER 5

END-TO-END APPLICATION FOR SPORT EPIDEMIOLOGY

5.1 Introduction

The goal of athlete monitoring is to capture how athletes are coping with imposed stressors in hopes to combat maladaptation. Non-invasive, functional monitoring methods such as jump-based assessments, have been utilized by sport practitioners for decades. With the increased availability of technology such as contact mats, contact grids, or force plates, jump analyses has boomed in the sport science literature. Jump-based tasks have been studied to help understand neuromuscular monitoring [136, 68], endocrine monitoring [146], inter-limb asymmetries [18], and risk factor identification [116]. While there are many jump assessments available, the countermovement jump has become a custom in today's athlete monitoring practices, possibly because of its superior reliability, factorial validity [133] and implementation ease. Deciphering complex jump kinetic profiles over time could support sport practitioners' when planning the chronological sequence of training, but the challenge goes beyond interpreting the highly relational data. Building complex inference pipelines entails significant technical investment that, unfortunately, can easily drop in priority for decision-makers in the field.

[153] claims that over 95% of effort into maturing a machine learning project is engrossed in software technicalities and less than 5% is dedicated to addressing true scientific questions at hand. This leads to a toilsome translational hurdle between useful inference techniques and any substantial athlete benefit that could be realized from improved decision support. What is desired is a seamless framework that encapsulates all major aspects of a

real-world athlete monitoring life cycle into one humming gestalt.

The purpose of this chapter is to thread each contribution from Chapters 2-4 into an end-to-end application for sport epidemiology. While each chapter's contribution can operate in isolation for their respective purposes, the comprehensive utility is to create and showcase a musculoskeletal injury forecasting pipeline. This chapter highlights the applicability of the proposed graph mining pipeline for nomothetic injury prediction and idiographic monitoring through an illustrative example of an authentic athlete monitoring dataset. To the best of my knowledge, the proposed longitudinal time-to-event graph mining pipeline is the first coherent effort to demonstrate the viability of a pragmatic injury prediction solution under the theory of complex systems.

The structure of this chapter deviates from the chapters above. The following section describes where the athlete monitoring dataset originated, how the covariate information was collected, and what constituted a valid injury from the Electronic Medical Records. Section 5.3 navigates how the dataset flows through the graph mining pipeline and declares how the injury forecasting experiment was validated and computationally implemented. Section 5.4 examines the forecasting discriminative performance and offers a case conceptualization of the highest risk athlete predicted from the hold-out test set. Section 5.5 concludes the chapter.

5.2 Student-Athlete Dataset

The present athlete monitoring dataset is authentic and intended to serve as a real-world exemplar. 891 student-athletes (male = 317, female = 574; mean age = 20.3 ± 0.64 years) from 14 different Division I National Collegiate Athletics Association (NCAA) teams

regularly assessed bilateral countermovement jump as a part of routine health and well-being surveillance. The retrospective dataset extracted injury instances and corresponding covariate information over the course of the 2017–2020 academic years.

5.2.1 Injury Criteria

De-identified injury data were retrieved from the University of Iowa Sports Injury Management Systems (FlanTech, Inc., Iowa City, IA, USA) HIPAA-compliant database. In order to homogenize classification scheme, each of the following criteria were to be adhered to before an injury instance was entered into dataset: 1) soft-tissue, non-contact, 2) of lower extremity, 3) occurred within practice or competition, and 4) resulted in time loss. For an injury to properly fit this criteria, the student-athlete must have been previously monitored from the bilateral countermovement jump described below. A total of 140 injury occurrences fit the aforementioned criteria.

Retrospective, de-identified data extraction was approved by University of Iowa Institutional Review Board #201812762.

5.2.2 Covariate Collection

Student-athletes performed a standardized warm-up before each testing session, which included dynamic stretching and progressively intensified locomotive patterns to physically prepare student-athlete for maximal jump testing. While starting in a standing position with feet placed hip-to-shoulder width and arms akimbo, student-athletes were instructed to stand with weight equally distributed on dual force platforms. Upon verbal command, student-athletes violently dropped into a countermovement position to a self-selected depth followed by a maximal effort vertical jump landing in an athletic position back onto the

dual force platforms. Each student-athlete performed three independent trials of the above instructions. For the present analysis, only the best singular jump per calendar day for each athlete was recorded on the basis of Jump Height via Impulse-Momentum [125, 145]. Aforementioned testing procedures were performed in sync with the natural progression of the respective team's training and competitive season.

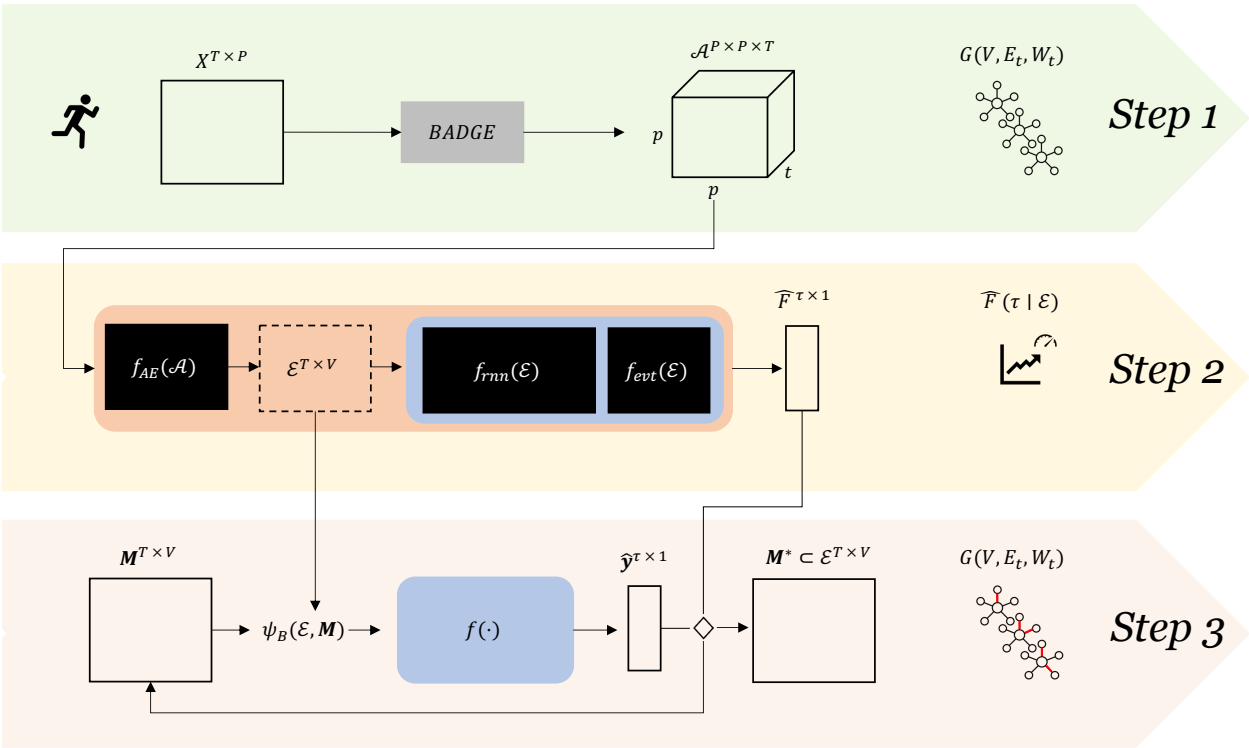
All countermovement jumps were performed on ForceDecks FD4000 Dual Force Platforms (ForceDecks, London, UK) sampling at 1,000 Hz. Commercially available ForceDecks software was used to analyse all countermovement jumps to calculate conventional performance variables. The kinetic parameters under consideration are supplied as a data dictionary in Appendix **??**. Covariates derived from initiation of movement to take-off were considered; landing forces were excluded due to poor reliability when take-off is unconstrained [2, 142].

5.3 Experimental Pipeline

This section describes the athletes' journey through the time-to-event graph mining pipeline, illustrated in Figure 5.1. An athlete's inference life cycle revolves around constructing individualized time-evolving graph sequences from raw temporal covariate observations, dynamic injury forecasting from resultant graph sequences, and estimating important edges over time contributing to the provisioned injury risk forecast. Each module involves all the intricacies of preparing, modeling, and tuning, and are interdependent for the logical flow to operate successfully. The end goal is to supply complementary levels of epidemiological inference under the complex systems theoretical constructs: 1) within-person idiographic inference by parameterizing dynamic graphs that characterize athlete-specific temporal evo-

lution, 2) nomothetic inference of injury risk by training a longitudinal time-to-event model from the retrospective dynamic graphs, and 3) bi-level inference by highlighting temporally important edges per athlete based upon the risks provisioned from the trained time-to-event model.

Figure 5.1. Graph Mining Pipeline



Step 1.

The first operation is to estimate athlete-specific time-evolving graphs from the raw temporal force plate observations. The raw data was extracted retrospectively from a real-

world collegiate setting which employed an athlete-centric monitoring philosophy. While team training sessions usually possess a stable weekly structure, measurements are still likely to be recorded in irregular time intervals within and between athletes. Capturing force plate measurements every Monday and Wednesday throughout pre-season training is an example of irregular intervals within athletes (i.e., two days, five days, two days, etc). If an athlete skipped a team training session – perhaps due to academic engagements – would then create irregular time intervals between athletes. But under the idiographic perspective, irregular time intervals between athletes is not of concern as the athlete-specific graph estimation process does not share information between athletes. The sparsity and irregularity of measurements may be seen as problematic for time-series modeling, but the proposed graph mining architecture was pragmatically developed with these issues in mind for real-world scenarios.

Time intervals for graph estimation can vary in practice given what a practitioner deems as meaningful in their context. For the sake of this project, graph estimation intervals were set to seven days for each athlete using the dynamic graph estimation under a Bayesian framework from [195], which was validated on similar sized datasets in Chapter 2. This equates to a weekly snapshot of an athlete's kinetic jump profile over time. The variational inference of the time-varying precision matrices in [195] temporally adapts the magnitude of regularization when estimating an athlete's sequential covariance matrices, and thus, is robust to an athlete's potentially sporadic data collection behavior.

The global time frame for dynamic graph estimation followed the natural course of data collection per team. Mandatory team training begins with an established pre-season

time block which contiguously transitions into the competitive season. Graph estimation was thus synchronized from the start of pre-season to the end of the competitive season for each academic school year per athlete.

Intra-individual dynamic graphs can effectively represent an athlete's temporal evolution throughout their season using Step 1. An athlete's topology can yield hypotheses for practitioners regarding potential mechanisms of coping strategies. Practitioners can begin to infer how to tailor heterogeneous interventions between athletes given someone's graph-level indices. But such graphs are unable to indicate risk for injury at this stage. We now port these dynamic graphs to Step 2 to learn which time-varying jump kinetic interactions lead to musculoskeletal injury occurrence.

Step 2.

A small amount of pre-processing is required before training the time-to-event model. All athletes were arranged based on their start of pre-season data collection to synchronize the time for comparing risk forecasts provisioned at different time points. Specifically, the time at which measurements were recorded and that at which injuries occurred was defined as the amount of time elapsed since the start of the pre-season; everyone began at time zero and increased thereafter. Final censoring times for athletes who remained injury-free were the end of their respective season's data collection period.

The format of the data at this point is an array of three-dimensional adjacency tensors; k amount of athletes each with T^k amount of $P \times P$ graph snapshots. Each t time slice is accompanied with the athlete's respective timestamp, injury or censoring time, and a binary label indicating injury status.

The first segment sequentially embedded the non-Euclidean dynamic graphs into a latent Euclidean vector space using the autoencoder architecture from Section 3.4.1. The semi-supervised objective function in Equation 3.2 idiographically transforms each adjacency tensor to a temporal neighborhood $T^k \times P$ matrix. Upon appending the k matrices, the final $kT \times P$ matrix of dynamic neighborhood embeddings was then randomly split by athlete into 80% training and 20% testing set. The training set was further partitioned by withholding 10% as a validation set to mitigate over-fitting during hyper-parameter optimization.

The temporal resolution for dynamic injury forecasts were set on a daily scale. Such granularity was enabled due to the high frequency of force plate jump monitoring. Dynamic forecasting was fixed *a priori* to seven days (7+), fourteen days (14+) and thirty days (30+) for consistent evaluation. These time scales are flexible to adjust to any dataset's resolution.

Completing Step 2 empowers practitioners to leverage historical data to forecast injury risk of current athletes in a dynamic online fashion. A coach can now identify who is at most risk within a meaningful temporal resolution. But when humans are confronted with something unexpected we instinctively ask why. We now port these dynamic neighborhood embeddings with their corresponding injury forecast vector to Step 3 to satisfy practitioners' curiosity and estimate which temporal edges are contributing to someone's provisioned injury risk.

Step 3.

This final phase revolves around the procedures discussed in Section 4.4. Holding the training and testing sets fixed from Step 2, the optimization scheme uses the provisioned temporal risk as the response label to estimate a reduced matrix of neighborhood embeddings

for each athlete in the hold-out test set.

Specifically, every athlete's graph snapshot over time t is labeled a risk in the form of cumulative hazard $\hat{F}(\cdot)$ from Step 2. Forecasts from an athlete's original $T \times P$ matrix are iteratively compared to the forecasts from a temporally perturbed embedding matrix. The optimization objective searches to minimize the error between the original forecasts and the reduced embedding matrix forecast reproduction. The designed loss function informs a masking matrix which conceals edges that do not impact the model's forecasts, and thus preserves temporally important edges.

Effectiveness of the loss function works in two ways. The total loss is heavily influenced by forecast comparisons which initially learns which edges are non-contributory. After sufficient iterations a penalty term begins to polarize the masking matrix toward the most temporally relevant edges.

Step 3 concludes the graph mining pipeline. Practitioners are now supplied with 1) intra-individual temporal dynamics from a graph-theoretic perspective, 2) forecasted injury risk of an athlete's time-evolving topological trajectory based on historical data, and 3) athlete-specific explanations for why their dynamic graph is expressing high (or low) injury risk in the near horizon. This resultant pipeline appreciates the theory of complex systems, supplies both levels of nomothetic-idiographic inference, and honors the longitudinal time-to-event problem formulation.

5.3.1 Validation

The ground truth does not exist for dynamic graph topologies in Step 1, nor the temporally important edges in Step 3. Performance of these two phases of the pipeline

relied on their cross-validation and neural network training epochs, respectively. But the ground truth was available for the time-loss injuries and censorship dates. Therefore, the performance and prognostic utility are judged solely on the longitudinal time-to-event model of Step 2.

The value of time-to-event models in the real-world should be evaluated predominantly by how the model discriminates between risk forecasts and secondly by the model's calibration. If a national meet is three weeks ahead, the coach and support staff want to know which of their athletes may warrant extra therapeutic attention to increase the odds of remaining healthy for the important day. Perhaps a harsh training phase is planned for next week and the physical preparation coach wants to know who is expressing vulnerable maladaptive signals. These narratives convey the types of real-world decisions being made and demonstrate the importance of correctly discriminating and prioritizing athletic care by means of risk. But if the risk forecasts are ill-calibrated – if large discrepancies exist between forecasts and ground truth outcomes – the model will disappoint as a prognostic tool.

Longitudinal time-to-event forecast accuracy in Step 2 of the experimental pipeline is evaluated using the time-dependent concordance index (CI_t) from Section 3.5.3 using Equation 3.4. Additionally, the time-dependent Brier score, $\mathrm{Br}_t(t, \Delta t)$, is used to evaluate calibration performance [77]. This metric follows the same temporal extension as CI_t but implies the mean squared error adjusted for Type I right-censoring. Using the forecasted injury risk given an athlete's dynamic graph embedding trajectory, $\hat{F}(\cdot)$, the time-dependent Brier score is defined as:

$$\mathrm{Br}_t(t, \Delta t) = \mathbb{E}[(\hat{F}(t + \Delta t \mid \mathcal{E}) - \mathbb{1}(T < t + \Delta t))^2]$$

where t indicates the prediction time which is the time when the prediction is made to incorporate dynamic predictions and Δt denotes the evaluation time which is the time elapsed since the prediction is made. CI_t is interpreted as higher the better whereas Br_t is interpreted as lower the better.

5.3.2 Implementation

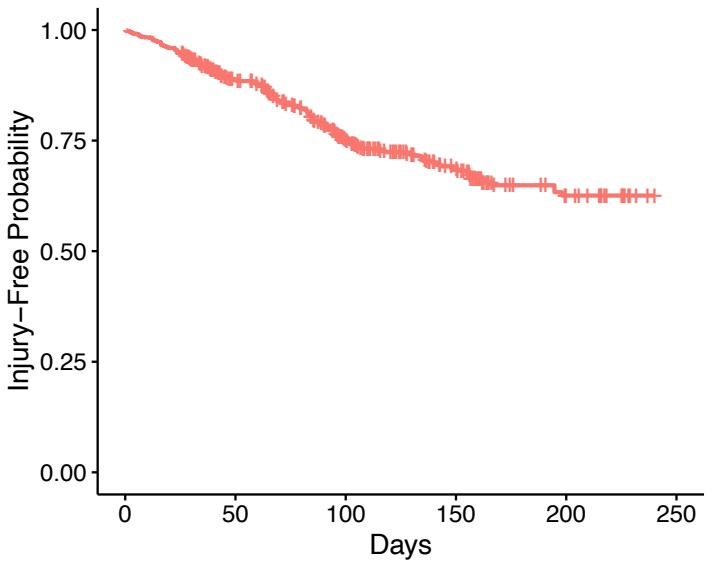
All experiments were computed on the University of Iowa's Argon High Performance Computing Cluster. The graph estimation procedures in Step 1 and dynamic neighborhood embeddings in Step 2 operate independently between athletes which creates an embarrassingly parallel workload. Therefore, Step 1 and the first segment of Step 2 were computed in a parallel design across 50 cores on a 56-core 512GB CPU configuration. Deep learning hyper-parameters of the network structures in Step 2 (i.e., activation functions, coefficients, number of nodes and hidden layers, etc), were selected via random search [16]. Table 3.1 lists the permissible values of the hyper-parameter search for each architecture, respectively.

5.4 Discussion

Figure 5.2 displays the naïve injury distribution of the entire dataset before train/test split via the Kaplan-Meier estimator. No confidence intervals are included as this is not intended for inference. Notice the times for censorship were restricted to at least 25 days. This heuristic enforced an athlete's longitudinal force plate monitoring to last a minimum of two weeks prior to entering the dataset. The ebb and flow thereafter implies either censorship or lower extremity, non-contact, time-loss injury occurrence during sanctioned practice or

competition. Censorship becomes more sparse when $t > 175$ as the number of athletes dwindled in such long durations; most sport season lasted less than 150 days. Recall that the dataset is an accumlation of almost four calendar years, synchronized by the start of the respective team's pre-season of that year.

Figure 5.2. Injury Distribution



5.4.1 Forecast Performance

The discriminative and calibration performance of the present experiment are reported in Tables 5.1 and 5.2, respectively. Mean and standard deviations of CI_t and Br_t were obtained by 10 random trials as described in Section 5.3. The prediction and evaluation times were held constant throughout the tests for equal comparisons.

Table 5.1. Discriminative Performance
Mean CI_t ($\pm$ standard deviation).

Pred Time	**Eval Time**		
	+7	+14	+30
$t = 60$	0.54 ($\pm$ 0.07)	0.55 ($\pm$ 0.08)	0.53 ($\pm$ 0.07)
$t = 105$	0.66 ($\pm$ 0.06)	0.62 ($\pm$ 0.06)	0.60 ($\pm$ 0.07)
$t = 150$	0.68 ($\pm$ 0.05)	0.69 ($\pm$ 0.05)	0.68 ($\pm$ 0.05)

The behavior of the forecasting window expressed an unsurprising inverse relationship: increasing the temporal proximity declines the predictive power. What is more amusing is the systematic increase in discriminative and calibration performance when prediction time was further into the season. First, because the dataset only considered the first event of time-loss injuries, the sample size inherently decreases through time. The naïve intuition is that performance could potentially suffer with fewer samples to learn from at these longer time periods. But the behavior suggests that the long-range dynamics improve the forecasting performance and that appending within-person information can strengthen acute event forecasts. Secondly, the competitive Br_t deep into the season ($t = 150$) demonstrates the prognostic strength of the event probabilities with long temporal graph trajectories; athletes provisioned a low cumulative hazard were more likely to be censored and athletes provisioned a high cumulative hazard were more likely to sustain a time-loss injury.

By virtue of the graph mining pipeline structure, whenever a new athlete measurement is accrued, a new graph snapshot can be constructed in Step 1, dynamically provision new risk forecasts in Step 2, and estimate important edges in Step 3, in a fully online fashion. This online characteristic preserves the pragmatism for sport practitioners to flexibly model their athletes according to their context-specific monitoring environment.

Table 5.2. Calibration Performance
Mean Br_t ($\pm$ standard deviation).

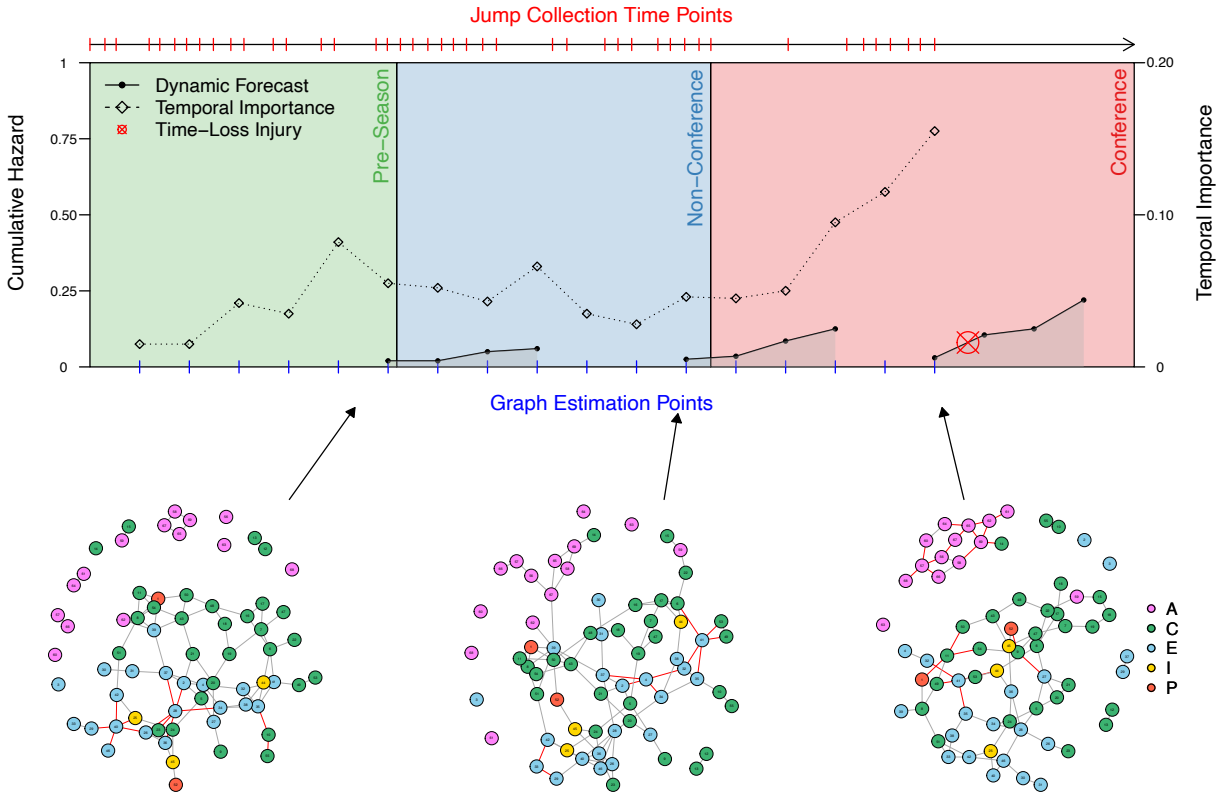
Pred Time	**Eval Time**		
	+7	+14	+30
$t = 60$	0.22 ($\pm$ 0.03)	0.22 ($\pm$ 0.03)	0.22 ($\pm$ 0.04)
$t = 105$	0.16 ($\pm$ 0.06)	0.15 ($\pm$ 0.06)	0.16 ($\pm$ 0.05)
$t = 150$	0.07 ($\pm$ 0.05)	0.08 ($\pm$ 0.06)	0.09 ($\pm$ 0.05)

5.4.2 Novel Idiographic Inference

This subsection explores a unique case conceptualization and exhibits a novel inferential apparatus for sport epidemiology. Upon completing the experiment, one student-athlete was drawn from the hold-out test set based on the highest provisioned +30 risk within the last prediction time period. The criteria was chosen owing to the superior calibration performance witnessed at the longer duration. Figure 5.3 includes many dimensions of the experimental results and will be carefully elaborated upon.

The red hashes spread across the top panel indicate when the raw force plate measurements were collected. Each hash represents a calendar day where the student-athlete performed routine jump surveillance before a team-based training session (i.e., sport practice or conditioning). Only one out of the three jumps were kept for the present analysis as described in Section 5.2.2. The blue hashes spread across the bottom panel indicate when the dynamic graph snapshots were estimated from Step 1. The colored background segments correspond to the sport's competitive period for the particular student-athlete. This type of information is not included in the model but was placed in the graphic for contextual

Figure 5.3. Student Athlete Dynamic Risk Profile



relevance.

The last graph snapshot of each competitive period, as well as the final snapshot recorded, were chosen to show meaningful evolutionary time points across the season. The grey shaded regions represent the forecasted injury risk at +0, +1 week, +2 weeks, +3 weeks, provisioned to the respective graph snapshots under consideration. The numeric values of the forecasted risk correspond to the left y-axis scale and were estimated from Step 2 of the pipeline. The diamonds connected by a dotted line represent the temporal importance

[107] assigned to each snapshot when making the final forecast. To be clear, the right-most blue hash is the final snapshot, which accepted the entire dynamic graph trajectory when provisioning the forecast. The diamonds are the relative contribution of each snapshot during this estimation and coincide with the right y-axis. The sum of this vector equals 1.

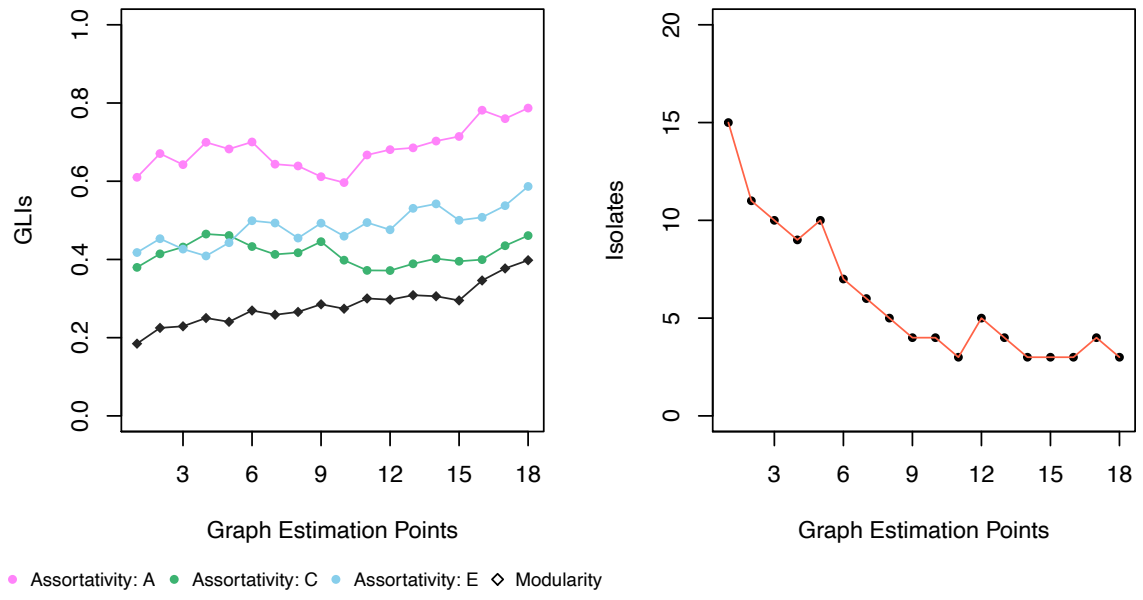
Lastly, the three structures below the panel are the aforementioned graph snapshots. Each node represents a force plate variable, listed in Appendix **??**, where colors segment the nodes by jump phase. Edges connecting nodes are partial correlations, where a lack of an edge between nodes denotes independence. Red edges were selected from Step 3, which imply contributory information to the provisioned forecast.

There are a number of prominent characteristics upon visual inspection which will now be synthesized. First to note is the topological evolution through time. This athlete's raw high-dimensional kinetic jump profile modified enough to warrant the covariance configuration to adapt. Referring to Figure 1.2, external load experienced by an athlete directly influences the evolution of internal load responses. Extrinsic stimuli, such as external loads, were not captured in the present experiment and thereby prohibits causal analysis into the covariance evolution. I acknowledge this limitation is the study design, but remaining agnostic between the diverse team settings – by being inclusive of the internal covariate stream – was of higher priority than incorporating complicated and heterogeneous external loads. But the empirical forecasting results, however, dynamically updated from the evolutionary trajectory independent of external covariates.

The dynamic forecast behavior is another characteristic to note. We see that the graph snapshot at the end of pre-season produced a forecast with minimal cumulative hazard,

slightly increased at the end of the non-conference schedule, but jumped at the final graph snapshot five weeks into the conference schedule. Our only indication as to why resorts to the edge importance estimation per snapshot. This brings our attention to the right-most topology corresponding to the final recorded snapshot, which contains a dense cluster of asymmetry covariates with many edges selected as important. Such cluster does not appear in the earlier snapshots, which alludes to a hallmark concept of complexity theory: emergence. Asymmetry covariates began in an archipelagic formation with many isolates, slowly attached edges during non-conference, and emerged as a dense cluster within the final weeks.

Figure 5.4. Graph-Level Indices Over Time.

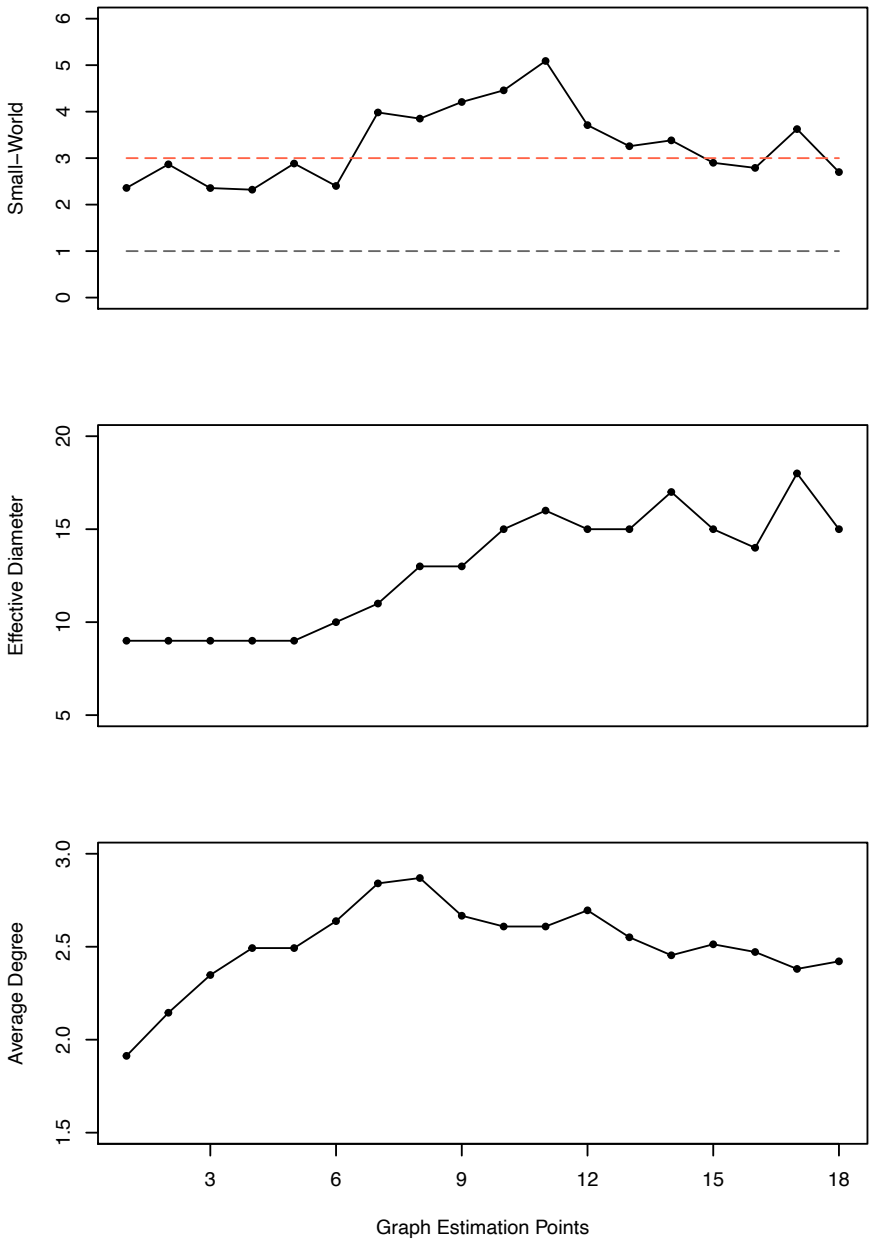


The graph-level indices in Figure 5.4 quantitatively substantiate this emergence concept. Displaying the assortativity over time of the three major node groupings (left; concentric, eccentric, asymmetry) helps parse the emergent behavior. The strong homophilic effect for asymmetry in the first half is perhaps inflated by the number of isolate nodes (shown on the right), which dropped precipitously. Witnessing the slow assortative increase, once the isolates seemingly plateaued, indirectly conveys that more asymmetry edges were being formed between each other. The gradual upward drift in modularity also corroborates the emergent cluster, especially in the final weeks.

It is interesting to note how the observed dynamic graph portrays a small-world structure, as opposed to Erdős-Rényi or scale-free behavior. Small-world topologies tend to possess high transitivity, dense local hubs, and few edges connecting distant portions of the graph [187]. The implication for this structure in the present context implies that most information – encoded as partial correlations – flows through a small number of highly connected hubs [141]. This small-worldness concept is quantified and statistically categorized using the S^{Δ} index [100] over time, and visually displayed in Figure 5.5. Grey and red horizontal lines indicate small-world thresholds for borderline ($1 \leq S^{\Delta} \leq 3$) and significant ($S^{\Delta} > 3$), respectively [100]. Intriguingly, the effective diameter dynamics in Figure 5.5 follows the inverse of the shrinking diameter theory from [120], but the empirical graph does follow the densification theory from [120], depicted by the increasing connectedness via average degree [52] in Figure 5.5.

We shall take a moment to reflect on the inferential language expressed hitherto. The dynamic injury forecasts and temporal edge selection may appear idiographic as they are

Figure 5.5. Structural Behavior Over Time.



estimated per an individual's dynamic graph. However, these estimates were derived from a trained model that had previously observed over 700 athletes; a nomothetic mechanism. Such interplay bestows a unique bi-level inference: within-person dynamic graph behavior

renders idiographic insight, while dynamic injury forecasts and temporal edge selection are nomothetic functions applied to an individual.

The novelty of this inferential language also complicates the linkage to prior literature. For example, the present time-to-event model did not accept raw force plate data, but instead, learned from time-evolving covariance structures. And more prevalent use of intra-individual graph mining is warranted before justifying this case study's idiographic generalizations of asymmetry cluster emergence as long-range risk patterns preceding injury. But perhaps graph parameterization dynamics simply offer, to a certain extent, novel shapes of well-documented concepts. In force plate literature for example, there has been heavy research activity regarding jump asymmetries as risk factors for injury [18, 91, 3]. [145] nomothetic topology inductively discovered that non-contact injury was conditionally independent given the magnitude of eccentric braking force asymmetry the week prior, and eccentric peak force asymmetry the week of jump measurement. Parallel inferences acquired from vastly disparate mechanics.

One final characteristic to note is the temporal importance mechanism. Recall that the diamonds connected by dotted lines in Figure 5.3 quantifies each graph snapshot's relative contribution to the final injury forecast. The most recent snapshot was indeed the most important when provisioning injury risk, which fell dramatically as the chronological proximity lengthened. This may sound obvious, and is the pivotal theory behind the landmarking strategy [177, 178]. But the present recurrent neural network architecture accepts the entire historical trajectory when provisioning risk, and is demonstrated by the case study's non-zero temporal importance in the long-range dynamic graph.

5.4.3 Limitations

Please note that the graph mining pipeline herein is not exclusive to force plate monitoring. The ideology and application are agnostic to any internal load monitoring tools available to a coach or sport practitioner. The force plate dataset was graciously granted from the University of Iowa Department of Intercollegiate Athletics to serve as an operational example. While the analysis supported epidemiological inference, the emphasis was on the graph mining pipeline contributions.

The observed forecasting discriminative performance I hypothesize that by widening the scope of internal load covariates would increase performance. Modern athlete monitoring environments capture numerous systems of an athlete. Relying on injury forecasts from one monitoring apparatus in isolation, as presented in this chapter, is conceptually restricted in most real-world applications.

The mechanics of dynamic Gaussian graphical model estimation produced small-world topologies. Such topological behavior perhaps was conducive for the longitudinal time-to-event architecture to focus on sparse evolutions, whereas hierarchical evolutions from scale-free environments may have been more challenging. Future work is needed to investigate the longitudinal time-to-event performance under varying graph geometries.

5.5 Conclusion

Machines will never replace a coach's judgment. But in the spirit of data-supported decision-making, this chapter's contributions empower practitioners with a longitudinal time-to-event graph mining pipeline inspired by complexity theory – so coaches can spend more time on the applied problems, and methodologists can concentrate on the technical barriers.

I presented an end-to-end graph mining pipeline and illustrated its capability for revealing nomothetic and idiographic inference for sport injury mitigation in real-world environments. The pipeline entailed three steps, 1) dynamic graph construction per athlete, 2) a longitudinal time-to-event model trained from the dynamic graphs, and 3) athlete-specific temporally important edge estimation. An authentic collegiate student-athlete monitoring dataset was fully ingested through the pipeline as a pragmatic experiment. The discriminative and calibration performance of the time-to-event model was moderate for short-range prediction times, but rapidly improved when more data was accrued per individual. Further, the forecasting window width and predictive power were inversely related. A novel inferential apparatus was then presented by blending all three steps together to derive innovative idiographic insight.

CHAPTER 6

CONCLUSIONS

Injury is pervasive to sport and is an important yet challenging forecasting problem with high practical value. Injury prediction algorithms have proliferated but with scant progress. This thesis was devoted to contributing a novel, well-formulated approach for injury forecasting while appreciating athlete-specific physiological complexities. The broader impact of such work is to mend the gap between machine learning capabilities and their actual effectiveness for sport-related musculoskeletal injury mitigation. Chapter 1 justified my arguments that forecasting from internal load (biomarkers) should enable idiographic inference under a systems biology worldview. Such propositions situated this thesis within the theory of complex systems and formulated injury forecasting as a graph-theoretic longitudinal time-to-event problem. Such objective was decomposed into three Specific Aims that were addressed throughout this thesis as follows.

Specific Aim #1 set out to address the intra-individual dynamic graph construction problem, detailed in Chapter 2. Techniques for estimating graph structures from raw data matrices were formally surveyed. A range of challenging synthetic datasets were then simulated to validate the temporal edge reconstruction performance of the most competitive solutions which assume temporal smoothness. Results expressed that variational approximations are an efficient estimation strategy for both small and high-dimensional datasets. The Bayesian covariance selection strategy exhibited competitive false discovery rate compared to maximum likelihood-based methods. However, dynamic topology inferences systemati-

cally suffered from severe bias; the source of error primarily arose from the inability to fit true edges. The dimensionality of raw observations should be carefully considered before embarking on dynamic graph estimation.

Specific Aim #2 sought to contribute a longitudinal time-to-event forecasting model from dynamic graphs as inputs, detailed in Chapter 3. Unlike Chapter 2, there was considerable dearth in the time-to-event modeling regarding dynamic graphs. The proposed solution interlaced deep learning and graph mining domains into a custom time-to-event formulation. Two longitudinal time-to-event benchmark algorithms served to validate the present forecasting framework under a realistic simulation experiment, conditional on graph-theoretic pre-processing. The proposed solution exhibited superior discriminative forecasting performance, especially with long-range graph trajectories. The major limitation was the lack of available graph-theoretic benchmarks, which perhaps biased the performance results. Future studies should scrutinize the present advantages and disadvantages in greater detail.

Specific Aim #3 focused on selecting temporally important edges to help explain the forecast provisioned to a particular dynamic graph, detailed in Chapter 4. Local interpretability methods for temporal data has received far less research attention compared to the static counterpart. A temporal perturbation was formulated inside an optimization scheme, which iteratively minimized the perturbed forecasts against the original forecast. Extensive simulations tested whether the local temporal perturbations could detect chronologically relevant edges compared to local static methods and global temporal methods. The proposed solution outperformed with respect to Type II error, whereas Type I error was less distinguishable amongst the benchmarks. The shortcoming was the required runtime to

query one dynamic graph instance.

Chapter 5 coalesced the three Specific Aims into an end-to-end application for sport epidemiology. An authentic athlete monitoring dataset was retrospectively extracted from University of Iowa Department of Intercollegiate Athletics. Athlete-specific dynamic graphs were constructed from longitudinal force plate data using the methodology from Chapter 2. The longitudinal time-to-event model from Chapter 3 was trained and tested to forecast non-contact musculoskeletal injury from the dynamic graph dataset. Athlete-specific injury forecasts were then explained using the architecture from Chapter 4, where novel idiographic inferences into the temporal interactions leading to injury were unveiled.

6.1 Future Work

To what extent time-to-event graph mining serves as an exhaustive solution for injury prediction remains to be seen; there are certain injury types that clearly fit the framework better than others. Forecasting acute injuries with well-delineated onset appears reasonable. It is less obvious how such formulation could forecast the genesis of steadily emerging symptoms from injuries deemed as chronic. These common musculoskeletal ailments are likely to characterize temporal interactions as well, but potentially involve evolutionary processes that unfold in hierarchical time scales. Whether such injuries are conformable to a graph-theoretic approach, and what such a graph would look like, is therefore a question for future investigation.

Extending this applied note, the present formulation and corresponding dataset considered forecasting the time-to-event of an athlete's first time-loss injury; an athlete was deemed healthy preceding the injurious event and was removed from the dataset thereafter.

But athletes may encounter multiple, less severe injuries within the duration of one season. This would be a recurrent event formulation, and was not incorporated into the present problem. Expanding this additional formulation is an important avenue for future research attention.

The only covariate information fed to the longitudinal time-to-event model is an array of intra-individual dynamic graph topologies. However, there may be cases when practitioners wish to include static variables into the model, such as comprehensive baseline measurements, or fixed attributes such as anatomical restrictions. Incorporating this level of flexibility to handle both static and temporal data for graph construction methodology is also of practical importance.

6.2 Closing Thoughts

The first steps in substantiating complexity theory were already taken in the form of preliminary narratives that framed the injury prediction problem via athlete monitoring as an abstract construct [19, 168, 60]. The graph-theoretic longitudinal time-to-event model, as advanced here, complies to the mature theoretical framework and offers a coherent problem formulation for forecasting non-contact musculoskeletal injuries. Assuming the use of time-to-event graph mining will become increasingly abundant in time, the second empirical step is to connect individual differences in topological trajectories to individual differences in maladaptive states. The next step would be to incorporate external covariates that instantiate certain graph connectivity patterns into the modeling framework. This progression shall allow us to effectively coordinate heterogeneous treatment modalities tailored to an athlete's maladaptive graph signals.

Predictive connections between dynamic graph signals and injuries discerned from my sequence of prior work, as well as the present insights from Chapter 5, are arguably hackneyed insights; lack of sleep coupled with high perceived stress days prior [147], elevated eccentric braking asymmetry the week prior [145], or the emergence of a modular asymmetry cluster within weeks prior (Section 5.4.2). Perhaps these connections are rooted in very ordinary physiological processes. And this is captivating because it means that injuries may not be ill-understood transcendental events, where the nature of which is yet to be uncovered by future esoteric researchers. Rather, the fact that we can capture sets of basic symptoms, and intuitively understand many of the relations between them, means that we already have a reasonable mental model for why injuries occur and how to mitigate them.

If so, our preceding lack of progress of injury forecasting from empirical data may not have resulted from small sample sizes or noisy instruments, as is typically supposed. Instead, we may have simply lacked a coherent problem formulation under a sound theoretical framework to effectively learn from the available data.

APPENDIX A

SUPPLEMENTARY MATERIAL

A.1 Covariate Data Dictionary

Table A.1. Glossary of jump-based athlete monitoring dataset.

Covariate	Unit	Group	Node ID
Jump Height Impulse-Momentum	cm	P	1
Braking Phase Duration:Concentric Duration	%	E	2
Braking Phase Duration:Contraction Time	%	E	3
Braking Phase Duration	s	E	4
Concentric Duration	ms	C	5
Concentric Impulse-50ms	Ns	C	6
Concentric Impulse-100ms	Ns	C	7
Concentric Impulse	Ns	C	8
Concentric Mean Power / BM	W/kg	C	9
Concentric Peak Force / BM	N/kg	C	10
Concentric Peak Velocity	m/s	C	11
Concentric RFD-50ms	N/s	C	12
Concentric RFD-100ms	N/s	C	13
Concentric RFD-200ms	N/s	C	14
Concentric RFD / BM	N/s/kg	C	15
Concentric RPD-50ms / BM	W/s/kg	C	16
Concentric RPD-50ms	W/s	C	17
Concentric RPD-100ms / BM	W/s/kg	C	18
Concentric RPD-100ms	N/s	C	19
Concentric RPD / BM	W/s/kg	C	20
Concentric RPD	W/s	C	21
Concentric Time to Peak Force	ms	C	22
Contraction Time:Eccentric Duration	%	C	23

CMJ = Countermovement jump, s = seconds, ms = milliseconds, W = Watts, N = Newtons, kg = kilograms, m/s = meters per second, cm = centimeter.

Covariate	Unit	Group	Node ID
Contraction Time	ms	C	24
CMJ Depth	cm	I	25
Eccentric:Concentric Duration Ratio	%	E	26
Eccentric:Concentric Mean Force Ratio	%	E	27
Eccentric Acceleration Phase Duration	s	E	28
Eccentric Braking Impulse	Ns	E	29
Eccentric Braking RFD-100ms / BM	N/s/kg	E	30
Eccentric Braking RFD-100ms	N/s	E	31
Eccentric Braking RFD / BM	N/s/kg	E	32
Eccentric Deceleration Impulse	N/s	E	33
Eccentric Deceleration Phase Duration	s	E	34
Eccentric Deceleration RFD / BM	N/s/kg	E	35
Eccentric Duration	ms	E	36
Eccentric Mean Braking Force	N	E	37
Eccentric Mean Deceleration Force	N	E	38
Eccentric Mean Force	N	E	39
Eccentric Mean Power / BM	W/kg	E	40
Eccentric Mean Force / BM	N/kg	E	41
Eccentric Peak Velocity	ms	E	42
Force at Peak Power	N	C	43
Force at Zero Velocity	N	I	44
Mean Eccentric/Concentric Power Time	W/s	I	45
Minimum Eccentric Force	N	E	46

CMJ = Countermovement jump, s = seconds, ms = milliseconds, W = Watts, N = Newtons, kg = kilograms, m/s = meters per second, cm = centimeter.

Covariate	Unit	Group	Node ID
P1 Concentric Impulse	N/s	C	47
P2 Concentric Impulse	N/s	C	48
Peak Net Takeoff Force / BM	N/kg	C	49
Peak Power / BM	W/kg	C	50
Positive Impulse	N/s	C	51
RSI Modified	ms	P	52
Takeoff Peak Force / BM	N/kg	C	53
Velocity at Peak Power	ms	C	54
Peak Takeoff Acceleration	ms	C	55
Concentric Impulse-100ms Asymmetry	% LR	A	56
Concentric Impulse Asymmetry	% LR	A	57
Concentric Peak Force Asymmetry	% LR	A	58
Concentric RFD Asymmetry	% LR	A	59
Eccentric:Concentric Mean Force Asymmetry	% LR	A	60
Eccentric Braking RFD-100ms Asymmetry	% LR	A	61
Eccentric Braking RFD Asymmetry	% LR	A	62
Eccentric Deceleration Impulse Asymmetry	% LR	A	63
Eccentric Mean Force Asymmetry	% LR	A	64
Eccentric Peak Force Asymmetry	% LR	A	65
Force at Peak Power Asymmetry	% LR	A	66
Force at Zero Velocity Asymmetry	% LR	A	67
Positive Impulse Asymmetry	% LR	A	68
Takeoff Peak Force Asymmetry	% LR	A	69

CMJ = Countermovement jump, s = seconds, ms = milliseconds, W = Watts, N = Newtons, kg = kilograms, m/s = meters per second, cm = centimeter.

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